December 22, 2025

Spectrum of Energy

The Spectrum of Energy aims to consolidate and assess events, data and forecasts that affect the broad 'spectrum' of global energy, considering all sources, technologies, policies, and financial implications as we transition to a lower carbon environment.

Front Section – In Focus:

- **Looking back at 2025, the first year of the Trump II era,** part I = *Pump for Trump:* Opec adds hugely to supply which *deflates* oil prices and allows the administration to put the screws on key adversaries (p. 4)
- **Part II = No Peace, Geo-politically, It's:** year 4 of Tsar Putin's war on Ukraine, year x of Xi Jinping's expansion of the middle kingdom's reach; and a historic one for the Mideast and it births the "***Donroe Doctrine"*** (p. 5)
- **Part III** Looking back on the US there's the knee-capping of green-subsidies and -projects in favor of fossil fuels freshly included in a new energy-strategy of "all of the above" and American "dominance" (p. 6)
- **Part IV highlights** green shoots for nuclear, batteries, EVs, solar power to AI and American LNG (p. 7)

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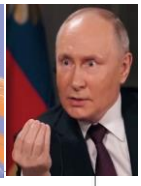
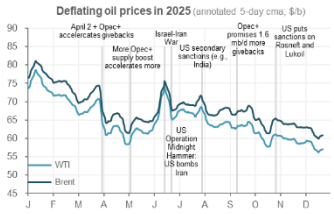
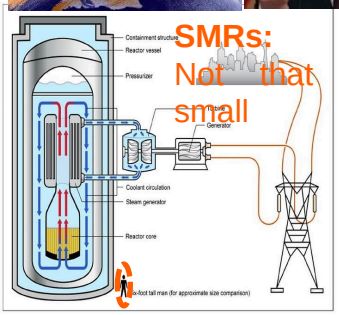
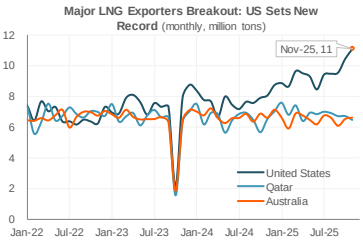
Performance of our global-energy portfolio
We track 600+ names across energy sub-sectors

Sector	MTD	YTD	MCAP $b
Integrated Oils, Ref, Marketing	-3.1%	13.4%	$3,842
New Energy Vehicles	2.8%	12.8%	$2,829
Mining (e.g. Metals, Coal, etc.)	5.0%	57.6%	$1,788
Power Generation	-4.4%	17.4%	$1,160
Infrastructure & MLPs	-2.1%	4.3%	$859
Oil & Gas Exploration & Prod	-5.6%	-4.9%	$526
Wind	-0.6%	41.3%	$443
Solar	-2.3%	47.0%	$398
Battery Storage	-2.4%	35.4%	$373
Oilfield Services	1.0%	10.9%	$249
Power Infrastructure Services	-4.6%	62.1%	$233
Hydrogen	-3.4%	44.0%	$28
Energy Portfolio	**-2.2%**	**25.7%**	**$12,728**









Thank you for your support
Wishing you Great Holidays and a Happy New Year

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In the News and in Focus – Key Themes

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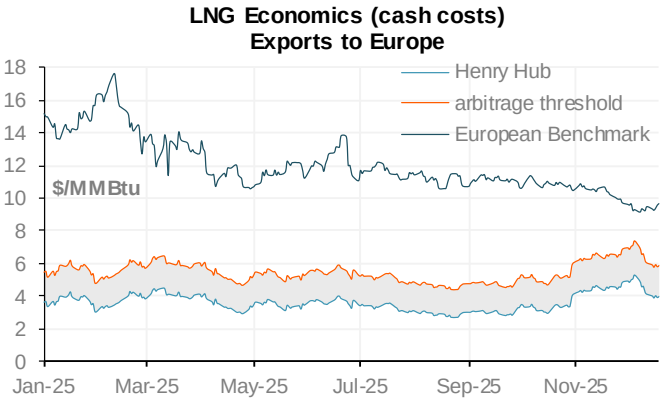
With this being the last "Spectrum" of 2025, we felt it worthwhile to look back and highlight key events in the energy and geopolitics space, because the first year of the new Trump administration was – politely put -- *eventful* for energy and geopolitics as well.

Whether that was Opec's decision to *Pump for Trump*; no peace in Ukraine, the eviscerating of Pres. Biden's climate policies or the renaissance of coal, dominance of LNG exports or China's going '*va banque*' on EV exports. We have segmented our review into four parts on the next four pages.

The Donroe Doctrine comes alive: Last week, President Trump imposed a military blockade to stop sanctioned oil tankers from leaving Venezuela, claiming that "it will only get bigger". That should worry Maduro but also Cuba's rulers …

The LNG supply glut is here: Energy Transfer this month threw in the towel on its LNG export project, suspending development of its 16.5 mtpa (~2.2 bcf/d) Lake Charles export terminal to focus instead on "…allocating capital to its significant backlog of natural gas pipeline infrastructure projects [which] provide superior risk/return profiles." One of those pipeline deals is the Desert Southwest system, due online in Q4-2029. ET will spend more to expand Desert Southwest from 1.5 bcf/d to 2.3 bcf/d.

As for LNG, its spot prices are falling going into winter, when normally its s/d gets tight …



*European Benchmark = NBP from Jan-'14 to Feb-'21: TTF from Mar-'21 to present

Pessimism and uncertainty were the prevailing themes of the latest ***Dallas Fed Survey*** (link). Whingey O&G executives griped about the uncertainty in government policies, Opec, geopolitics (e.g. peace or no peace deal in Ukraine). One even went as far as to say that "there is no way the U.S. oil production data [that show surprising growth] is correct." Most respondents expect WTI to range from $60-$65 by the end of 2026. On the services side, respondents mentioned that steel tariffs continue to drive well costs up, but there is also talk of activity picking up next year.

Permitting reform is starting to take shape The House recently voted to pass the **PERMIT Act**, which targets a provision within the Clean Water Act that governors have previously used to halt pipelines viewed as threats to water supplies and the environment. The other bill on the agenda, known as the **SPEED Act**, would expedite agency environmental reviews by reshaping the 1970 National Environmental Policy Act, which has also historically contributed to project delays.

A win for nuclear and a loss for LNG Japan's Kashiwazaki-Kariwa plant, at 8.2 GW it's the largest in the world, cleared a final electoral hurdle and will resume partial operations next year, when it'll restart one 1.4 GW unit and another in 2030. The plant was among the 54 shut down following the Fukushima disaster in 2011. Since then, Japan has restarted 14 out of the 33 that remain operable.

Source: PSC Research, Bloomberg

WHAT'S IN OUR FOCUS – 2025: LOOKING BACK, PART I

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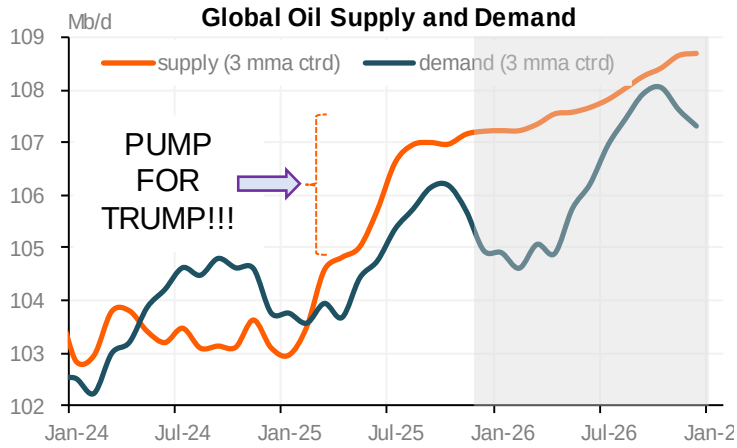
It seems longer ago, but it was *only on January 20th* that President Trump took office for his second term. And it was only on April 2nd, not even 9 months ago, that Trump showed he was serious not just about cleaning house in Washington (remember DOGE?) but about ushering in a new global trading system. Tariffs became reality.

Even allies would need to pay up to do business with and in America. To us, observers of global energy and policy, the most surprising was that on that day, Opec announced it would pay the piper and begin to Pump for Trump …

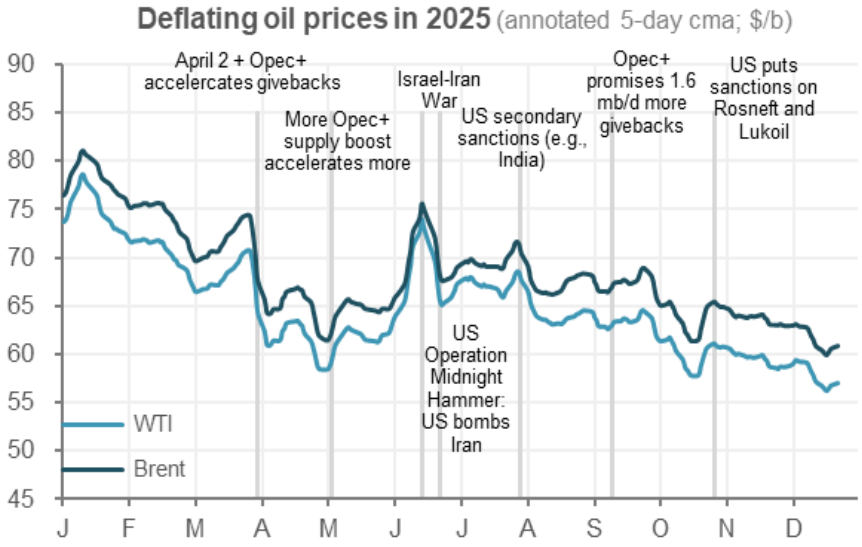
Global oil supply grew +3.5 mb/d

- By Q4, global oil supply had grown 3.5 mb/d y/y – dwarfing +1.1 mb/d of demand growth
- Because non-Opec grew +2 mb/d (from the US, Brazil, Canada, Guyana, etc) …
- Opec did not need to add 1.5 mb/d of its own
- But it did …



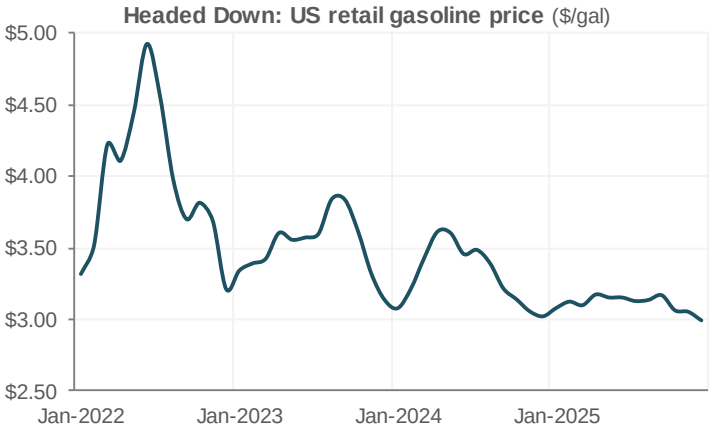
Historic Opec move is political

- The big Mideast sovereign producers remind the world (and their competition) that *oil is policy*
- Opec heavyweights need Trump, America, its protection, AI chip-stacks, civilian nukes, arms, etc
- Predictably, oil prices deflated …
- Stunningly **while**, the Mideast itself nearly blows up, Russia's war (and sanctions) grind on, a new theatre of war is opened up … and still prices deflated …
- And China gets to build up its SPR



Oil (and energy-) prices deflate

- Maybe not all by design,
- But gasoline prices, global LNG prices, even coal prices (of late) – all quoted mostly in US dollars – all have deflated
- Supply abundance is a thing, and America un-apologetically is at the core of it …



Source: PSC Research, Bloomberg, EIA

WHAT'S IN OUR FOCUS – LOOK-BACK PART II: GEO-POLITICS

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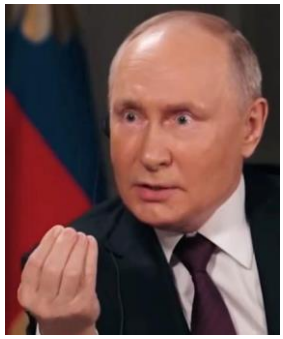
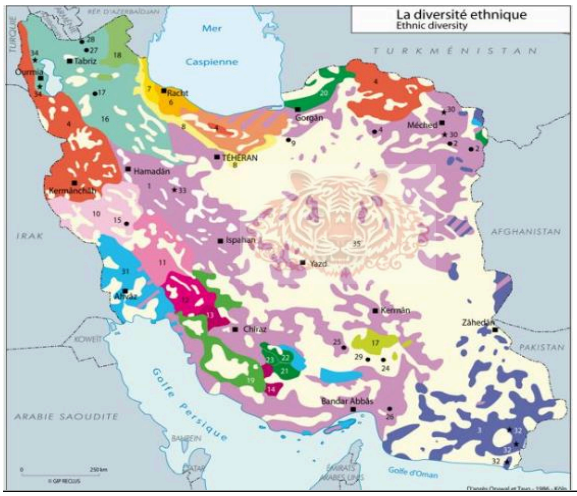
Gone is the global Post-WWII order of things. Gone are 7-decades of pretend multilateralism and selectively applying international law in a theory in which all nation states are equal. With the defeat of the Democrats was swept aside the US-centric old world. With Trump II definitively arrives an era of multipolarity, with spheres of influence for those willing to fight for them.

Trump and his team aspire to morph the US presidency into *an imperial* presidency. It would run things 'more efficiently' domestically, while also lording it over the western hemisphere, including Latin America and Greenland. To describe that hemispheric ambition we gladly borrow from the WSJ the moniker of ***The Donroe Doctrine***, which emerged at the end of only year 1 of Trump II [link]…

We do not have a peaceful ending to eight wars – we do have historic events: In 2025 at least three of them mattered greatly to energy markets,

- **America is starting a new war, with Venezuela** (and Cuba and others?)
- **War in the Ukraine rages on. For a fourth year** all talk of an armistice & peace is so far just that, talk. Vladimir Putin is the protagonist that has not yielded (and in our view will not yield …)
- **The Middle East is the one big success** within reach. Not a done deal, but once-unimaginable progress has been made by Israel, which soundly defeated Iran's 40-yr old strategy of regional influence & power through proxy (link).
- **Iran** acted like a paper tiger and is hopelessly complex. ***It may not be done***.





Source: PSC Research, The Wall Street Journal, Bloomberg

WHAT'S IN OUR FOCUS – LOOK-BACK, PART III GREEN IS OUT

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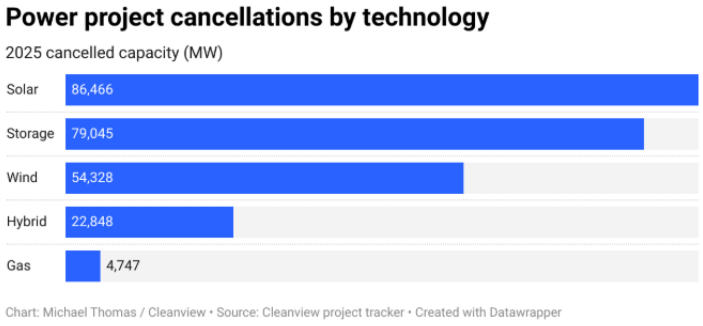
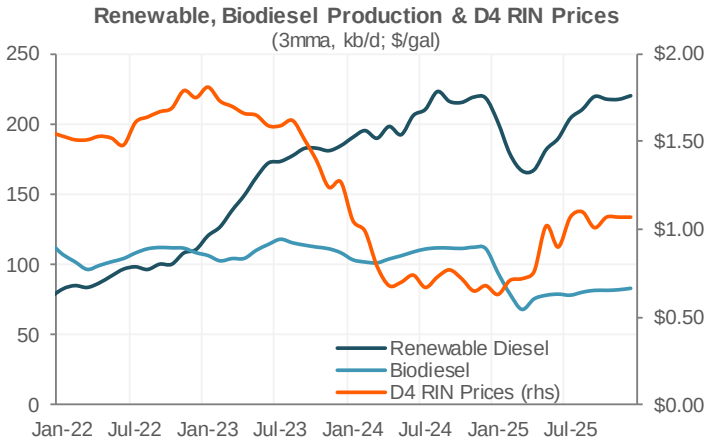
Not a friend of DEI, progressives, the environment, regulation or the 'science' of climate-change, it's not surprising that President Trump's signature *One Big Beautiful Bill* involved the knee-capping, of green subsidies and alternative energy projects.

It was going to be "all of the above" except not those things that cost tax-payer money and/or those things President Trump simply hates (e.g. windmills, especially those offshore) … here are some key *high* lights and *low* lights:

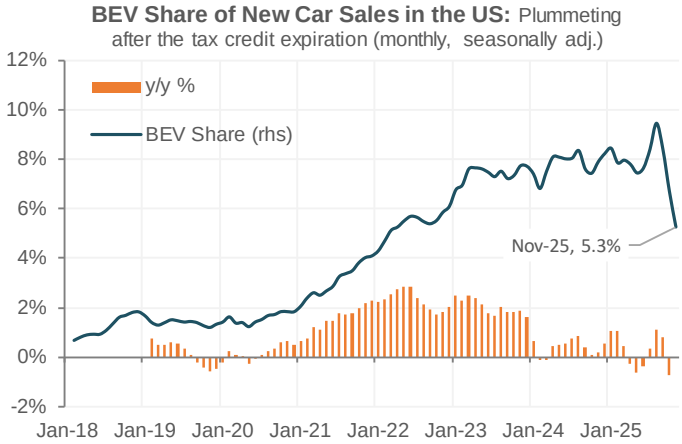
The list is long of renewable energy project cancellations the OBBB eviscerated some $560 billion in tax credits for green energy, which we wrote about in May. According to Cleanview, a renewable energy project tracker firm, nearly 2,000 projects representing 266 GW of combined capacity were cancelled this year with the largest cuts involving utility-scale solar (86 GW), battery storage (79 GW), and wind projects (54 GW).



Biofuels may have a good year in 2026 given the surprisingly huge renewable volume obligation (RVO) the EPA rolled out back in August. That mandate and shift to the 45Z tax regime should incentivize more domestic production and help RIN prices recover to improve profitability. **But this year was a dog**. Biodiesel production has yet to reach back to last year's levels while renewable diesel continues to trend sideways. D4 RIN prices still track well below 2022-23 levels. See pp. 34-35 for data and prices.



Without EV subsidies, those sales plummeted (see below), which combines with relaxing of fuel efficiency standards and assaults on the regulatory agencies to rewrite the future of the American automobile. **Significantly,** for instance, Ford took a $19.5 billion charge and scotched plans for electrifying its large truck and SUV segments – which are of course Americas best selling vehicle classes…



We offer an award to anyone able to explain the President's hatred of offshore wind. But whatever it is, he put that sub-segment into a world of hurt. We wrote about this back in March following an executive order signed on Trump's first day in office, but even now at the end of the year, the Dept of the Interior suspended leases for 5 large-scale offshore wind farms under construction, on frankly dubious 'radar-interference' claims..

Source: PSC Research, EIA, FactSet, Cleanview, Wards

WHAT'S IN OUR FOCUS – LOOK BACK, PART IV: BOOMING BITS

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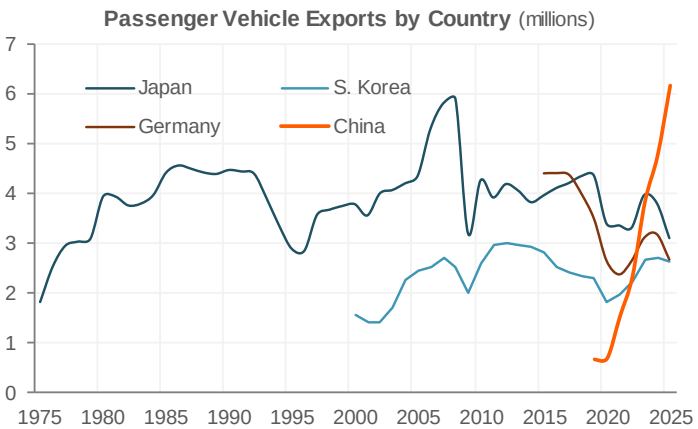
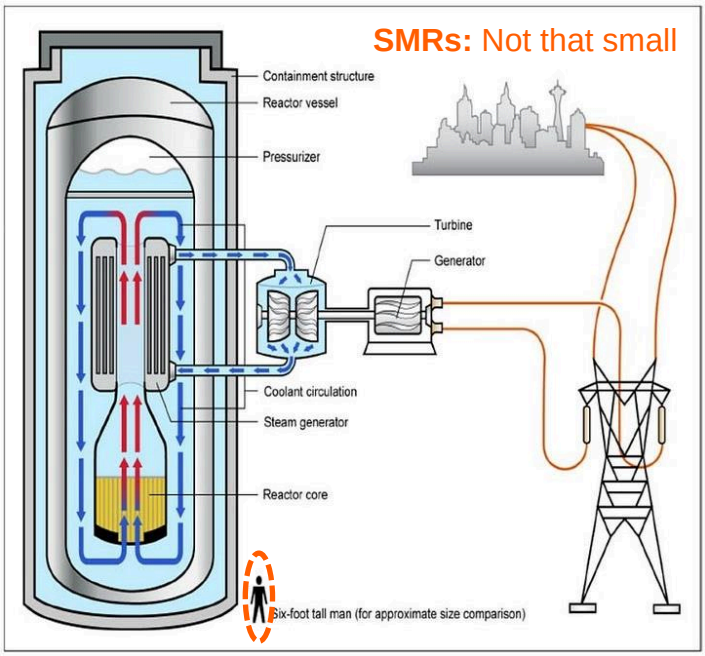
More broadly, there were some outstanding stories/themes of **extraordinary growth**, e.g. American LNG exports; **nascent outright dominance** of an entire global manufacturing sector, i.e. China's insane automobile export growth; **early innings growth story**, AI and data-center electricity demand and of course **the global renaissance** of one of our favorite low-emissions electricity source: nuclear energy (both conventional and of the SMR type).

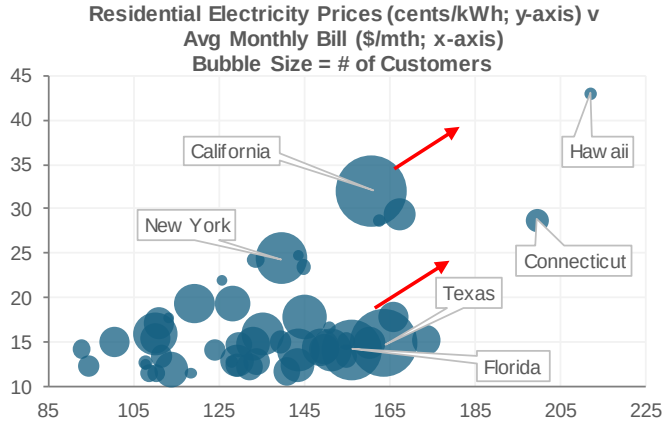
EVs anyone? The most telling and frightening chart is this: the orange line plots the actual volume growth of China's automobile exports, which began in 2019. And this year those were the greatest ever for any exporter. The other lines track once dominant exporters and their volumes back to 1975 …

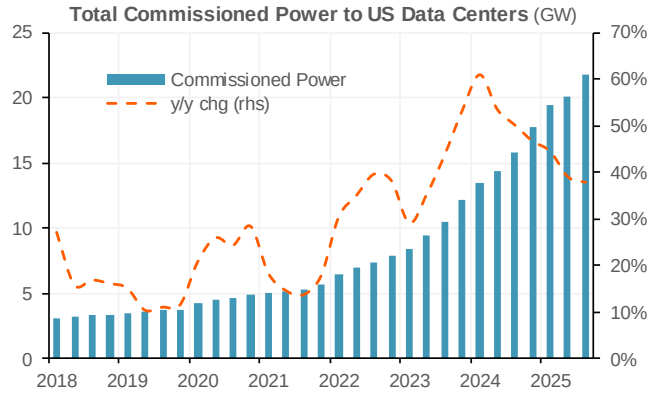


2025 was another big year for the **nuclear** industry, for one because DC policy is now in high-gear backing mode (see string of Executive Orders: June). The advent of SMRs which promise faster scalability for next decade, with which gov't also got involved …

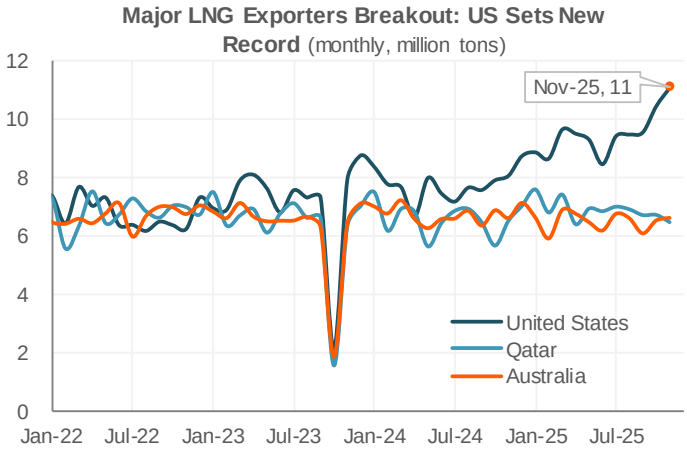


Not as much growth, but still: Data centers and their electricity needs have been *the investor interest story of the year*. Their proliferation, however, risks blowing out consumer-prices in ever more states:





And finally in 2025, the US became officially the fastest growing and largest exporter of LNG in the world. Now please, who becomes the fastest growing user of the stuff …



Source: PSC Research, Bloomberg, FactSet, EIA

TACTICAL MACRO SECTOR PREFERENCES

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These are ***our*** sub-sector preferences. We base these on **shorter term factors, i.e. this is tactical and not always thematic (i.e. not always aligned with the leanings of our fundamentals analyst)**

Again, this is tactical and as such about the shorter-term (~3 mths):

Sub-Sector	Tactical direction / Preference	Rank	Prior	Delta
US Refining	Refining has rallied a lot but has more room to run into 2026: Lower feedstock costs, no sign of weakening demand and tight middle distillate fundamentals should continue to benefit those refiners that can keep their plants running	1	2	*+1*
Integrated Oils	The major integrated oil & gas companies are solid earners with fortified balance sheets and strong dividends, if the global macro does improve, they'll also appear under-valued from their refining to their upstream and even their chemical segments	2	3	*+1*
Oil Services - International / Offshore	Investors are now shifting from expecting international capex to fall next year to expecting slight growth - from a rate-of-change standpoint, international/offshore services may outperform -- *exploration is back*	3	6	*+3*
E & P Oil	We like oily E&Ps over gassy E&Ps as we believe there is more upside price risk for oil than gas -- and the oily e&ps have been unloved for too long	4	2	*-2*
E & P Natural Gas	the gassy e&p segment has had a great run, but you have to believe in much higher mid-cycle prices than we do to continue to like them	5	5	*+0*
Power-Infrastructure Services	Has had a good run and will likely still perform but relative relative we're less enamoured with the sub-segment	6	1	*-5*
Oil Services - Lower 48	After earnings we're even more convinced that this is not the place to be	7	7	
Utility-Scale / Large Levered Renewables	Broadly still like the sector for growth that and the defensiveness of power producers is attractive given the residual recession risk and the outlook for lower interest rates	8	8	
Biofuels	In theory 2026 is looking better with some tailwinds - mandates increasing materially and improved technical setup following significant underperformance	9	12	*+3*
Chemicals	Still a few ticks too early for that proper industrial upturn later next year	10	9	*-1*
Autos & New Energy Vehicles	the US segment of this sector will take some time to get over the hangover of dissappearing subsidies, look at Alex Potter's new coverage of China's producers maybe …	11	10	*-1*
Consumer Levered Renewables	keep praying for more rate-cuts	12	11	*-1*

Source: PSC Research, Bloomberg

WHAT'S IN OUR FOCUS

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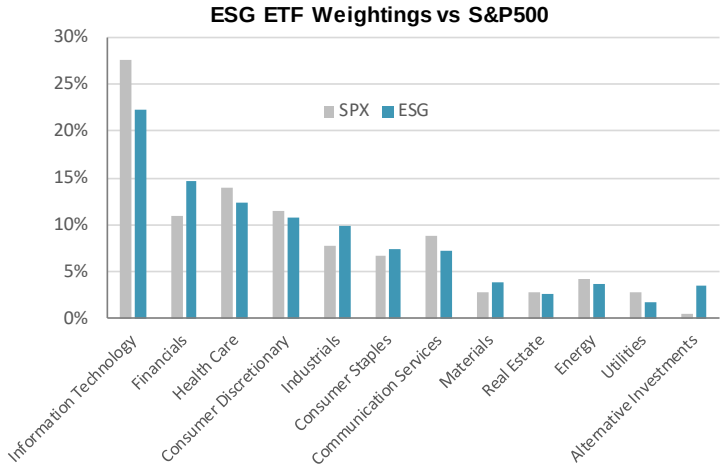
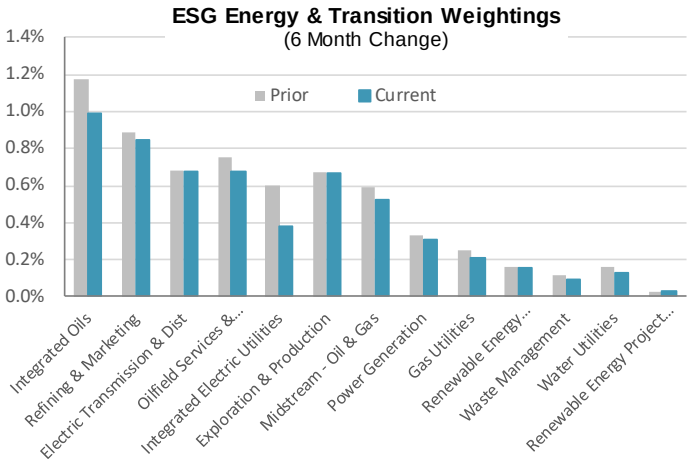
Largest Exchange Traded Funds (ETF) for ESG, ex-Fossil Fuels, and Clean Energy

Ticker	Security Name	Fund Total Asset Value (TAV) 12/31/20	12/31/21	12/30/22	12/29/23	12/31/24	12/18/25	YTD Perform	Total Chg	YTD Change TAV Perfmnce	Inflows / (Outflows)	Inflw/TAV
ESGU	ISHARES ESG AWARE MSCI USA	$ 13,424	$ 25,695	$ 19,587	$ 13,393	$ 13,207	$ 15,041	15.4%	$ 1,834	$ 2,028	$ (194)	-1%
ESGD	ISHARES TRUST ISHARES ESG AW	$ 3,986	$ 7,622	$ 6,722	$ 7,624	$ 8,013	$ 10,753	24.3%	$ 2,740	$ 1,944	$ 797	10%
ESGV	VANGUARD ESG US STOCK ETF	$ 2,981	$ 6,399	$ 5,659	$ 7,446	$ 10,059	$ 11,852	15.0%	$ 1,793	$ 1,508	$ 285	3%
ESGE	ISHARES INC ISHARES ESG AWAR	$ 6,133	$ 6,225	$ 3,746	$ 4,069	$ 4,278	$ 5,522	30.4%	$ 1,244	$ 1,301	$ (57)	-1%
ICLN	ISHARES GLOBAL CLEAN ENERGY	$ 4,700	$ 5,605	$ 5,124	$ 3,073	$ 1,421	$ 1,869	44.6%	$ 448	$ 635	$ (187)	-13%
LIT	GLOBAL X LITHIUM & BATTERY T	$ 1,984	$ 5,564	$ 3,228	$ 2,083	$ 1,088	$ 1,432	57.6%	$ 343	$ 627	$ (284)	-26%
SUSA	ISHARES ESG OPTIMIZED MSCI U	$ 2,298	$ 4,826	$ 3,228	$ 5,343	$ 3,726	$ 3,714	14.5%	$ (12)	$ 539	$ (551)	-15%
DSI	ISHARES ESG MSCI KLD 400 ETF	$ 2,628	$ 4,202	$ 3,582	$ 4,086	$ 4,792	$ 5,034	16.5%	$ 242	$ 791	$ (550)	-11%
SUSL	ISHARES ESG MSCI USA LEADERS	$ 2,826	$ 4,246	$ 2,878	$ 1,121	$ 987	$ 1,012	17.2%	$ 25	$ 170	$ (146)	-15%
USSG	XTRACKERS MSCI USA SELE EQTY	$ 2,851	$ 4,143	$ 2,961	$ 983	$ 944	$ 499	17.3%	$ (445)	$ 163	$ (608)	-64%
VSGX	VANGUARD ESG INTL STOCK ETF	$ 1,584	$ 3,175	$ 3,010	$ 3,580	$ 3,946	$ 5,460	25.2%	$ 1,513	$ 994	$ 519	13%
TAN	INVESCO SOLAR ETF	$ 3,632	$ 2,729	$ 2,388	$ 1,758	$ 801	$ 952	48.6%	$ 151	$ 389	$ (238)	-30%
QCLN	FIRST TRUST NASDAQ CLEAN EDG	$ 1,999	$ 2,824	$ 1,573	$ 1,139	$ 548	$ 544	35.6%	$ (5)	$ 195	$ (200)	-36%
NULV	NUVEEN ESG LARGE-CAP VAL ETF	$ 677	$ 1,272	$ 1,548	$ 1,484	$ 1,625	$ 1,899	14.5%	$ 274	$ 236	$ 38	2%
SPYX	SS SPDR S&P 500 FOSS FL ETF	$ 893	$ 1,407	$ 1,209	$ 1,473	$ 2,054	$ 2,443	16.6%	$ 389	$ 341	$ 48	2%
PBW	INVESCO WILDERHILL CLEAN ENE	$ 2,175	$ 1,564	$ 741	$ 475	$ 295	$ 661	57.8%	$ 366	$ 171	$ 195	66%
NUSC	NUVEEN ESG SMALL-CAP ETF	$ 534	$ 1,112	$ 894	$ 1,057	$ 1,226	$ 1,269	8.0%	$ 43	$ 98	$ (55)	-4%
SNPE	XTRCKR S&P 500 SCOR&SCRN ETF	$ 402	$ 851	$ 677	$ 975	$ 1,638	$ 2,263	16.4%	$ 625	$ 268	$ 357	22%
ACES	ALPS CLEAN ENERGY ETF	$ 784	$ 892	$ 564	$ 298	$ 115	$ 110	28.8%	$ (5)	$ 33	$ (38)	-33%
KRMA	GLOBAL X CONSCIOUS COMPANIES	$ 403	$ 674	$ 640	$ 605	$ 628	$ 166	13.3%	$ (462)	$ 84	$ (546)	-87%
USXF	ISHARES ESG ADVANCE MSCI USA	$ 81	$ 583	$ 599	$ 824	$ 1,291	$ 1,176	14.9%	$ (115)	$ 193	$ (308)	-24%
EFIV	SS SPDR S&P 500 ESG ETF	$ 93	$ 455	$ 643	$ 1,109	$ 1,726	$ 1,133	16.6%	$ (593)	$ 286	$ (879)	-51%
ERTH	INVESCO MSCI SUSTAINABLE FUT	$ 466	$ 459	$ 305	$ 232	$ 149	$ 141	17.5%	$ (8)	$ 26	$ (34)	-23%
DMXF	ISHARES ESG ADVANC MSCI EAFE	$ 63	$ 323	$ 396	$ 569	$ 772	$ 642	16.4%	$ (130)	$ 127	$ (257)	-33%
FAN	FIRST TRUST GLOBAL WIND ENER	$ 402	$ 348	$ 278	$ 218	$ 145	$ 199	38.1%	$ 54	$ 55	$ (1)	-1%
EFAX	SPDR MSCI EAFA FOSSIL FUEL	$ 171	$ 276	$ 238	$ 246	$ 303	$ 439	26.2%	$ 136	$ 80	$ 56	18%
NUDM	NUVEEN ESG INTL DEVEL MRK EQ	$ 82	$ 167	$ 299	$ 389	$ 452	$ 592	22.5%	$ 140	$ 102	$ 39	9%
IQSI	NYLI CANDRI INTER EQ ETF-USD	$ 133	$ 215	$ 184	$ 203	$ 207	$ 214	22.5%	$ 7	$ 47	$ (39)	-19%
ETHO	AMPLIFY ETHO CLIMATE LEADERS	$ 119	$ 201	$ 160	$ 184	$ 183	$ 159	11.0%	$ (24)	$ 20	$ (44)	-24%
NUEM	NUVEEN ESG EMERGING MARKETS	$ 67	$ 135	$ 209	$ 233	$ 280	$ 308	21.7%	$ 28	$ 61	$ (33)	-12%
EEMX	SPDR MSCI EM FSL FL RSV FREE	$ 139	$ 139	$ 92	$ 80	$ 78	$ 87	29.8%	$ 8	$ 23	$ (15)	-19%
CHGX	STANCE SUSTAINABLE BETA ETF	$ 33	$ 116	$ 108	$ 132	$ 131	$ 134	11.5%	$ 3	$ 15	$ (12)	-9%
LDEM	ISHARES ESG MSCI EM LEADERS	$ 835	$ 75	$ 50	$ 33	$ 30	$ 32	26.5%	$ 2	$ 8	$ (6)	-21%
EMXF	ISHARES ESG ADV MSCI EM ETF	$ 13	$ 30	$ 44	$ 71	$ 82	$ 117	23.3%	$ 36	$ 19	$ 17	21%
	TOTAL ESG STYLE ETF	$ 59,592	$ 94,549	$ 73,566	$ 66,588	$ 67,222	$ 77,867					
			59%	*-22%*	*-9%*	*1%*	*17%*					
SPY	SPDR S&P 500 ETF TRUST	$ 332,336	$ 463,713	$ 354,633	$ 493,539	$ 626,343	$ 692,603	15.9%	$ 66,260	$ 99,720	$ (33,461)	-5%
IWM	ISHARES RUSSELL 2000 ETF	$ 58,303	$ 69,137	$ 52,086	$ 67,850	$ 71,908	$ 75,217	13.4%	$ 3,309	$ 9,620	$ (6,311)	-9%
EEM	ISHARES MSCI EMERGING MARKET	$ 28,301	$ 28,977	$ 22,910	$ 17,857	$ 17,003	$ 20,466	28.6%	$ 3,462	$ 4,867	$ (1,405)	-8%
		$ 418,941	$ 561,827	$ 429,628	$ 579,246	$ 715,255	$ 788,286					
			34%	*-24%*	*35%*	*23%*	*36%*					

Energy & Transition Equity Holdings Across Aggregated ESG Exchange Traded Funds (ETF)





Source: PSC Research, Bloomberg, MSCI, Sustainalytics

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CALENDAR AHEAD: WHAT WE'RE WATCHING

Date	Time	Events, Economic, Earnings & Data Releases
12/30/2025		NIKOLA CORP EARNINGS
12/30/2025	2:00:00 PM	Federal Open Market Committee
1/4/2026	8:45:00 PM	RatingDog China Services PMI B
1/5/2026		US Auto Sales Total Annualized
1/13/2026	12:00:00 PM	DOE Short Term Outlook Current
1/21/2026		HALLIBURTON CO EARNINGS
1/22/2026		GENERAL ELECTRIC EARNINGS
1/23/2026		SLB LTD EARNINGS
1/23/2026		FREEPORT-MCMORAN INC EARNINGS
1/23/2026		NEXTERA ENERGY INC EARNINGS
1/27/2026		GENERAL MOTORS CO EARNINGS
1/29/2026		TESLA INC EARNINGS
1/29/2026		ABB LTD-REG EARNINGS
1/29/2026		VALERO ENERGY CORP EARNINGS
1/30/2026		BAKER HUGHES CO EARNINGS
1/30/2026		PHILLIPS 66 EARNINGS
1/30/2026		CHEVRON CORP EARNINGS
1/30/2026		EXXON MOBIL CORP EARNINGS
2/4/2026		ENPHASE ENERGY INC EARNINGS
2/4/2026		EQUINOR ASA-SPON ADR EARNINGS
2/5/2026		FORD MOTOR CO EARNINGS
2/5/2026		SHELL PLC-SPON ADR-A EARNINGS
2/5/2026		VESTAS WIND SYSTEMS A/S EARNINGS
2/6/2026		PEABODY ENERGY CORP EARNINGS
2/6/2026		ORSTED A/S EARNINGS
2/11/2026		SIEMENS ENERGY AG EARNINGS
2/11/2026		BP PLC-SPONS ADR EARNINGS
2/12/2026		QUANTUMSCAPE CORP EARNINGS
2/12/2026		VALE SA-SP ADR EARNINGS
2/16/2026		BHP GROUP LTD-SPON ADR EARNINGS
2/18/2026		EQT CORP EARNINGS
2/18/2026		DEVON ENERGY CORP EARNINGS
2/18/2026		OCCIDENTAL PETROLEUM CORP EARNINGS
2/19/2026		SOLAREDGE TECHNOLOGIES INC EARNINGS
2/19/2026		RIO TINTO PLC-SPON ADR EARNINGS
2/24/2026		COTERRA ENERGY INC EARNINGS
2/25/2026		RANGE RESOURCES CORP EARNINGS
2/25/2026		FIRST SOLAR INC EARNINGS
2/26/2026		EXPAND ENERGY CORP EARNINGS
2/26/2026		APA CORP EARNINGS
2/26/2026		STELLANTIS NV EARNINGS
2/27/2026		DAQO NEW ENERGY CORP-ADR EARNINGS
2/27/2026		EOG RESOURCES INC EARNINGS
2/27/2026		SUNRUN INC EARNINGS
3/3/2026		PLUG POWER INC EARNINGS

Source: PSC Research, Company Filings

US Natural Gas: Storage and underlying dynamics

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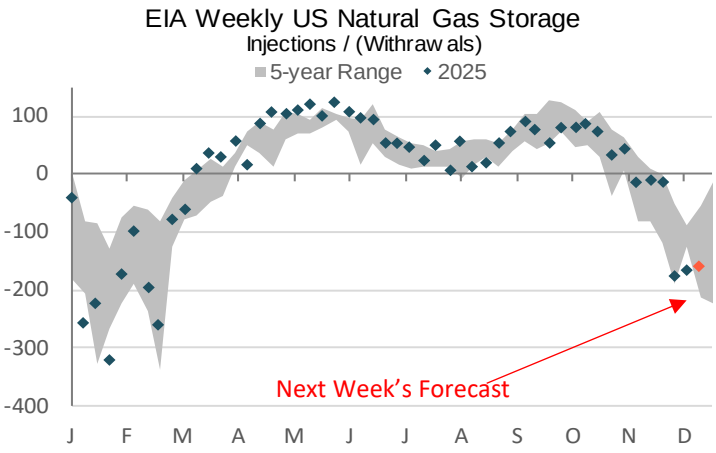
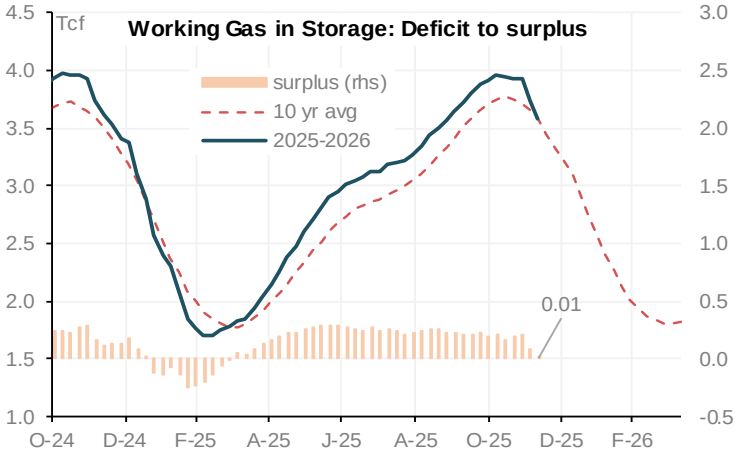
A nice bit of cold and poof, there goes the storage surplus … what will weather do next?

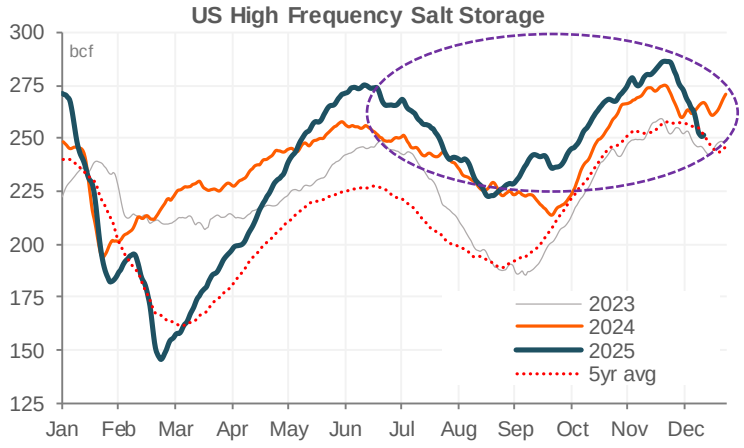
Already this week, storage is climbing back up as the second half of Dec is much less cold

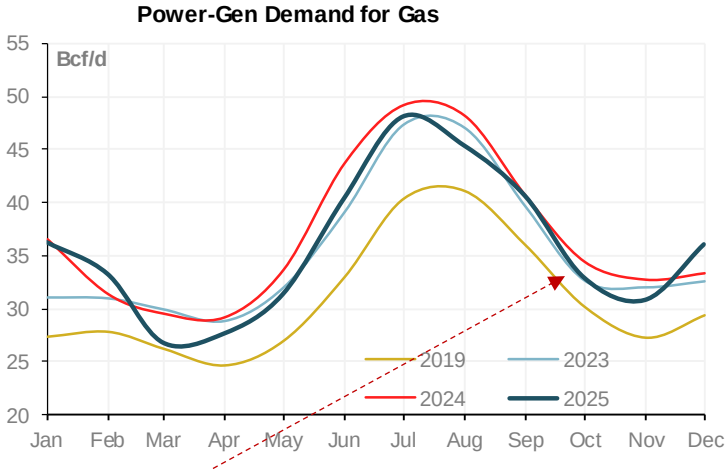
- Only the northeast stays cold through end December, but so far January forecasts are decidedly mild…



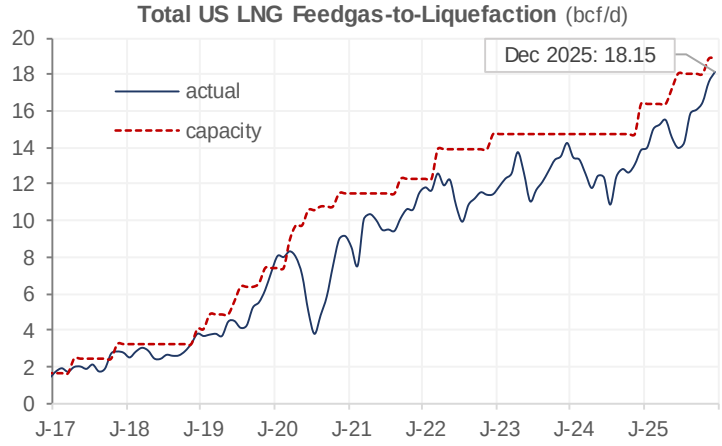


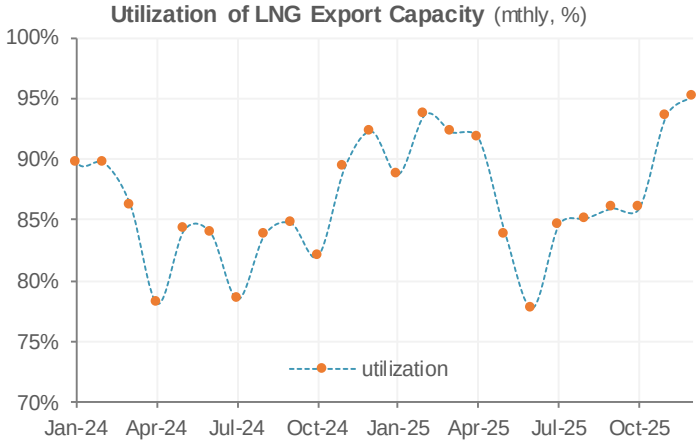
Salt storage echoed the drain on broader inventory, as demand for power and res-com ballooned





That power-gen dynamic is the stealth bullish factor that may be amplified with super-cold weather. The not so stealthy and secular growth story of **LNG** exports also helps – the storage surplus should not rebuild v soon





Source: PSC Research, Bloomberg, EIA

US NATURAL GAS: STORAGE FILL AND WEATHER

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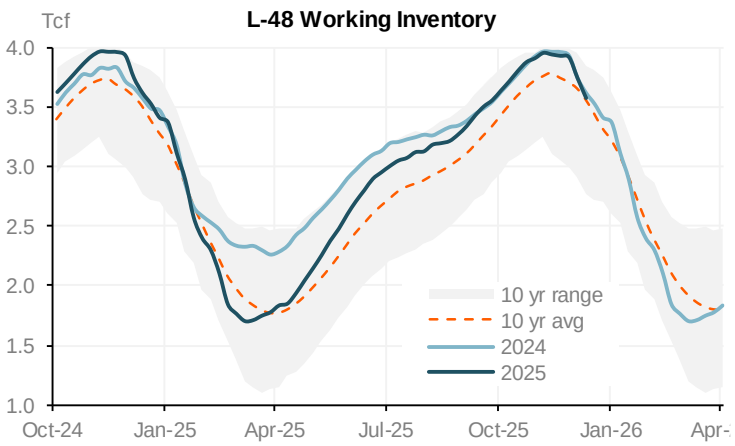
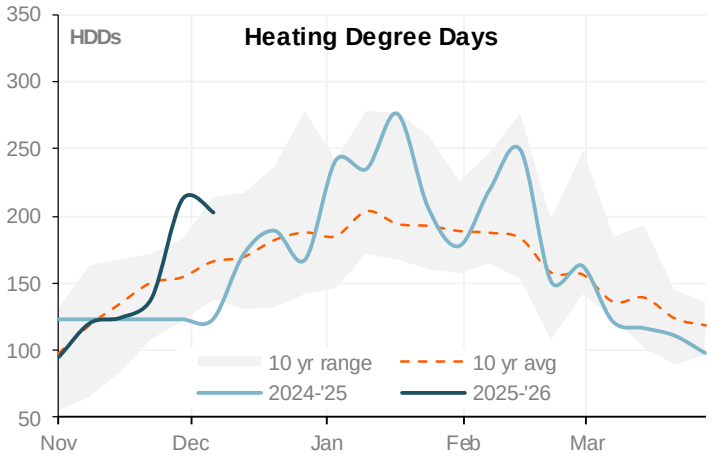
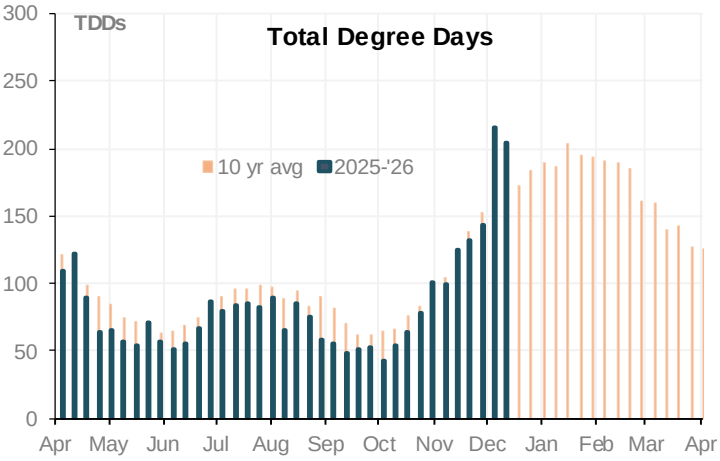
One of the colder Decembers of the last 10-yrs, right where it matters a lot (the northeast)

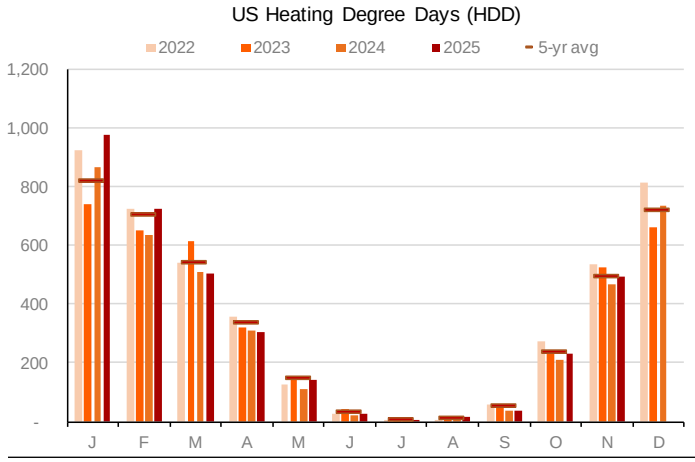
- Look at what a trifecta of strong weekly storage draws does to total inventory. And look at the run of good and cold weather for the first 20-days of December. Res-Com gas demand will likely set a 10-year record for the month of December
- But to get to super low EoS storage we need a run of cold weather right through Feb we last saw some such abnormally cold winter in 2013-'14

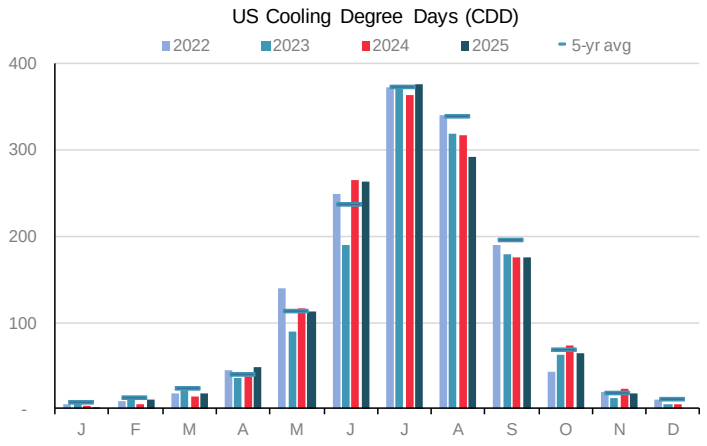






Gas Burn in EIA Res / Comm					
(Bcf/d)	nov	dec	jan	feb	mar
2015	22.9	30.3	44.5	38.0	24.2
2016	22.0	40.4	42.1	33.4	30.6
2017	26.3	42.1	49.4	39.7	34.6
2018	32.9	39.3	48.5	45.6	35.9
2019	32.5	38.9	42.2	40.6	27.8
2020	24.3	40.3	44.7	48.9	30.0
2021	27.2	34.3	48.5	44.7	31.3
2022	28.9	42.6	41.3	39.8	33.7
2023	28.2	34.5	47.5	36.6	27.5
2024	24.3	39.3	52.9	45.8	28.7
2025	26.7	43.5			
10 yr avg	**27.0**	**38.2**	**46.2**	**41.3**	**30.4**





Source: PSC Research, NOAA, BNEF

US NATURAL GAS: STORAGE MODEL EOS & ***PRICE***

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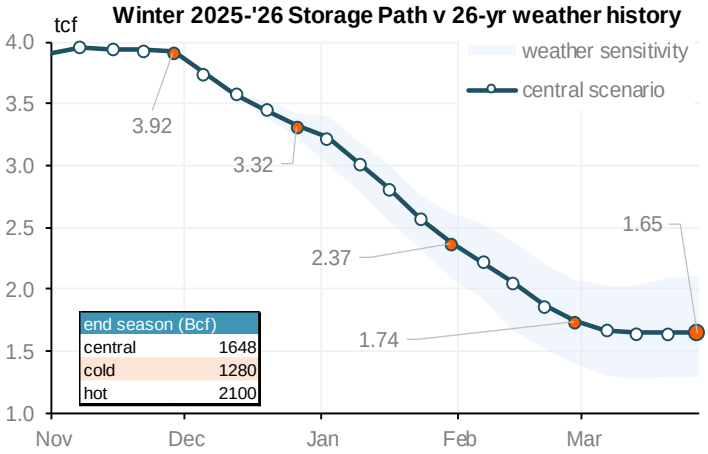
Please also see the key details of our new gas price-forecast here.

Thanks to December's run, our winter EoS working-gas-in-storage level fell to 1.65 Tcf, *mildly* bullish

Already winter weather is not boring. We should see more Polar Vortex events that could / should create freeze offs that threaten production disruptions – but we still do not expect a long run of extreme cold …

- We still have to model for normal weather – especially in a La Nina year [see 10/27 Spectrum]
- Normal weather in Q1 would mean that **production growth** runs only slightly below **all demand growth**

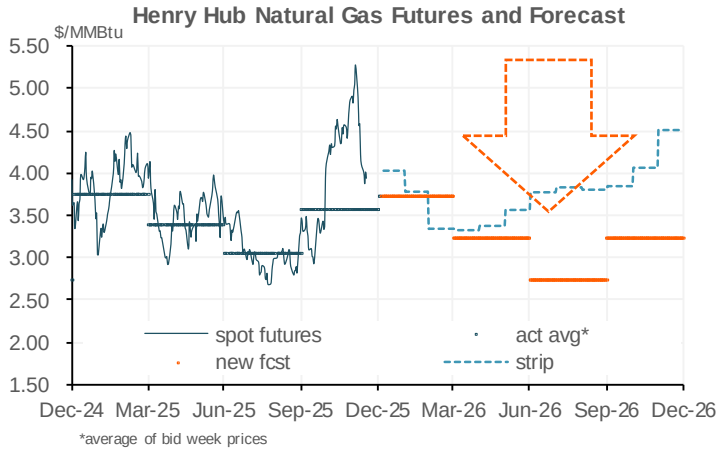


Winter 2025/26 vs 2024/25						
Bcf/d	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	**avg**
Residual = Prod-Dem-Exp	**0.2**	**-1.2**	**-2.3**	**-0.6**	**-0.5**	**-0.9**
Demand	**0.7**	**0.6**	**1.7**	**0.7**	**0.6**	**0.9**
Electric Power	0.2	0.2	1.1	0.2	0.1	0.3
Industrial	0.2	0.2	0.2	0.2	0.2	0.2
Other	0.4	0.3	0.4	0.4	0.2	0.3
Net Exports	**3.8**	**3.4**	**5.1**	**4.2**	**2.4**	**3.8**
LNG Net Exports	3.0	2.8	3.6	2.2	2.0	2.7
Mexico Net Exports	0.3	0.5	0.0	0.2	0.5	0.3
Canada Net Imports	-0.5	-0.1	-1.6	-1.8	0.1	-0.8
Production (Dry Gas)	**4.8**	**2.8**	**4.5**	**4.3**	**2.4**	**3.7**

Downgraded price forecast: We lowered targets by ~$0.75/MMBtu, to between $2.75-$3.75 thru '26

- Because **we model *normal* weather** and we expect that Q1 prices will deflate from ~$4.50 to $3.50/MMBtu
- Base case has fast-rising LNG exports as well as power-demand growth, and still we end winter with a comfortable 1.65 Tcf of working gas left in storage …
- and the '26 injection season sets up to overfill
- … which means that *Summer 2026 needs to price below Summer '25* to entice power-generators to burn more gas
- **Ample production growth** differentiates *our* view from consensus
- *Producer-discipline remains an oxymoron* … they're too good

We re-iterate our LT target of $3.50/MMBtu (~Haynesville BEP)



$/mmbtu		2019	2020	2021	2022	2023	2024	Q2-'25	Q3-'25	Q4-'25	2025	Q1-'26	Q2-'26	Q3-'26	2026	2027	LT
HHub	**PSC**	**$2.62**	**$2.08**	**$3.73**	**$6.55**	**$2.79**	**$2.30**	$3.40	$3.07	$3.59	**$3.46**	$3.75	$3.25	$2.75	**$3.25**	**$3.50**	**$3.50**
	latest/ old fcst									*$3.50*	$3.44	$4.50	$3.75	$3.75	$4.00	$3.50	$3.50
	Consensus										$3.56	$4.15	$3.75	$3.91	$3.93	$4.00	$4.00
	Strip										$3.45	$3.59	$3.30	$3.71	$3.74	$3.85	$3.67
JKM				$18	$34	$12	$13			$11	$12	$9	$9	$9	$9	$9	$9
TTF				$19	$23	$11	$12			$9	$11	$9	$8	$8	$8	$8	$8
memo: HHub y/y % chg		-14%	-21%	80%	76%	-3%	-18%	76%	39%	30%	51%	-1%	-4%	-11%	-6%	8%	

Source: PSC Research, Bloomberg, EIA

MORE ON LNG: EUROPE'S STORAGE AND DRIVERS OF DEMAND

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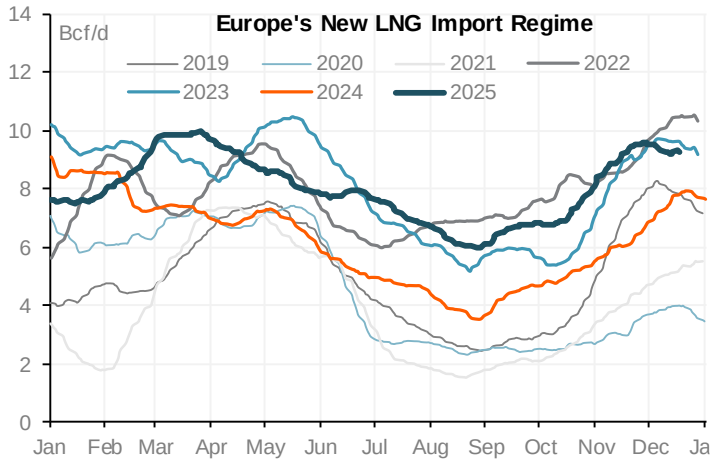
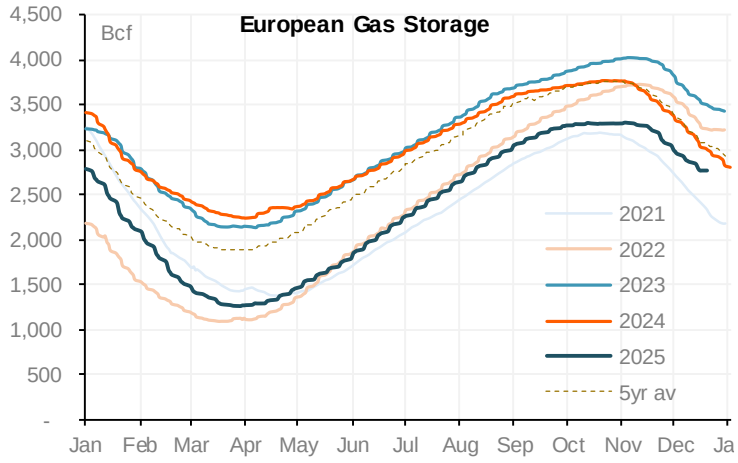
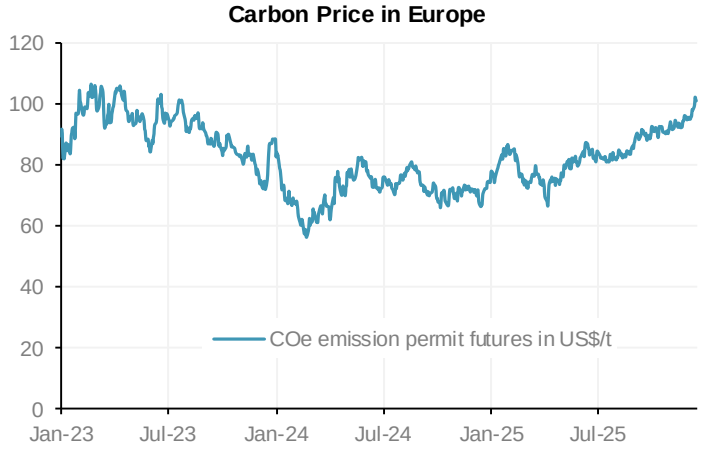
Surplus: Europe keeps pulling in more LNG and still LNG spot prices decline

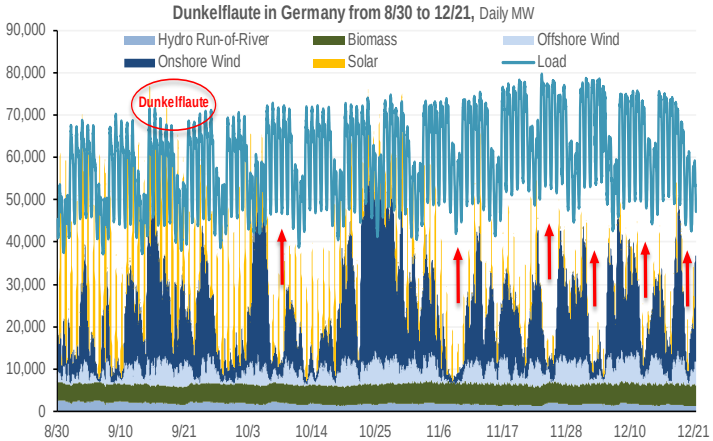
Europe's gas storage is less far below year ago and longer run averages, thanks to all that LNG coming in. but it's clear that only properly ***cold*** weather in ***both Europe and NE Asia*** can tighten up the LNG market.



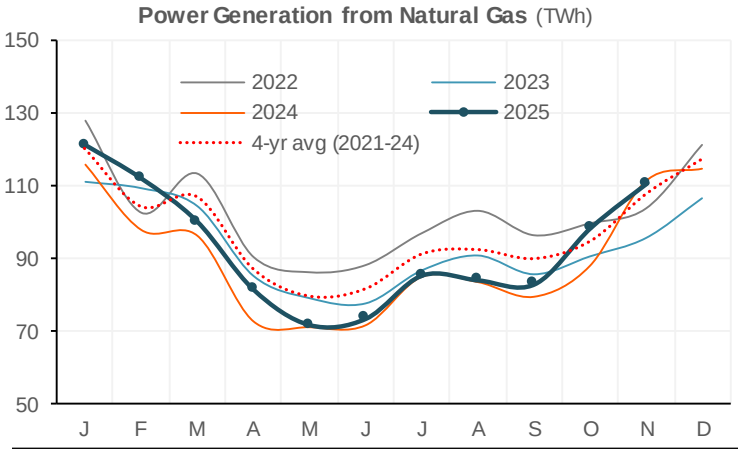


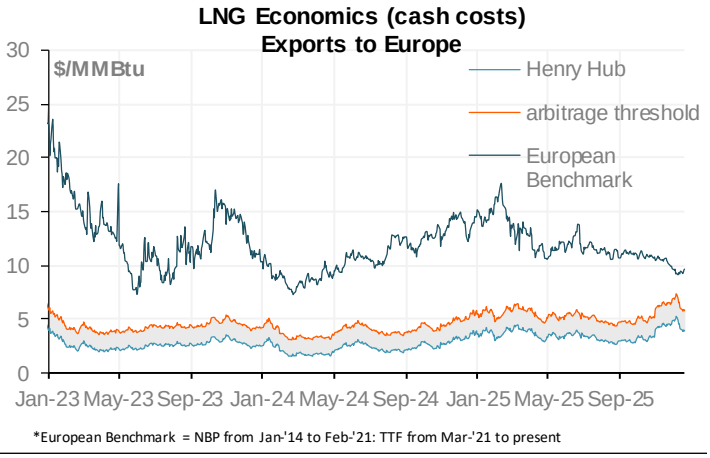
Europe's gas demand for power-gen blipped up in November, and anecdotally remains fairly elevated. Batches of "dunkelflaute" are doing their part, and now some cold air is coming ….





Carbon prices seem to breaking out and are approaching late 2023 levels. We think that has much to do with gas-fired power picking up and hopefully/perhaps is also driven by a nascent industrial recovery …





*European Benchmark = NBP from Jan-'14 to Feb-'21: TTF from Mar-'21 to present

Source: PSC Research, Bloomberg, Energy-Charts, Ember

GLOBAL LNG: ASIA'S IMPORTS PICK UP A TEENY, WEENY BIT

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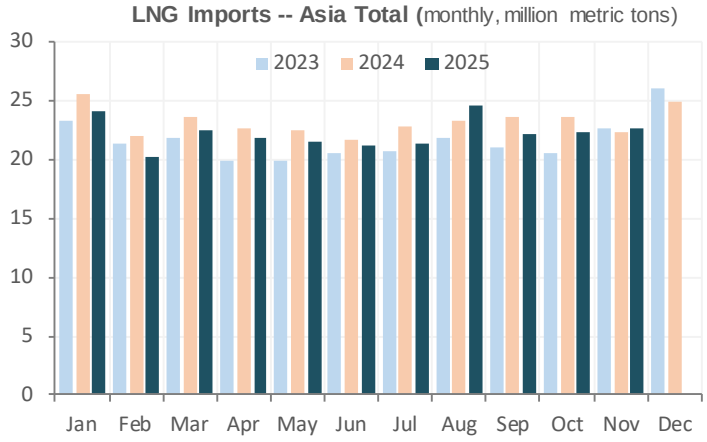
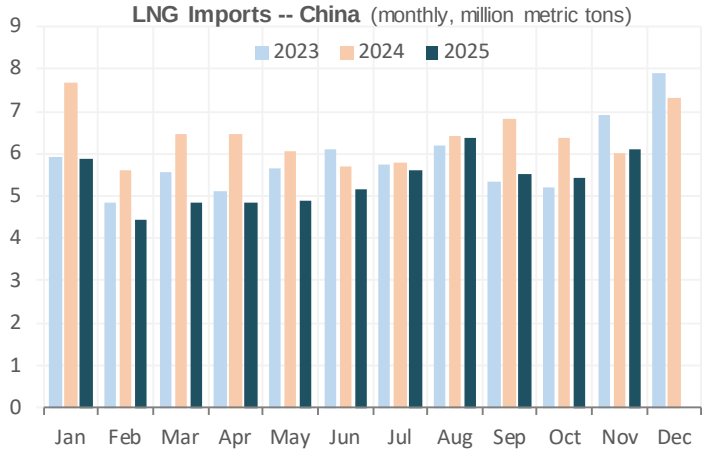
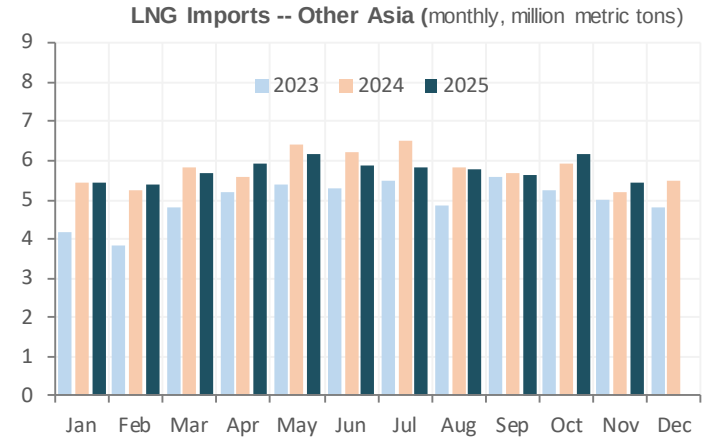
Asia's LNG imports rose 2% y/y in Nov and are a smidgen higher than Oct as well. China made much of a difference as its imports blipped up and were higher y/y as well (a first in 2025)

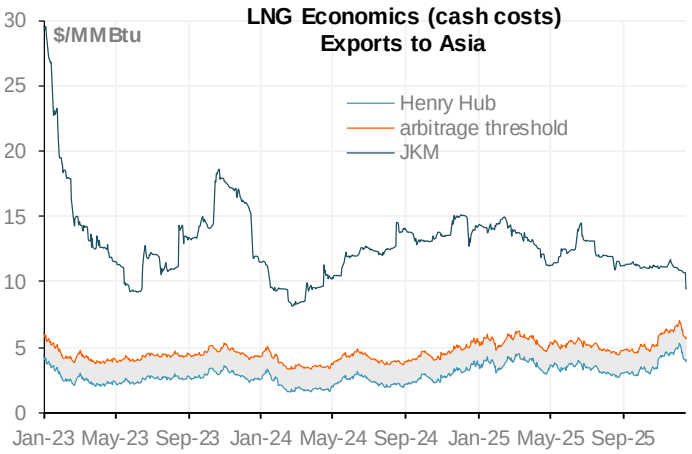
- Asia's LNG imports grew ~0.4 mmt (~0.6 bcf/d), y/y in Nov, but ***YTD imports are down -3% (1.3 bcf/d)***
- **China's imports** ytd average ~1 mmt/month less than last year and are down 15% (~1.5 bcf/d) YTD



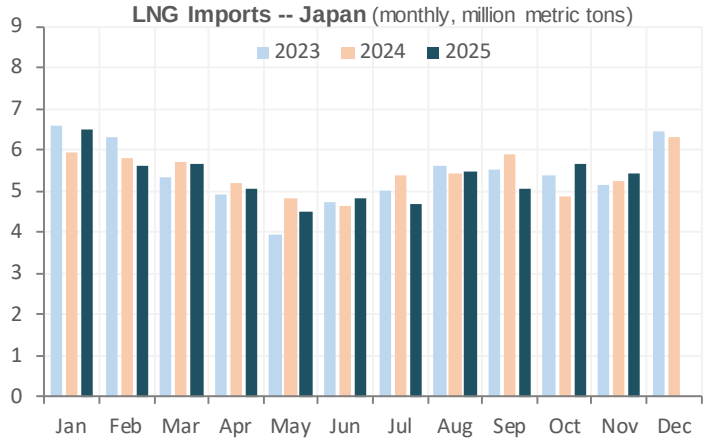


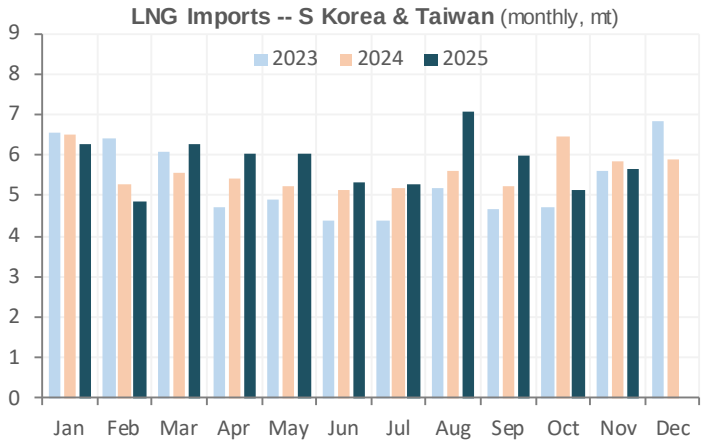
Ex-China, Asia's principal LNG importers were up 2% y/y in November, while imports into "Other Asia" (India, Thailand, Vietnam et al) were up 5% y/y.





Japan's imports grew by +4% y/y while S. Korea & Taiwan's combined imports fell 3% y/y.





Source: PSC Research, Bloomberg

GLOBAL CARBON EMISSIONS: *WHAT* DE-CARBONIZING?

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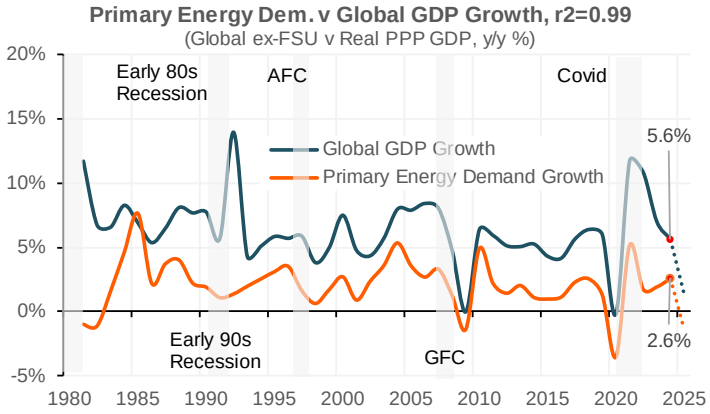
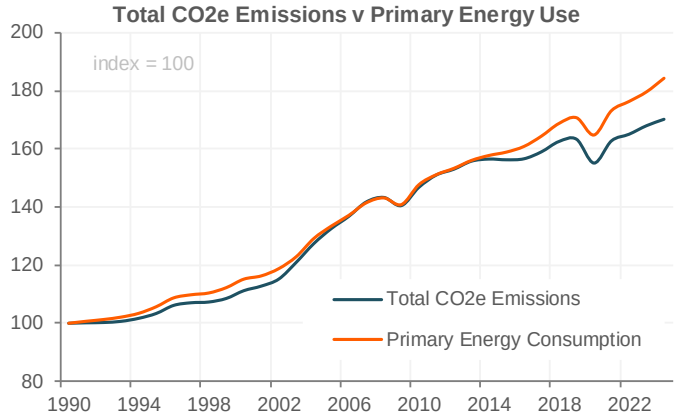
Glacial progress only (pun intended) as global econ growth = greater energy use

Ever Green Charts on the global energy system – updated with 2024 data – show:

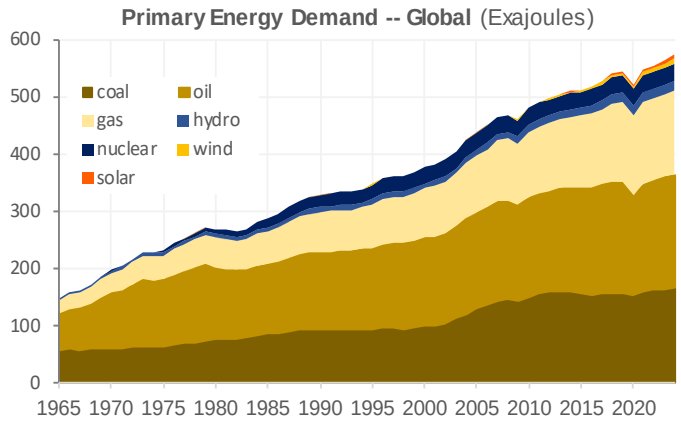
- That the uber-tight correlation of global economic growth and growing demand for energy holds firm;
- The carbon intensity of economic activity is declining, but really only because its energy intensity continues to decline – *there were there is the most economic activity …*

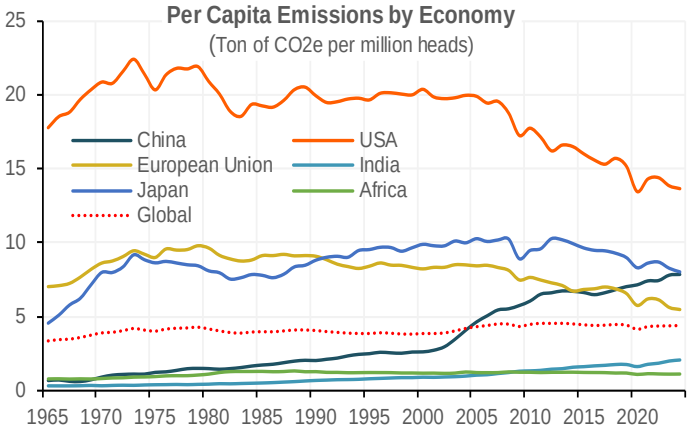




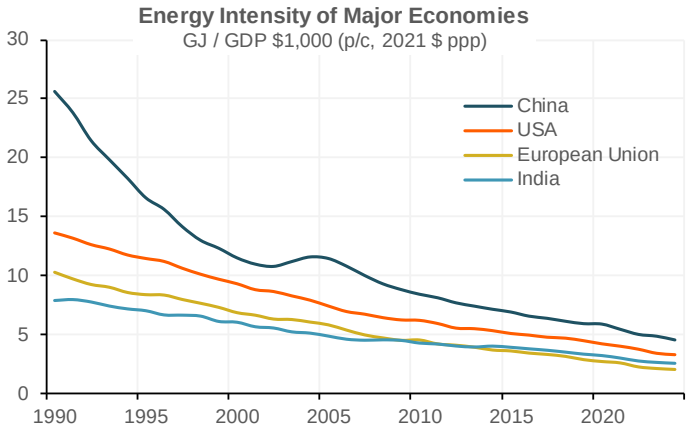
That's the problem: 'energy' still means fossil fuels, there's literally no getting around them yet.
Worse China and India – or 2.8 billion people – still have rising per capita emissions.

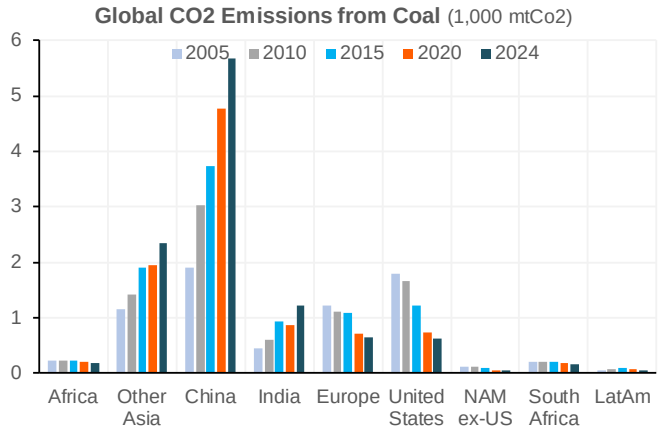
- *Whether we like it or not, their fossil fuel of choice is coal*





The world's biggest economies emit less (relatively), ***mostly because*** they become **more energy efficient**





Source: PSC Research. Global Carbon Project, Climate Watch Portal, BP Statistical Review, World Bank, Ember

GLOBAL ELECTRICITY-GENERATION: A STORY OF *GROWTH*

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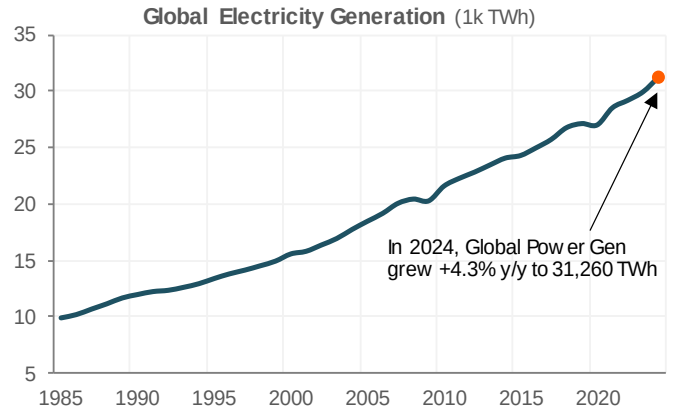
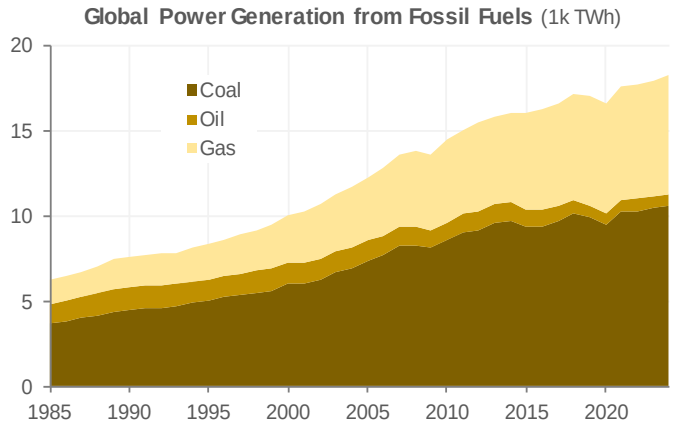
"Electrification" is a global thing; electricity demand grew +4.3% globally in 2024

Global power demand grew nearly twice as fast as primary energy use did

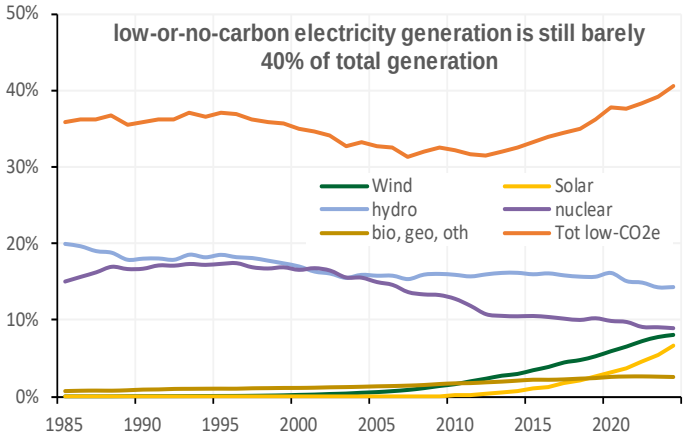
- As with all energy, **EM economies drive electricity demand growth too** (+5.5% y/y);
- **Electrification is a thing everywhere**, however, and DM economies grew their power-gen by +2% (also nearly 2x their 1% growth of primary energy demand in 2024).

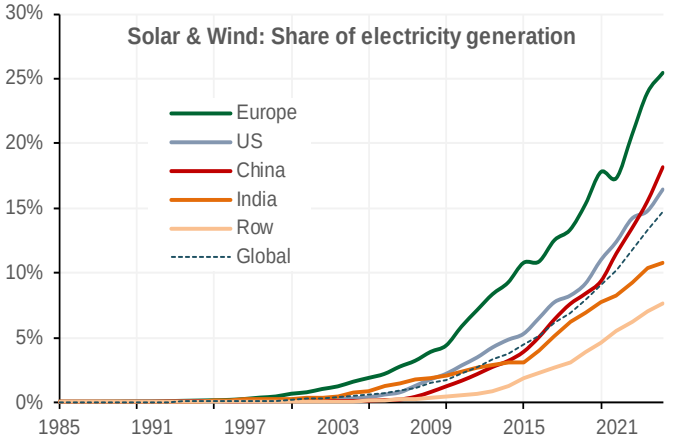


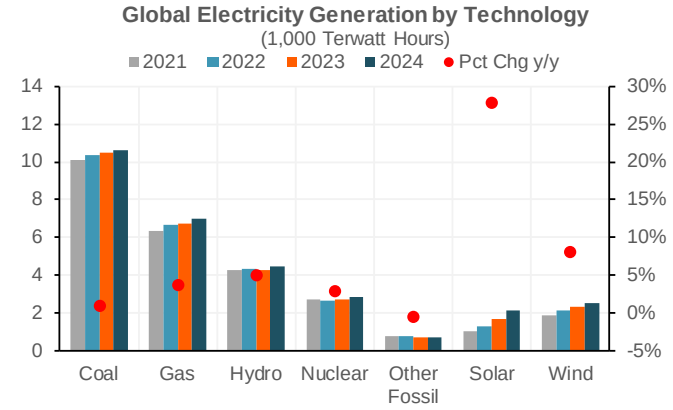


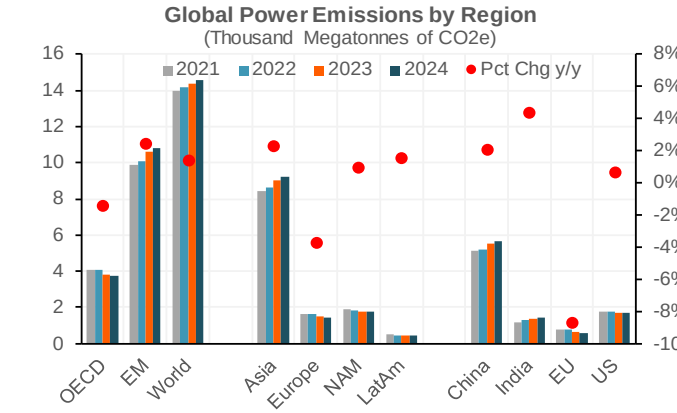
'Clean' Power Expands Really Fast – from a depressingly low base

Power generated by hydro + nuclear + solar + wind expanded by 8% in 2024 as well. Once again, EM growth hit +11%, thanks principally to the stupendous scale of China's solar & wind build out. nearly doubled that of the OECD (+6%). Europe's clean power generation grew by +7% once again.









Source: PSC Research, IEA, EI Statistical Review of World Energy, Ember

POWER GEN: THE NUCLEAR VALUE CHAIN

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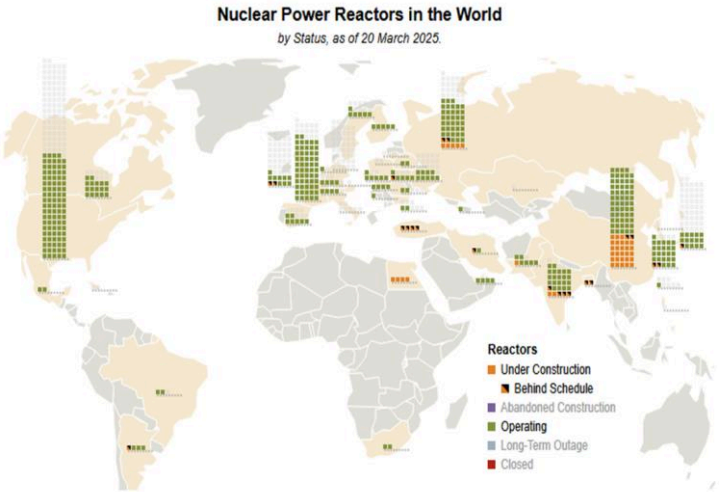
A renaissance of the nuclear power gen sector is coming, here in the US and is well underway in Europe. In China as well as across the Mideast, governments are pushing to build or expand the sector.

Globally it seems, **big money, big government and good new tech** is aligning to drive growth in a sector left to languish for decades. So, we suspect there'll be many more opportunities for investors across the value-chain.



Presenting our layout of the nuclear value chain, which comprises all parts starting from uranium mining all the way to the utilities operating nuclear plants [link]

Market Cap in million US$

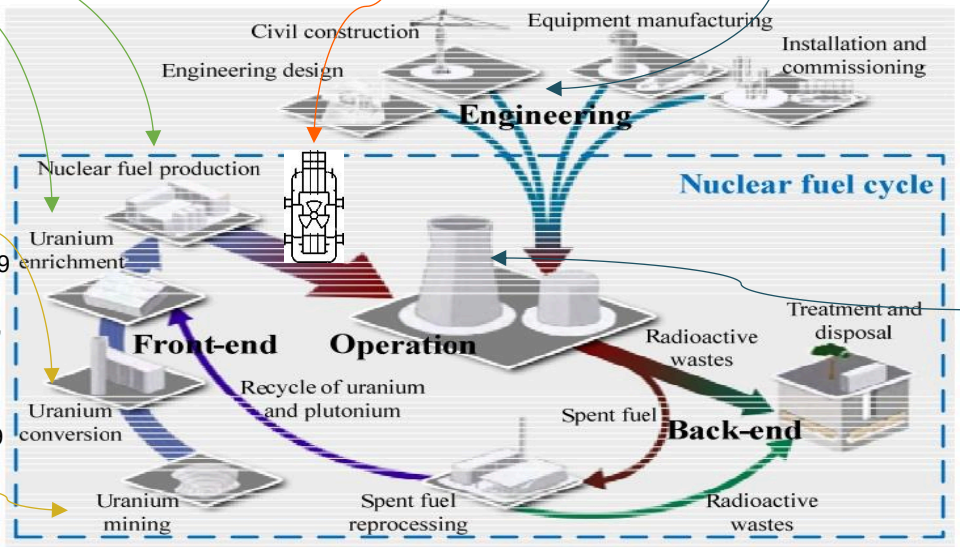
Nuclear Enrichment & Fuel Production:
Centrus (LEU), $1,930
ASP Isotopes (ASPI), $645
Orano, Private
Urenco, Private
General Matter, Private
Silex Systems (SLX AU), AUD 848
LIS Technology, Private
Global Laser Enrichment (Silex, Cameco JV), Private

SMR/MMR Designers/Builders:
NuScale (SMR), $8,600
Oklo (OKLO), $6,803
Nano Nuclear Energy (NNE), $1,345
TerraPower, Private
X-energy, Private
Kairos Power, Private
Rolls Royce Holdings (RYCEY), $100,110
Fabrication:
BWX Technologies (BWXT), $10,917
Global Nuclear Fuel (GEV/Hitachi JV), Private

Developers:
Bechtel, Private
Westinghouse, Private (owned by Brookfield Renewable, Cameco)
Fluor Corp (FLR), $6,490
Korea Electric Power Co (KEPCO), $14,157

Uranium Conversion:
Converdyn, Private

Uranium Mining/Extraction:
Cameco (CCJ), $25,549
Orano, Private
NexGen Energy (NXG), $180
Denison Mines (DNN), $1,443
Ur-Energy (URG), $299
Uranium Energy Corp (UEC), $2,744
enCore Energy (EU-CA), $397



Operators/Utes:
Constellation (CEG), $95,081
Vistra Corp. (VST), $54,201
NRG Energy (NRG), $30,200
NextEra (NEE), $142,025

Source: PSC Research, Bloomberg, World Nuclear Industry Status Report, Atlantis Press

EUROPEAN POWER-GEN: PERFORMANCE BY SOURCE

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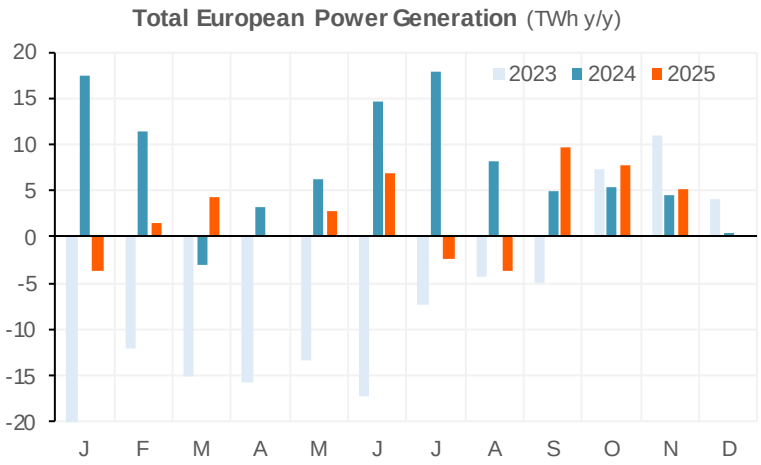
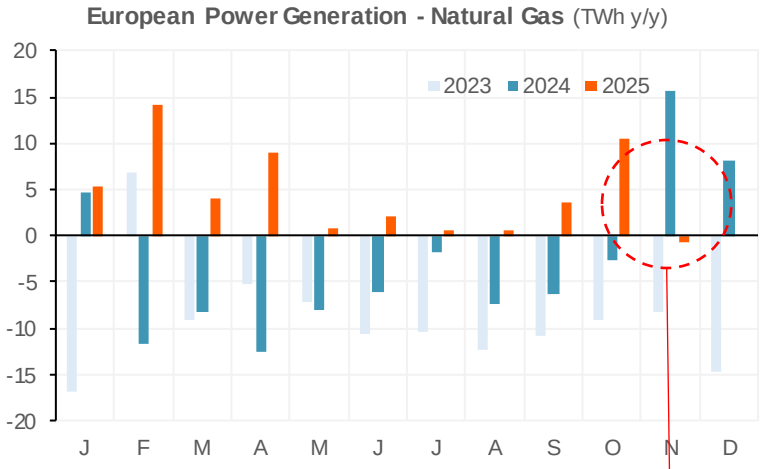
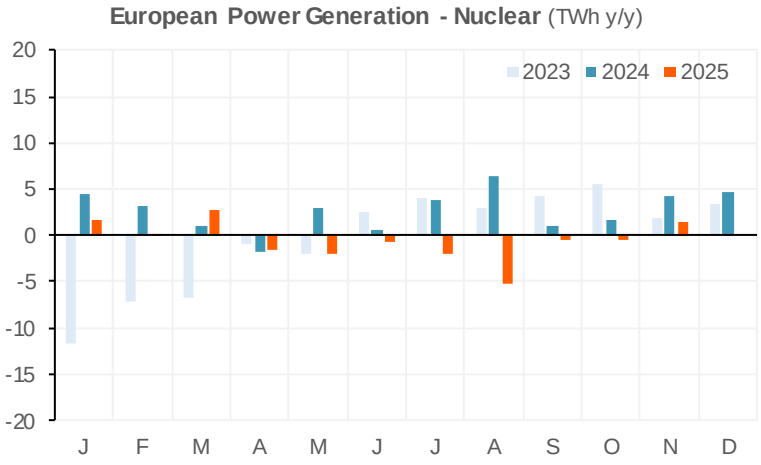
Europe's electricity demand in Nov. increased by +5.3 TWh, or 1% y/y. These data come courtesy of Ember – we now have super timely data for Europe, China and the US, which follow on p. 22 & 23).

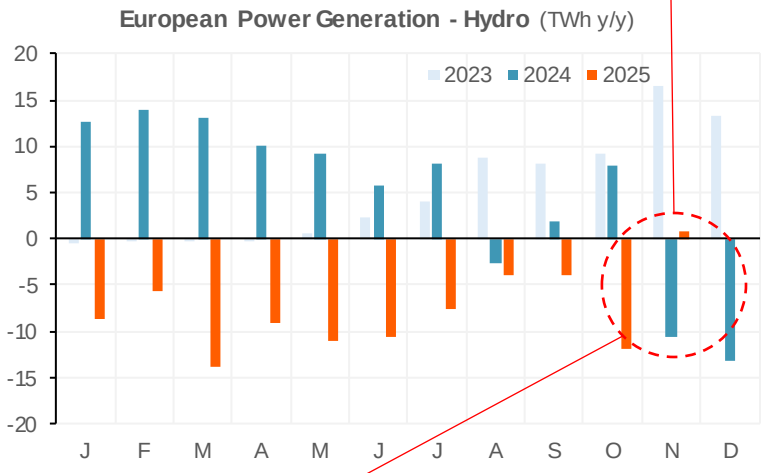
- **Natural gas burn ended its 12-months winning streak of y/y growth** with a -0.7 TWh y/y loss in Nov.; the mirror opposite is **hydro,** which snapped its 12-month losing streak with a +1 TWh y/y gain



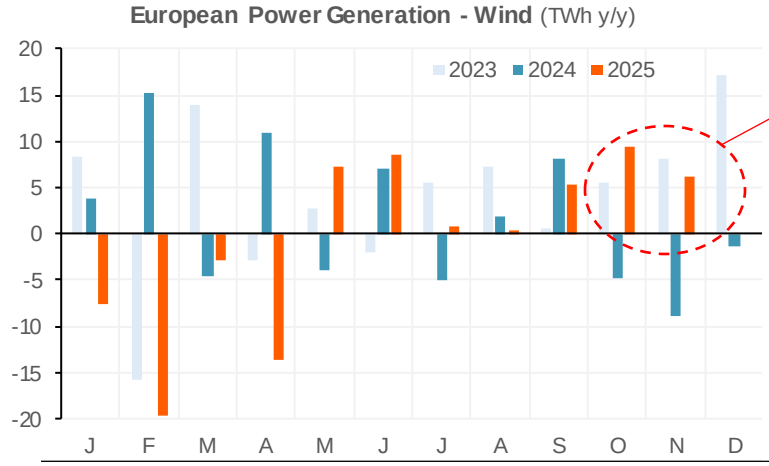


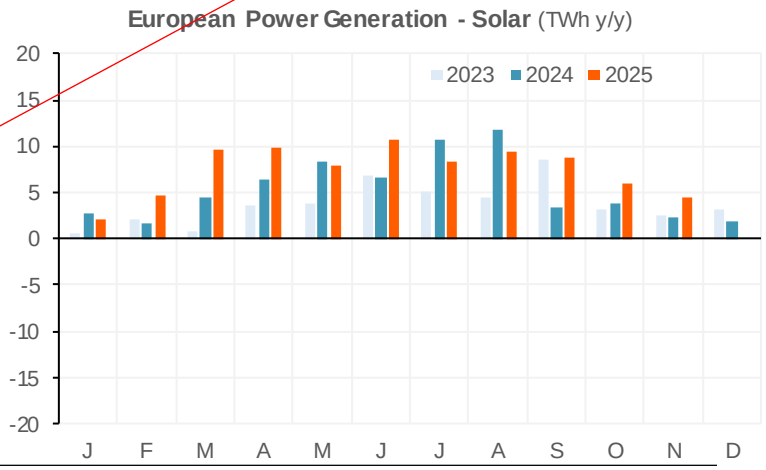
Nuclear ticked up slightly by +1.5 TWh, ending its 7-month streak of y/y losses





Wind is making *all the difference* compared to last year's abnormally low contribution in Oct & Nov





Source: PSC Research, Ember Energy

POWER-GEN IN CHINA: SHIFTING SOURCES OF POWER

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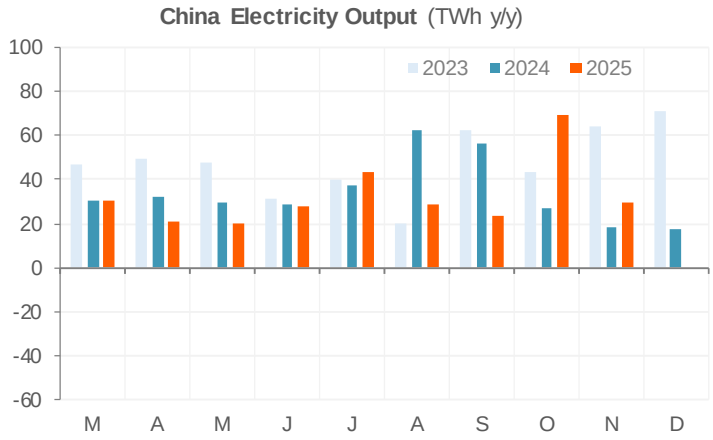
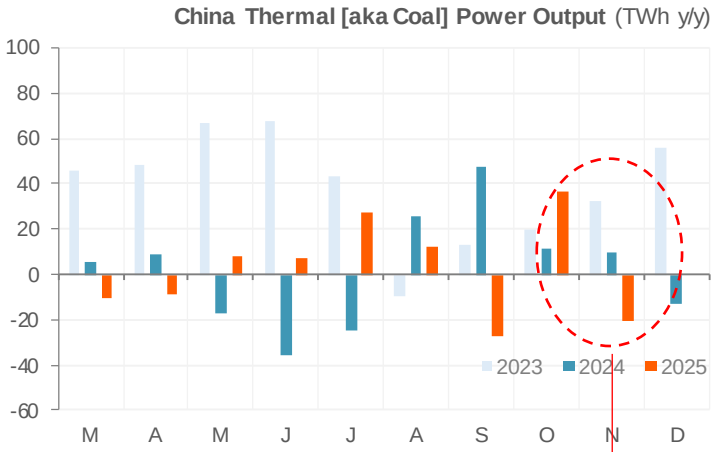
Power-Gen growth decelerated toward the year avg mean in November. Still robust at +30 TWh, or 4.0%, y/y, it was less than half the exceptional boost of Oct. That relative slowing jives with a slowing macro and lackluster industrial production growth. Power generation is tracking +4.2% ytd through November.

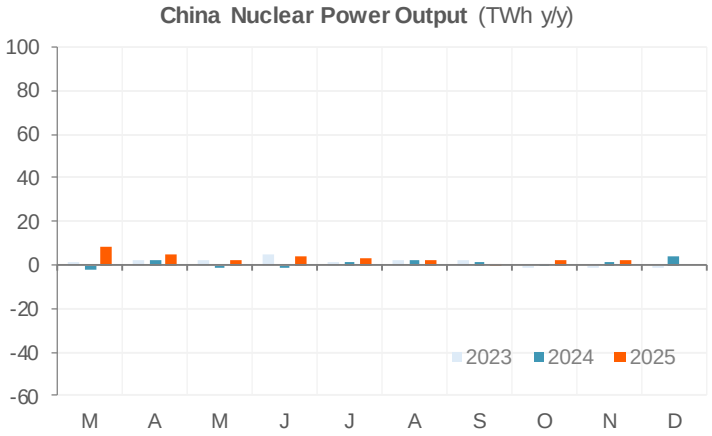
Among the power-gen sources, everyone excepting only thermal, recorded y/y gains in November.

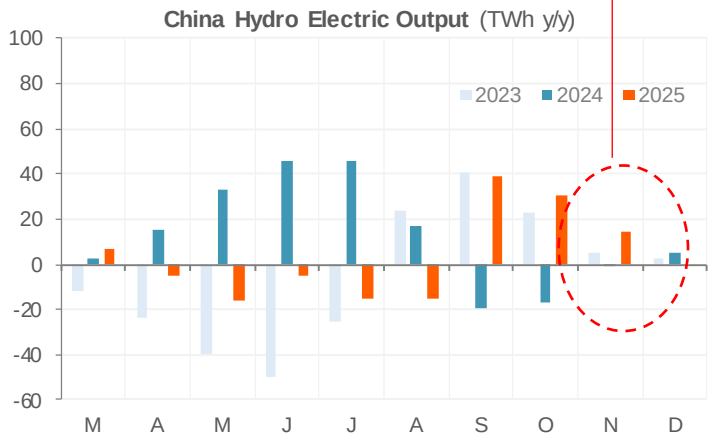
And after being down y/y last month, wind turned in the best performance in November, up +22.2 TWh while Hydro had another strong month at +14 TWh, for three straight months of solid y/y gains.



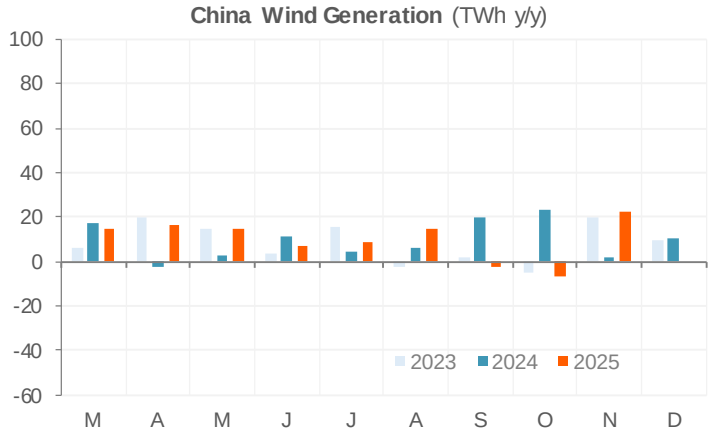


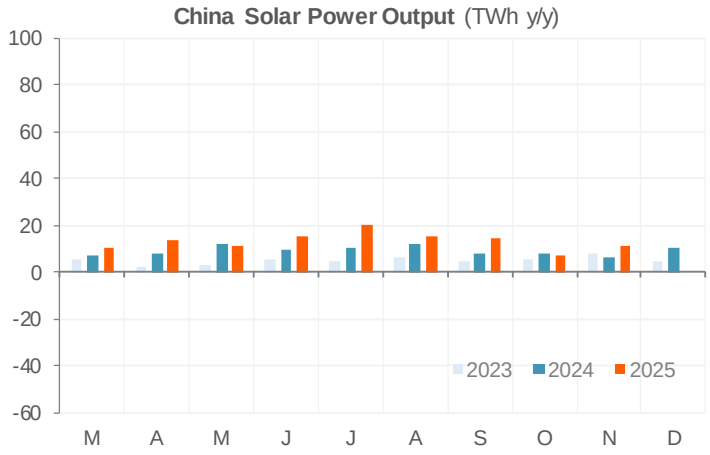
Gains for Hydro, solar and wind more than made up for the dip in coal-fired power-gen …





Squint to see Nuclear *gains of* 2 TWh; But a large upshift for Wind Generation is hard to miss





Source: PSC Research, China General Customs

US Power Sector – Power-Gen by source in November

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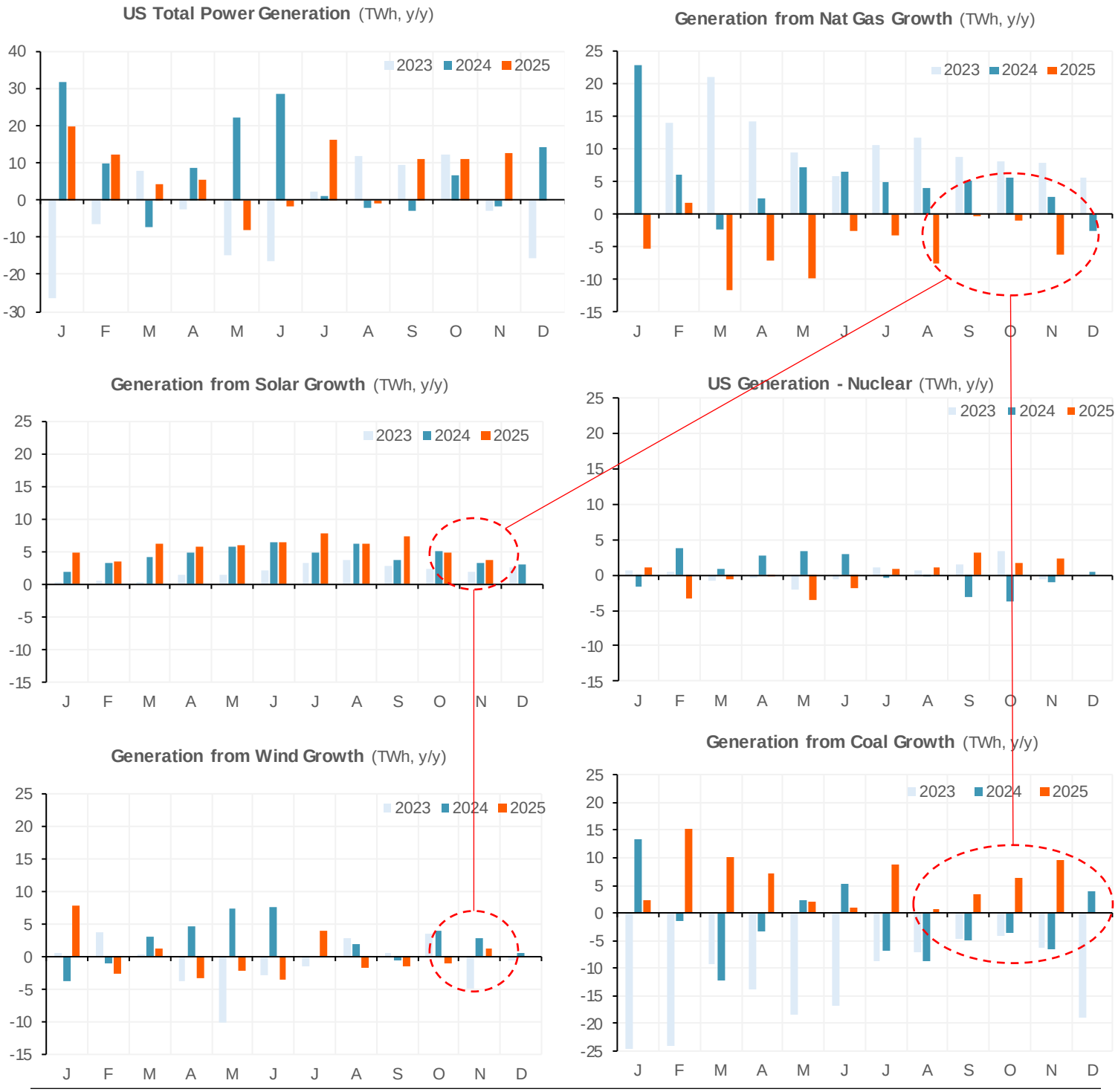
US power-generation extended its Autumn growth spurt into November (TWh)

Electric power generation *in November* increased by +13 TWh y/y, or a solid 4.1%. Fueling that growth were increases from Coal (+10 TWh), Solar (+4 TWh) and Nuclear (+2 TWh).

- Power generation *from natural gas was down -6 TWh* y/y, extending its losing streak to 9 months
- The expansion of solar generation remains the most consistent (~6 TWh per month, ytd)
- Wind generation was up +1 TWh y/y, breaking a 3-month losing streak



Source: PSC Research, EIA

US Power Sector: Regional Fuel Mix

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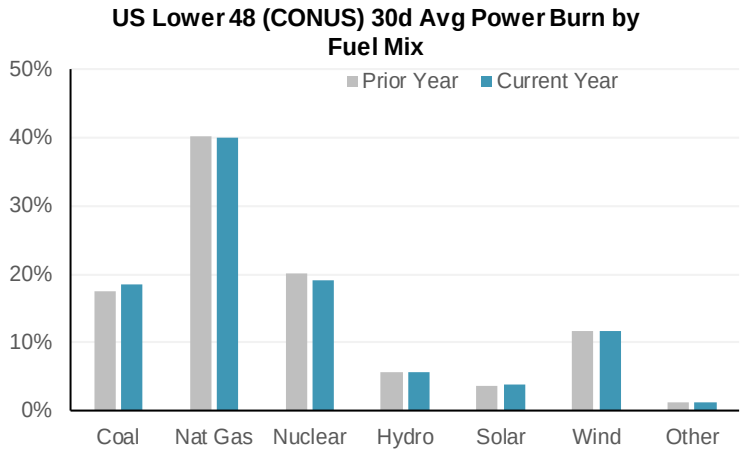
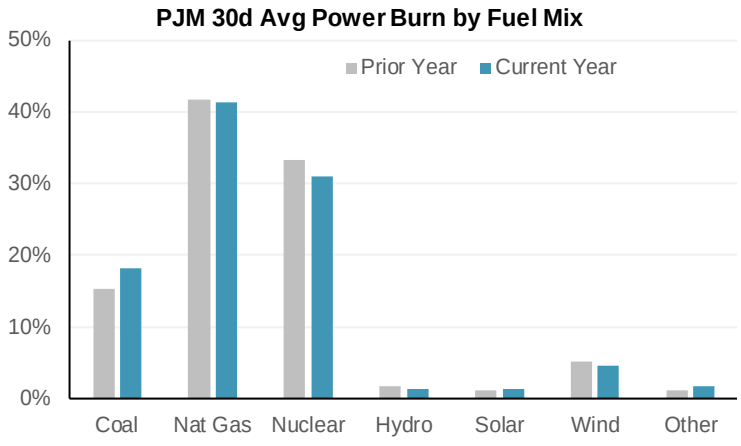
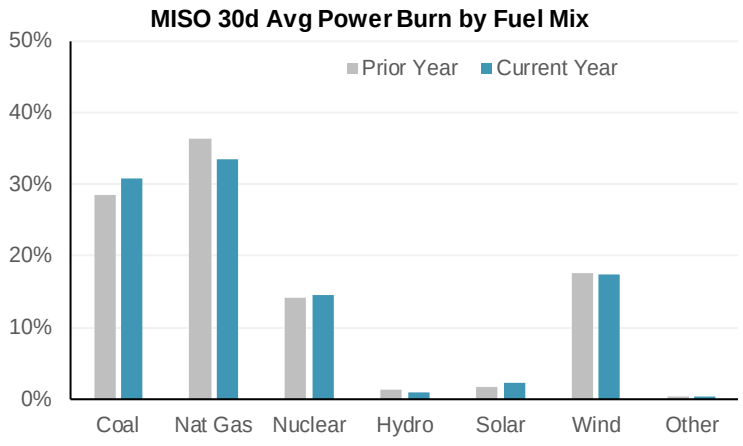
EIA-930 Report: Daily Power Burn by Fuel Mix (MWh)

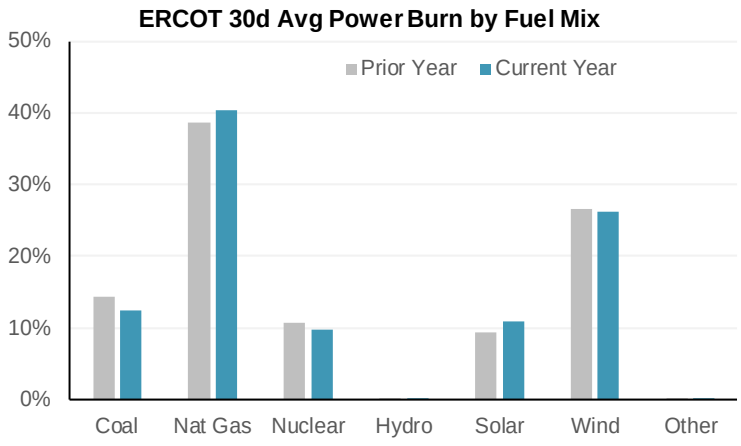
Daily 24 hr averages

	12/20/2025										(30d Average)						
											Current Period		Prior Year				2025 v 2024
	Cur Week	Prev Week	W/W Chg	Pct Chg	Share	Prev Year	Y/Y Chg	Pct Chg	Share	Y/Y Delta	Avg Daily	Share	Avg Daily	Share	Y/Y Chg	Pct Chg	Y/Y Delta
Fuel Type																	
Coal	93,639	99,665	(6,026)	-6.0%	18.5%	77,431	16,208	20.9%	17.0%	1.5%	91,994	18.4%	79,857	17.5%	12,137	15.2%	1.0%
Nat Gas	184,884	194,127	(9,243)	-4.8%	36.5%	175,265	9,619	5.5%	38.4%	-1.9%	199,714	40.0%	183,611	40.2%	16,103	8.8%	-0.2%
Nuclear	97,752	98,493	(741)	-0.8%	19.3%	94,770	2,982	3.1%	20.8%	-1.5%	95,087	19.1%	92,128	20.2%	2,958	3.2%	-1.1%
Petroleum	1,481	1,640	(159)	-9.7%	0.3%	403	1,078	267.9%	0.1%	0.2%	1,203	0.2%	533	0.1%	671	125.8%	0.1%
Hydro	30,370	26,016	4,354	16.7%	6.0%	27,350	3,020	11.0%	6.0%	0.0%	27,877	5.6%	25,390	5.6%	2,487	9.8%	0.0%
Solar	19,894	20,347	(453)	-2.2%	3.9%	15,205	4,689	30.8%	3.3%	0.6%	19,208	3.8%	16,853	3.7%	2,354	14.0%	0.2%
Wind	72,378	59,945	12,432	20.7%	14.3%	60,959	11,419	18.7%	13.3%	0.9%	57,852	11.6%	53,157	11.6%	4,696	8.8%	0.0%
Other	6,387	6,172	215	3.5%	1.3%	5,308	1,079	20.3%	1.2%	0.1%	5,968	1.2%	5,369	1.2%	599	11.2%	0.0%
Total Net Gen	**506,785**	**506,405**	**379**	**0.1%**	**100%**	**456,690**	**50,094**	**11.0%**	**100%**		**498,903**	**100%**	**456,898**	**100%**	**42,005**	**9.2%**	
Lower 48 Demand																	
California	30,467	31,492	(1,025)	-3.3%	6.4%	23,421	7,047	30.1%	4.8%	1.6%	30,120	6.4%	23,439	5.3%	6,682	28.5%	1.1%
Carolinas	27,595	30,284	(2,689)	-8.9%	5.8%	25,169	2,426	9.6%	5.2%	0.6%	26,032	5.6%	25,971	5.9%	61	0.2%	-0.3%
Florida	25,691	25,512	179	0.7%	5.4%	25,039	651	2.6%	5.2%	0.3%	25,775	5.5%	24,683	5.6%	1,093	4.4%	-0.1%
Mid-Atlantic	109,907	109,513	394	0.4%	23.3%	100,433	9,474	9.4%	20.7%	2.5%	103,058	22.0%	98,822	22.4%	4,235	4.3%	-0.4%
Midwest	88,772	87,532	1,239	1.4%	18.8%	72,022	16,750	23.3%	14.8%	3.9%	84,650	18.0%	76,929	17.4%	7,720	10.0%	0.6%
New England	14,991	15,331	(340)	-2.2%	3.2%	11,469	3,521	30.7%	2.4%	0.8%	14,374	3.1%	11,717	2.7%	2,657	22.7%	0.4%
New York	18,965	19,050	(85)	-0.4%	4.0%	16,130	2,835	17.6%	3.3%	0.7%	17,994	3.8%	15,390	3.5%	2,604	16.9%	0.4%
Northwest	42,840	42,228	613	1.5%	9.1%	45,930	(3,090)	-6.7%	9.5%	-0.4%	44,516	9.5%	44,336	10.0%	179	0.4%	-0.5%
Southeast	28,742	28,567	175	0.6%	6.1%	25,992	2,750	10.6%	5.4%	0.7%	26,747	5.7%	27,329	6.2%	(582)	-2.1%	-0.5%
Southwest	11,751	12,131	(379)	-3.1%	2.5%	15,201	(3,449)	-22.7%	3.1%	-0.6%	11,819	2.5%	14,150	3.2%	(2,330)	-16.5%	-0.7%
Tennessee	22,236	20,990	1,246	5.9%	4.7%	77,333	(55,097)	-71.2%	15.9%	-11.2%	34,084	7.3%	32,093	7.3%	1,990	6.2%	0.0%
Texas	50,750	49,168	1,582	3.2%	10.7%	46,943	3,807	8.1%	9.7%	1.1%	49,847	10.6%	46,918	10.6%	2,929	6.2%	0.0%









Source: PSC Research, EIA

EV Forecasting – How Much Oil Demand We'll Lose

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With the final segment of our global car fleet modeling now complete, we can ***forecast with significant confidence some 95% of global gasoline demand and 16% of diesel*** (equivalent to 30.6 million barrels a day of oil demand, or ~28% of all oil consumption in 2024)

- Shockingly, **global gasoline demand** associated with Light Duty Vehicle use most likely peaks in 2027, which is a good few years earlier than most expect / model
- **Out to 2030, the growth in conventional car fuel demand across Emerging Market economies will, we model, be less than the losses to NEVs in China, Europe and the USA**

Per the charts on the right, NEVs are currently displacing **~970 kb/d** of road fuel demand, and we estimate that will rise to **2.9 mb/d** by 2030 as adoption grows. This will mainly be driven by China, the US, and Europe but the *Rest of World* will play an increasingly important role too. In terms of oil demand, we estimate the US, China, and Europe will collectively decline by **1.3 mb/d** but interestingly grow by 1.0 mb/d in our *RoW* regions.

Uniquely our model uses historic fleet and/or sales data for 78 countries, which we marry to forecasts from our Auto and Tech team of **Alex Potter** and **Ben Johnson**, which they update at least 2x per year…

Earlier episodes in our NEV modeling series: the US [link], Europe [link], China [link], Western Hem. ex-US [link], APAC ex-China [link], and RoW [link].

From 2005 to 2019, global LDV fuel demand grew from 24 mb/d to 30 mb/d at a CAGR of 1.5%. The pandemic put that on hold, and since then growth slowed significantly.

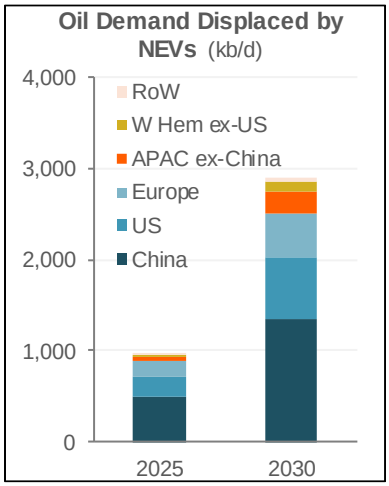
Through our forecast period from 2026-'30, that growth slows even further and flips to a decline rate of -0.3%.

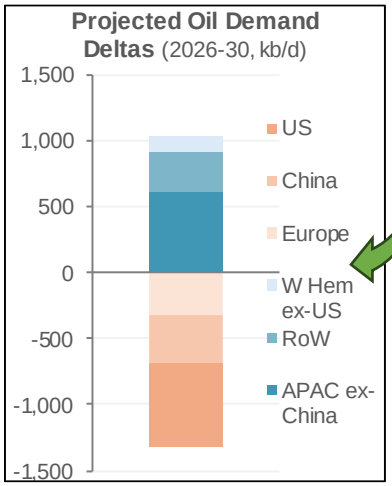
We model that the combined gasoline and diesel use of the global LDV fleet (every car and/or truck that weighs less than 15.4k lbs) will peak around 30.7 mb/d in 2027

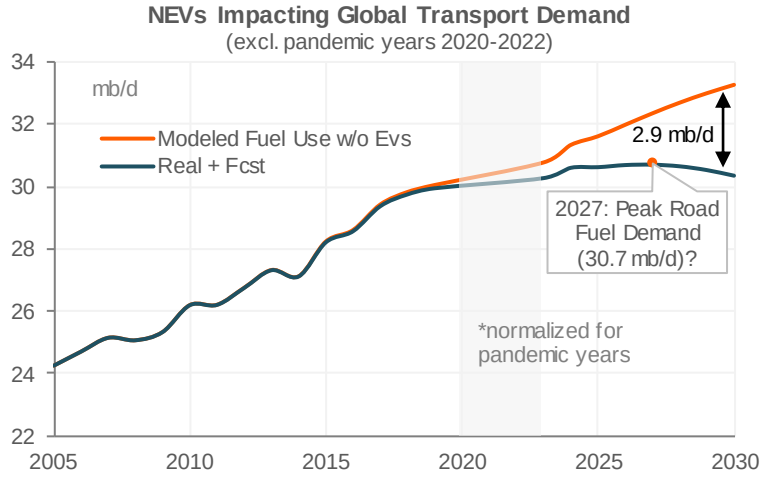
By 2030, we estimate road fuel demand to decline by ~300 kb/d to 30.4 mb/d

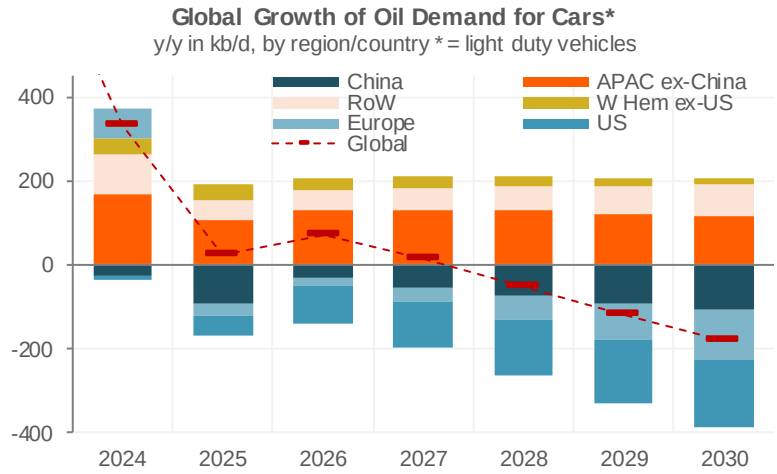
Without NEVs, global road fuel demand would have continued to grow at a CAGR of 1%, reaching ~33.3 mb/d by the end of the decade.

As such, the delta between real consumption and modeled fuel use without EVs translates to some 2.9 mb/d of demand displacement by 2030.









Source: PSC Research, IHS, Rystad

EV Forecasting (Cont'd): Principal Regions

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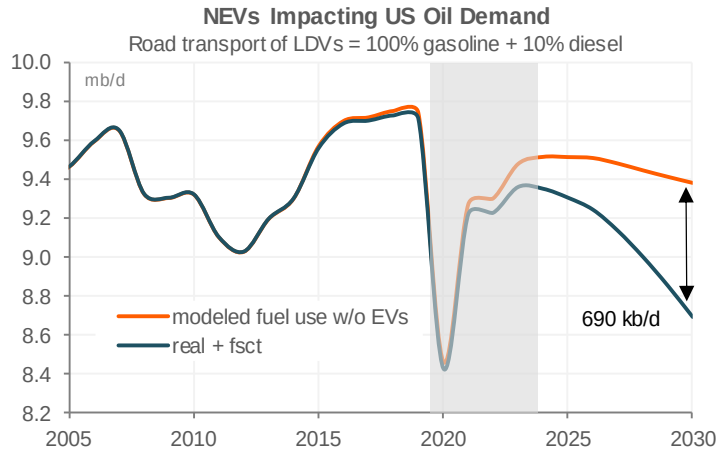
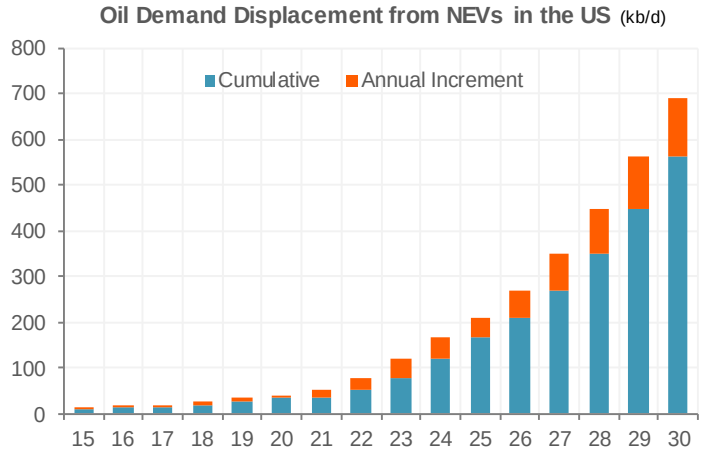
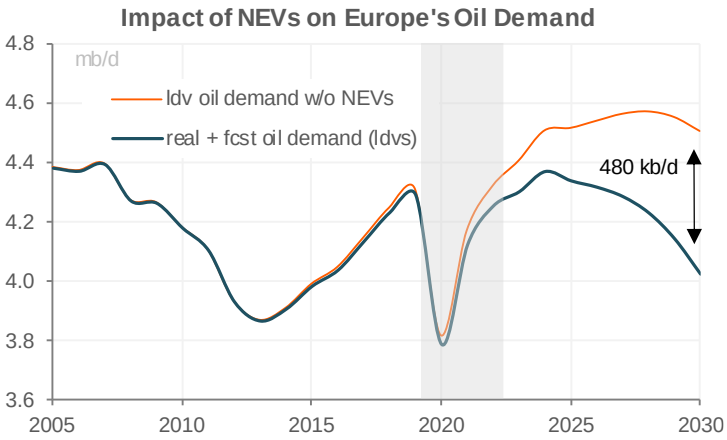
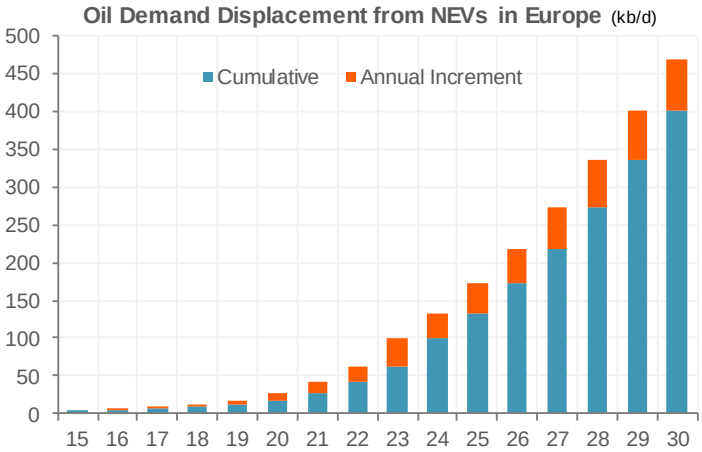
We model a US oil demand decline of ~700 kb/d from 2023's peak of 9.4 mb/d to ~8.6 mb/d *(gasoline only demand deflates to barely 8.0 mb/d).* **It's kinda cute** that even without any NEV sales, the US gasoline recovery would have peaked in 2026 and declined to 9.3 mb/d by 2030.



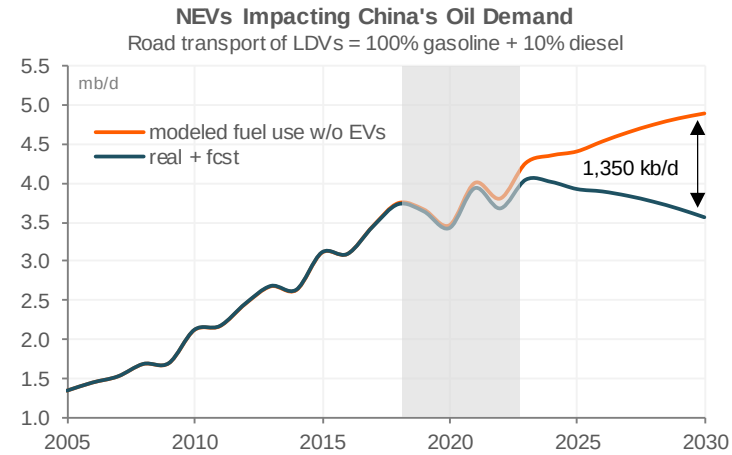


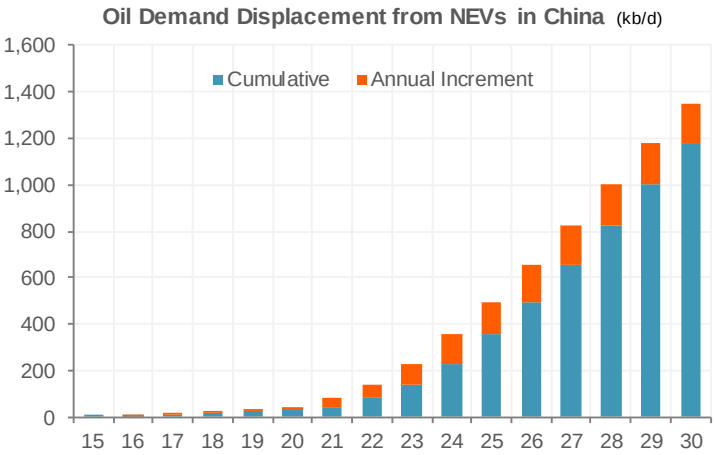
We model a European road-fuel demand decline of ~400 kb/d from 2024's 4.4 mb/d peak, with *gasoline and diesel demand deflating to barely 4.0 mb/d.* **Interestingly,** without NEVs, in theory, European road fuel demand would have grown to a new peak in 2028 (just shy of 4.6 mb/d).





We model a Chinese road-fuel demand decline of ~500 kb/d from 2023's 4.0 mb/d peak that *deflates to barely 3.6 mb/d.* **That is quite the 1.3 mb/d hit** to what would have amounted to 4.9 mb/d in road-fuel demand if NEVs weren't in the picture…





Source: PSC Research, Wards Intelligence, IHS Markits, Marklines

EV Forecasting (Cont'd): "RoW" Forecasting

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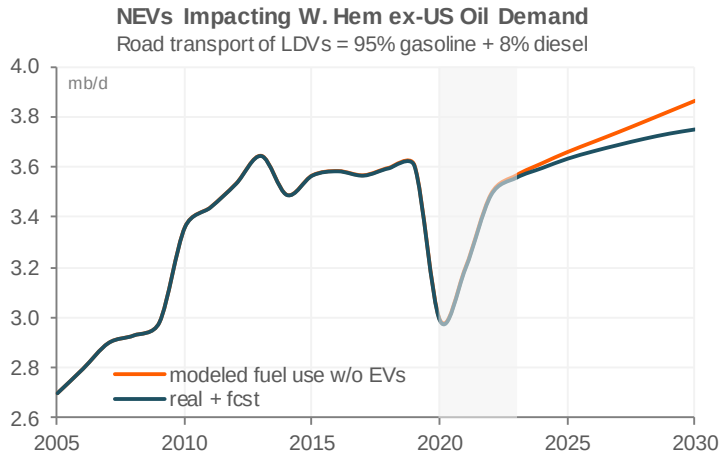
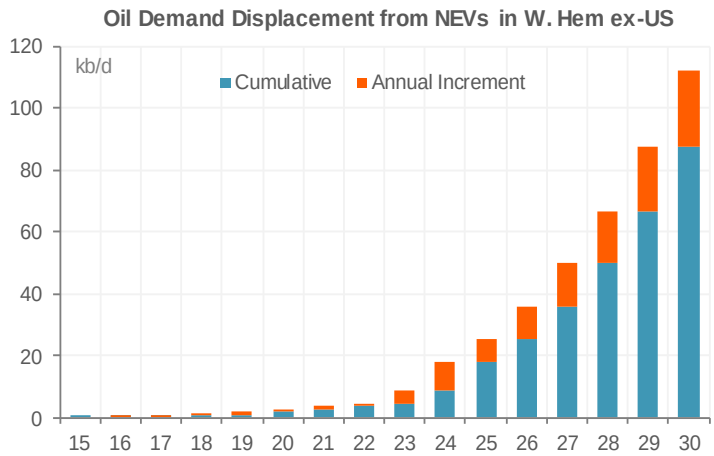
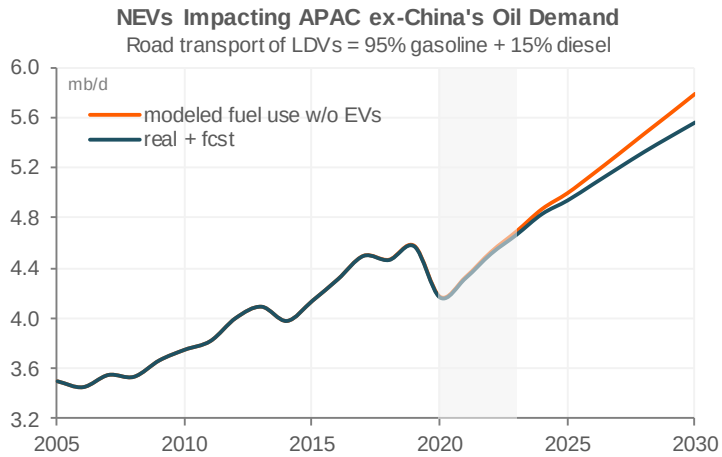
Here we introduce our differentiated *Rest of World* EV/Oil Demand forecast, which includes the **W. Hem. ex-US** (defined as **Canada**, **Brazil**, **Mexico**, **Argentina**, **Chile**, and **Peru**). Next is **APAC ex-China** (**Japan**, **S. Korea**, **India**, **Indonesia**, **Turkey**, **Australia**, **Malaysia**, and **Taiwan**). Lastly is the **Rest of World** (**Russia**, **Iran**, **Ukraine**, **S. Africa**, **S. Arabia**, **Egypt, Kazakhstan, and Israel**).

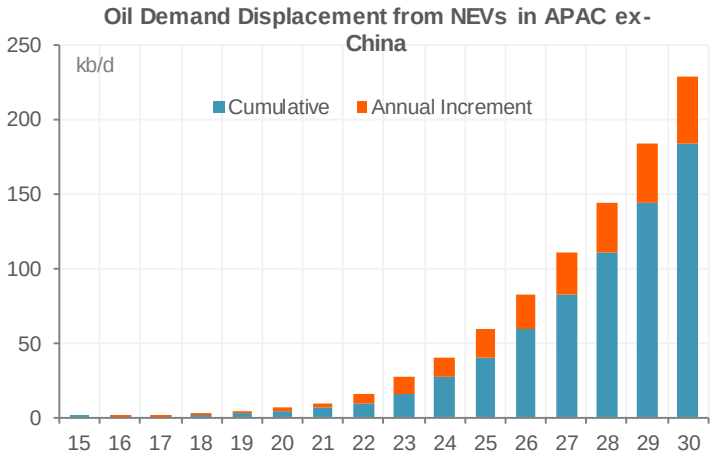
For the W. Hem ex-US, we model oil demand growing at a CAGR of 0.5% from 2026-2030 (3.6 to 3.7 mb/d). Without NEVs, road fuel demand would have grown at a 1% CAGR to 3.9 mb/d.



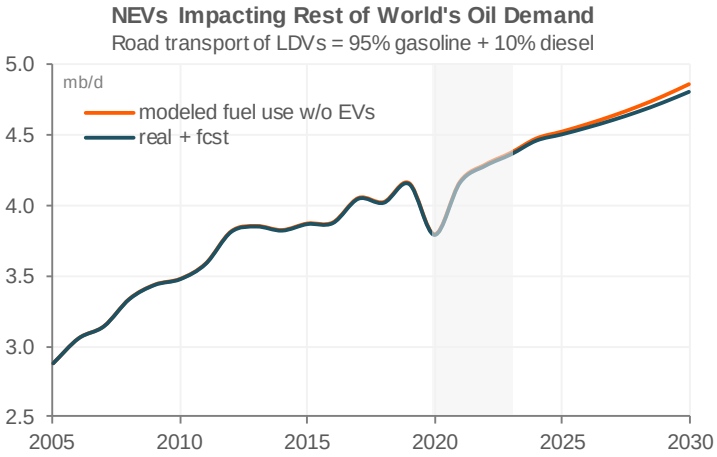


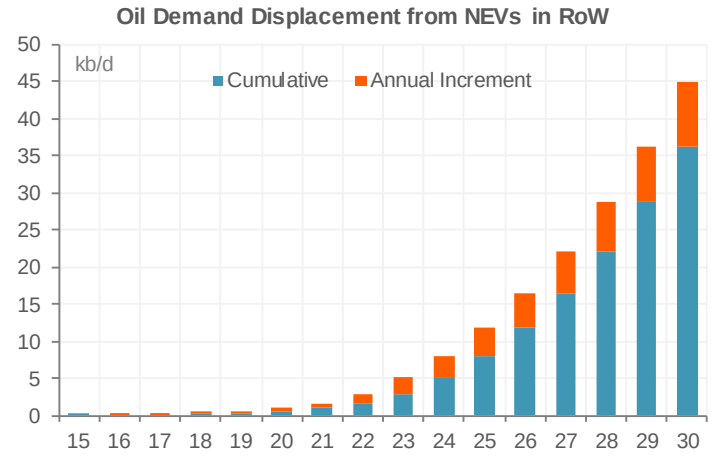
For APAC ex-China, we model oil demand growing at a CAGR of 1.9% from 2026-2030 (5.0 to 5.6 mb/d)**. Without NEVs,** that growth would have accelerated to 5.8 mb/d.





For the Rest of World, we model oil demand growing at a CAGR of 1.4% from 2026-2030 (4.6 to 4.8 mb/d)**. Without NEVs,** that growth would have accelerated to ~4.9 mb/d.





Source: PSC Research, Wards Intelligence, IHS Markits, Marklines

TRACKING THE LATEST NEV SALES - OCTOBER

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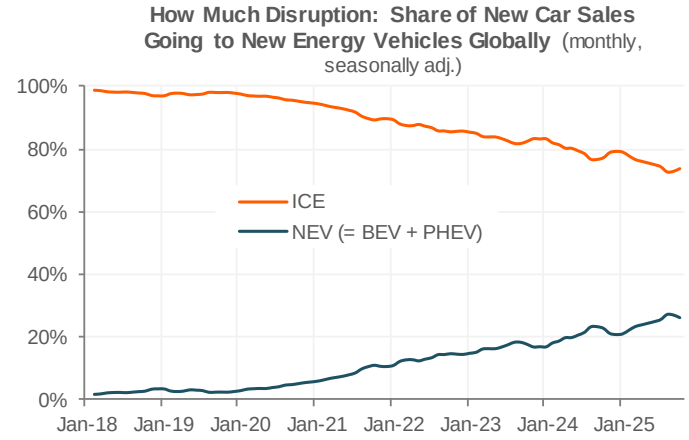
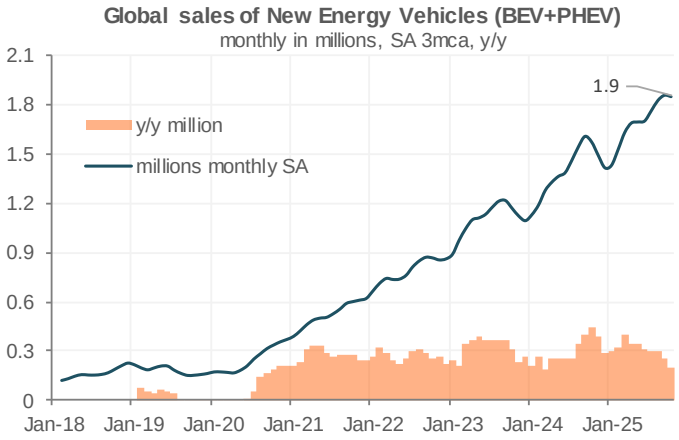
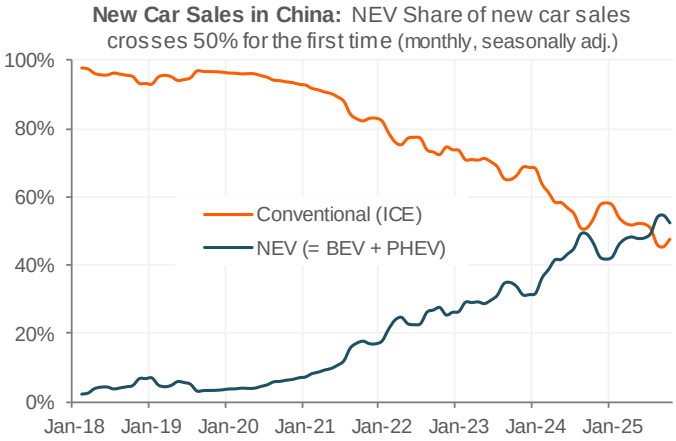
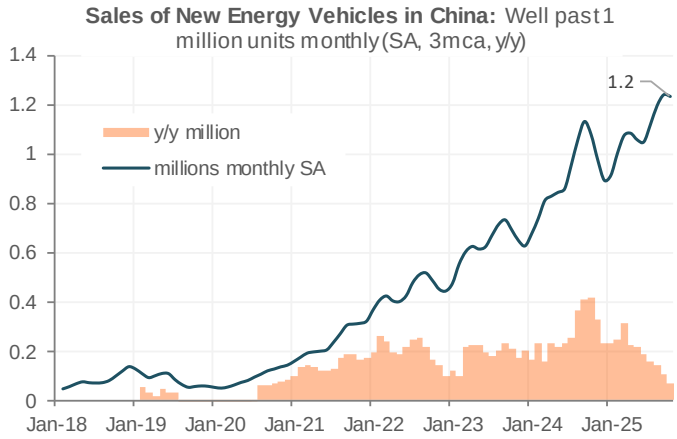
EV Sales Momentum: We track monthly EV sales globally and in key regions by taking 3-month moving averages and seasonally adjusting the data. We draw on expertise from PSC's Transportation gurus **Alex Potter** and **Ben Johnson** as well, especially their forecasting acumen (see p. 31).



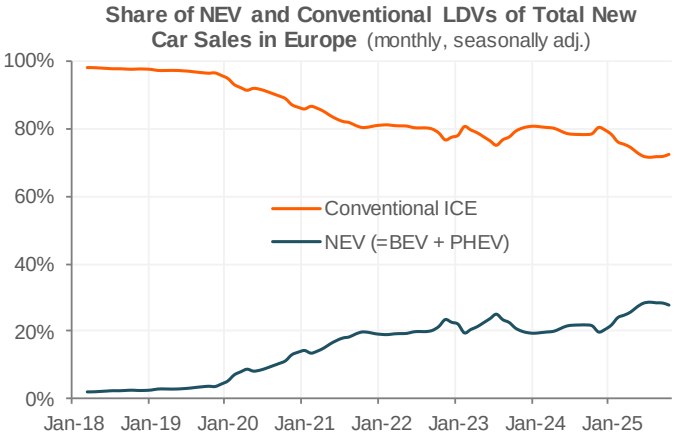


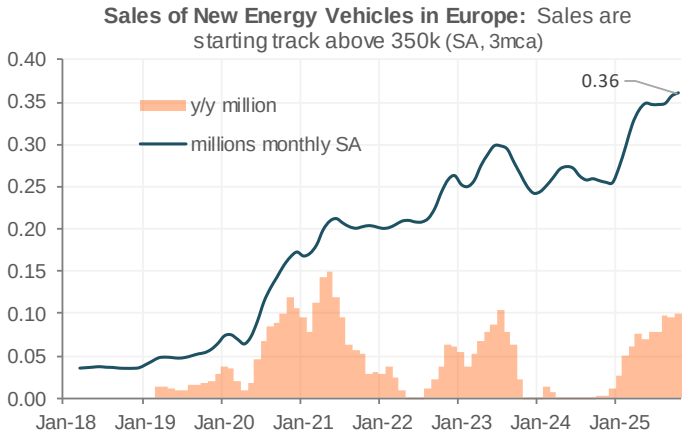
Global NEV sales continue to track at ~1.9m units, but its share of new car sales slipped to 26% last month. China continues to account for two-thirds of these sales and cemented its status as the driver of global EV adoption. However, NEV sales in China slipped last month but continue to maintain at ~1.2m units sold monthly. As such, ICEs gained back some share but NEVs continue to account for over half of new car sales.





NEV sales in Europe are starting to break past ~350k units after trending sideways for the last five months. The share of NEVs as a percentage of total new car sales also continues to hold at 28%. BEVs continue to remain popular in Europe, accounting for 67% of new NEV sales.





Source: PSC Research, Wards, Dataforce, Rystad, Gasgoo

TRACKING THE LATEST NEV SALES - OCTOBER

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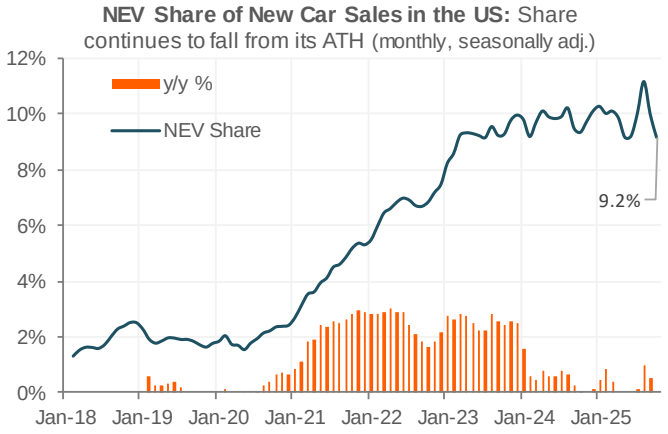
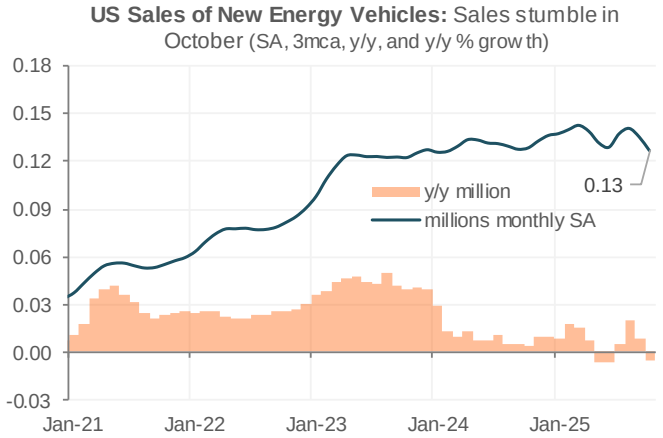
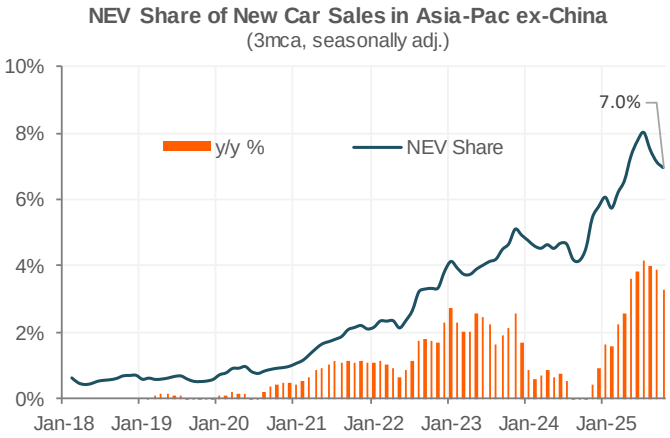
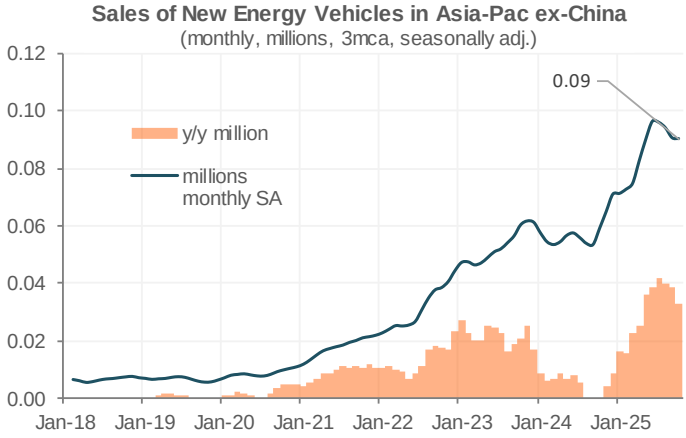
Global EV Sales momentum (cont'd): Here we focus on the US and our two newly defined "Rest of World" regions: the **Western Hemisphere ex-US** and **Asia-Pacific ex-China** (see definitions on p. 28). We apply the same process used on the previous page here as well.



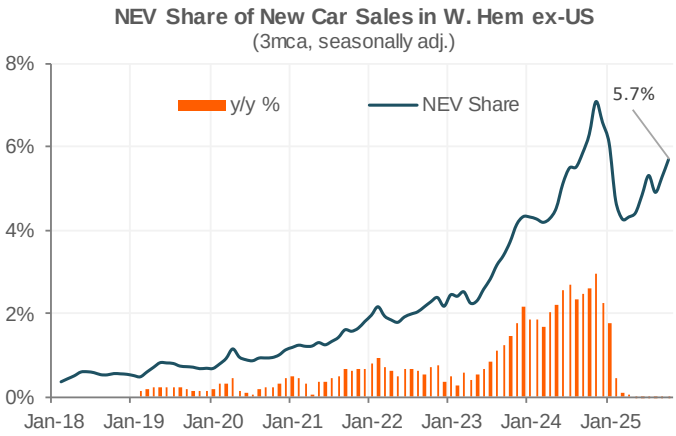


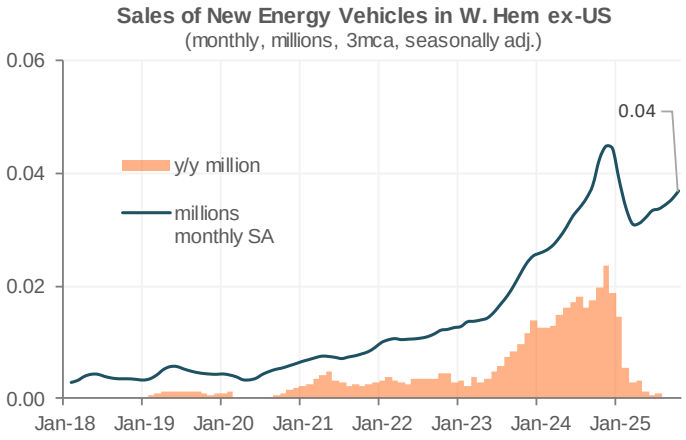
NEV sales in the US plummeted to 90k units in October, the first month after the federal EV credit's expiration. NEVs' share of new car sales also fell to 9.2%, which is a significant drop from its ATH of 11.4% in August. NEV sales in the US are on track to reach ~1.6m this year, similar to 2024.





APAC ex-China and the W. Hem. ex-US are starting to play increasingly important roles in the electrification of transportation. Sales in APAC ex-China continue to average 90k units sold per month and in the same territory as the US in terms of share of new car sales. Meanwhile, sales in the W. Hem. ex-US continue to average ~40k per month and are far from last November's peak…





Source: PSC Research, Wards, Dataforce, Rystad

GLOBAL AUTO ELECTRIFICATION – LDV FLEET AND SALES

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We have hugely detailed data-sets of global light-duty vehicle sales & car fleets – here's a summary

Revamped NEV Modeling

- We recalibrated our EV sales data to align our historicals with those of **Alex Potter**, who regularly publishes his brand new "Big Book of Automotive Trends".
- A material, massive re-acceleration of NEV sales is in the works (i.e. in the production/marketing plans of 23 top car manufacturers and sellers, which go out 3-yrs)
- We modeled out Europe where, similar to the US, total sales should decline by -3% in 2025 and extend a broadly flat sales trend through 2030.
- Total sales in China should grow 8% in 2025, then 2% per annum through 2030.
- This is about LDVs (private and commercial vehicles no heavier than 7 tons). "NEVs" include Battery Electric Vehicles (BEVs) and Plug-in hybrids (PHEVs)
- Alex's work primarily focuses on tracking sales data for the US, China, and Europe (which he defines as the EU + EFTA + UK) and using marketing and manufacturing plans down to the model level for a slew of the western world's most important OEMs.
- We look to complement his work by tracking the rest of the world outside of what we dub "The Big Three", in newly defined regions such as the "**Western Hemisphere *ex-US***", which consists of Canada, Mexico, Brazil, Argentina, Chile, and Colombia; and "**Asia Pacific *ex-China***", which consists of India, Malaysia, Indonesia, Taiwan, Japan, Australia, South Korea, New Zealand, and Turkey.

Sales by Economy (millions and y/y % growth)					Forecast (2025-'30)					
ICE Sales by Economy	**2019**	**2022**	**2023**	**2024**	**2025**	**2026**	**2027**	**2028**	**2029**	**2030**
United States	16.74	12.93	14.15	14.45	13.83	13.14	12.35	11.68	10.92	10.12
	-1%	-10%	9%	2%	-4%	-5%	-6%	-5%	-6%	-7%
Europe (EU + EFTA + UK)	17.53	10.26	11.59	11.91	11.00	10.40	9.65	8.95	8.22	7.51
	1%	-11%	13%	3%	-8%	-5%	-7%	-7%	-8%	-9%
China	23.18	17.22	16.69	14.73	14.37	12.58	10.44	8.70	6.93	5.12
	0%	-17%	-3%	-12%	-2%	-12%	-17%	-17%	-20%	-26%
West. Hemisphere ex-US	7.58	6.25	6.60	7.05	7.29	7.34	7.42	7.48	7.47	7.39
	5%	12%	6%	7%	3%	1%	1%	1%	0%	-1%
Asia Pacific ex-China	15.46	14.52	15.61	15.10	15.16	15.29	15.42	15.57	15.66	15.68
	-4%	5%	8%	-3%	0%	1%	1%	1%	1%	0%
Rest of World	3.76	2.67	3.22	3.86	3.83	3.89	4.03	4.22	4.42	4.65
	24%	-5%	21%	20%	-1%	1%	4%	5%	5%	5%
Total	**83.23**	**62.86**	**66.70**	**65.82**	**61.64**	**58.74**	**55.27**	**52.37**	**49.20**	**45.82**
	-1%	-9%	6%	-1%	-6%	-5%	-6%	-5%	-6%	-7%
NEV Sales by Economy	**2019**	**2022**	**2023**	**2024**	**2025**	**2026**	**2027**	**2028**	**2029**	**2030**
United States	0.32	0.93	1.46	1.58	1.66	2.12	2.98	3.73	4.49	5.21
	-3%	46%	56%	8%	5%	28%	41%	25%	20%	16%
Europe (EU + EFTA + UK)	0.56	2.70	3.17	3.09	3.76	4.49	5.50	6.51	7.50	8.46
	42%	15%	17%	-2%	22%	19%	23%	18%	15%	13%
China	1.00	5.42	7.53	11.01	13.31	15.93	18.63	20.96	23.32	25.73
	-6%	80%	39%	46%	21%	20%	17%	12%	11%	10%
Western Hemisphere ex-US	0.06	0.14	0.21	0.42	0.37	0.52	0.69	0.87	1.08	1.34
	28%	47%	55%	97%	-12%	41%	33%	27%	24%	24%
Asia Pacific ex-China	0.08	0.39	0.73	0.77	1.16	1.46	1.77	2.12	2.51	2.94
	-6%	70%	86%	5%	52%	25%	22%	20%	18%	17%
Rest of World	0.01	0.06	0.10	0.12	0.17	0.20	0.24	0.29	0.34	0.40
	1%	11%	17%	24%	17%	20%	24%	29%	34%	40%
Total	**2.03**	**9.69**	**13.27**	**17.11**	20.43	24.71	29.82	34.48	39.24	44.08
	6%	53%	37%	29%	19%	21%	21%	16%	14%	12%
Total Vehicle Sales	**2019**	**2022**	**2023**	**2024**	**2025**	**2026**	**2027**	**2028**	**2029**	**2030**
United States	17.06	13.86	15.61	16.03	15.49	15.26	15.33	15.41	15.41	15.33
Europe (EU + EFTA + UK)	18.08	12.96	14.76	15.00	14.76	14.89	15.15	15.45	15.72	15.97
China	24.18	22.64	24.22	25.74	27.67	28.50	29.07	29.66	30.25	30.85
Western Hemisphere ex-US	7.64	6.39	6.81	7.47	7.66	7.86	8.10	8.35	8.55	8.72
Asia Pacific ex-China	15.54	14.91	16.34	15.87	16.32	16.74	17.20	17.69	18.17	18.63
Rest of World	3.77	2.73	3.32	3.98	4.00	4.09	4.27	4.51	4.77	5.04
Total	**85.25**	**72.55**	**79.97**	**82.93**	**81.90**	**83.25**	**84.85**	**86.56**	**88.11**	**89.51**
Fleet ('Vehicles in Operation', million vehicles)										
Country/Region	**2019**	**2022**	**2023**	**2024**	**2025**	**2026**	**2027**	**2028**	**2029**	**2030**
United States	281	284	286	289	291	292	294	296	297	299
ICE	*280*	*282*	*283*	*284*	*284*	*284*	*283*	*281*	*279*	*276*
NEV	*1*	*2*	*4*	*5*	*7*	*8*	*11*	*14*	*18*	*22*
Europe (EU + EFTA + UK)	312	321	324	328	330	333	336	339	342	344
ICE	*311*	*316*	*316*	*317*	*317*	*316*	*315*	*312*	*310*	*306*
NEV	*1*	*5*	*8*	*10*	*14*	*17*	*21*	*26*	*32*	*38*
China	234	289	309	326	345	365	384	401	417	431
ICE	*232*	*279*	*290*	*299*	*308*	*314*	*318*	*321*	*321*	*320*
NEV	*2*	*10*	*17*	*26*	*38*	*51*	*65*	*80*	*96*	*112*
Western Hemisphere ex-US	141	145	147	150	153	157	160	164	167	171
ICE	*141*	*144*	*147*	*149*	*152*	*155*	*158*	*160*	*163*	*166*
NEV	*0*	*0*	*0*	*1*	*1*	*2*	*2*	*3*	*4*	*5*
Asia Pacific ex-China	261	280	290	299	308	318	328	338	349	359
ICE	*261*	*279*	*288*	*296*	*305*	*313*	*321*	*329*	*337*	*345*
NEV	*0*	*1*	*2*	*3*	*4*	*5*	*7*	*9*	*11*	*14*
Rest of World	131	135	136	138	140	141	143	145	148	150
ICE	*131*	*135*	*136*	*138*	*139*	*141*	*142*	*144*	*146*	*148*
NEV	*0*	*0*	*0*	*0*	*1*	*1*	*1*	*1*	*2*	*2*
Total	**1,360**	**1,454**	**1,492**	**1,529**	**1,568**	**1,607**	**1,645**	**1,683**	**1,719**	**1,755**
ICE	*1,355*	*1,435*	*1,460*	*1,484*	*1,505*	*1,523*	*1,537*	*1,548*	*1,556*	*1,561*
NEV	*5*	*18*	*30*	*46*	*63*	*84*	*108*	*134*	*163*	*194*

Source: PSC Research, Wards Intelligence, IHS Markits, BNEF, Rystad

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US Biofuels: Background 'rationale' & framework

The US Congress created the **Renewable Fuel Standard (RFS)** program to reduce greenhouse gas emissions and expand the nation's renewable fuels sector while reducing reliance on imported oil. This program was authorized under the Energy Policy Act of 2005 and expanded under the Energy Independence and Security Act (EISA) of 2007. Renewable Identification Numbers (RINs) are credits used for compliance and are the 'currency' of the RFS program. Arguably all this made sense back then, when we imported 12 mb/d of oil and smog was an issue. And after MTBE failed the test in the 1990s… *things make less sense now.*

Anyhoo, since no politician wants to lose the ag vote. We still have The Clean Air Act requiring the EPA to set the **Required Volume Obligations (RVO)** annually. Those volumes can of course be adjusted: EISA requires the EPA to evaluate and adjust the standards, which are to be finalized by Nov. 30 of the year prior to compliance. For cellulosic, the EPA sets volume requirements based on the EIA's projections, the EPA's assessment, and other factors. For biomass-based diesel, the EPA must determine the RVO at least 14 months prior to the year of the mandate.

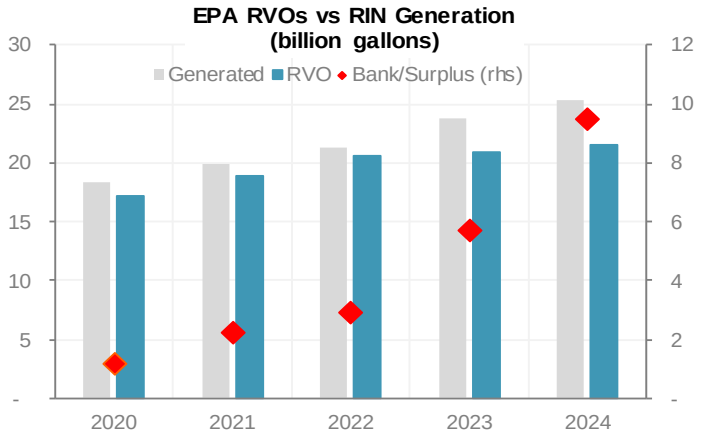
The EPA released their latest RVO proposal on June 13th. The proposal included a 10% increase in biomass-based diesel RVOs for 2026 and 2027 while reducing the RVOs for Cellulosic Biofuels. Additionally, the proposal included a 12% and 10% increase for Advanced Biofuels for 2026 and 2027, respectively. Factoring everything in, the RVO for total Renewable Fuels now stands at over 24 billion gallons per year, which is an 8% increase from 2025's final ruling.

The new RVO mandate means biofuels supplies have to rise by ~1,600 million gallons (104 kb/d) from current levels. **PSC's Ryan Todd** views this is not only bullish for biofuel margins, but also for RIN prices since limitations on foreign feedstock and biofuel imports would force domestic production to ramp up in order to balance the market [link].

Renewable Volume Obligations (RVO)

Billions of Gallons per Year	GHG Reduction	2021 Final	2022 Final	2023 Final	2024 Final	2025 Final	2026 Proposed	2027 Proposed
Cellulosic Biofuel (D3)	60%	0.56	0.63	0.84	1.09	1.38	1.30	1.36
Biomass-based Diesel (D4)	50%	2.43	2.76	2.82	3.04	3.35	5.61	5.86
Advanced Biofuels/Biomass (D5)	50%	5.05	5.63	5.94	6.54	7.33	9.02	9.46
Renewable Fuel (D6)	20%	18.84	20.63	20.94	21.54	22.33	24.02	24.46
Advanced		5.05	5.63	5.94	6.54	7.33	9.02	9.46
Conventional		13.79	15.00	15.00	15.00	15.00	15.00	15.00
Changes		1.71	1.79	0.31	0.60	0.79	1.70	0.44
Supplemental Standard (2016 Remand) ^			0.25	0.25	na	na	na	na
			0.96	1.52	4.91	3.99	3.65	4.09

Renewable Volume Obligations (RVO)	2025 Final	2026 Proposed	2027 Proposed
Cellulosic Biofuel (D3)	**1.38**	1.30	1.36
Biomass-based Diesel (D4)	**3.35**	5.61	5.86
Advanced Biofuels/Biomass (D5)	**7.33**	9.02	9.46
Renewable Fuel (D6)	**22.33**	24.02	24.46



Source: PSC Research, EPA

US Biofuels - Tracking RINs Production and "Margins"

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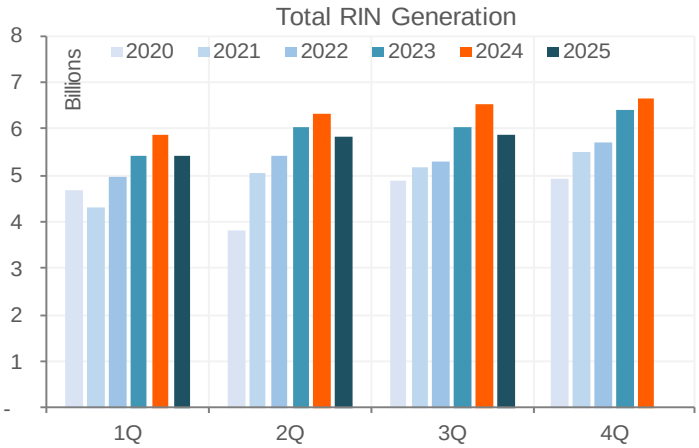
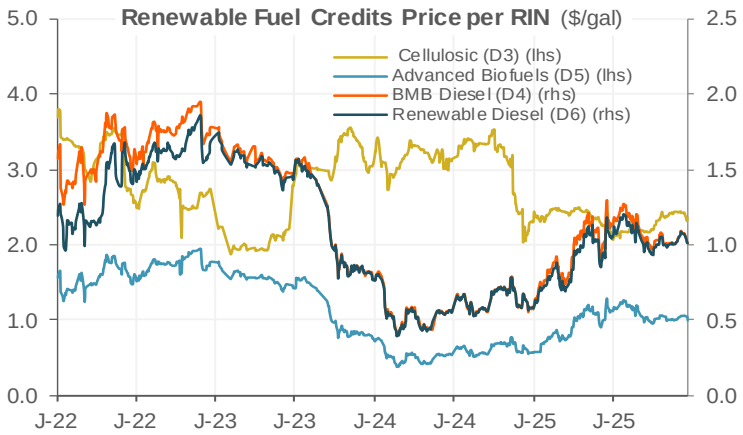
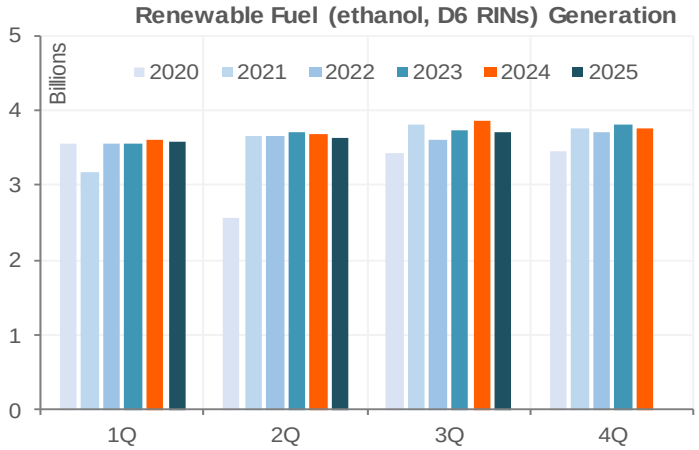
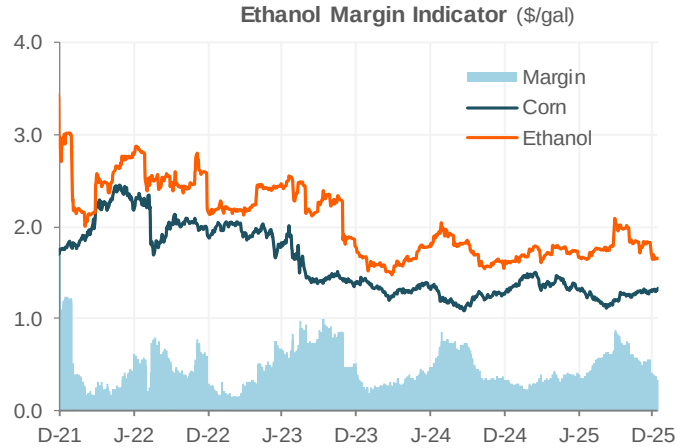
- Total RIN generation reached just over 17.1 billion units YTD, which is 9% below last year's total of 18.7 billion units through the same time period. However, given the EPA's recent RVO proposal, we expect this number to continue climbing. Look for credit prices to follow suit as well.



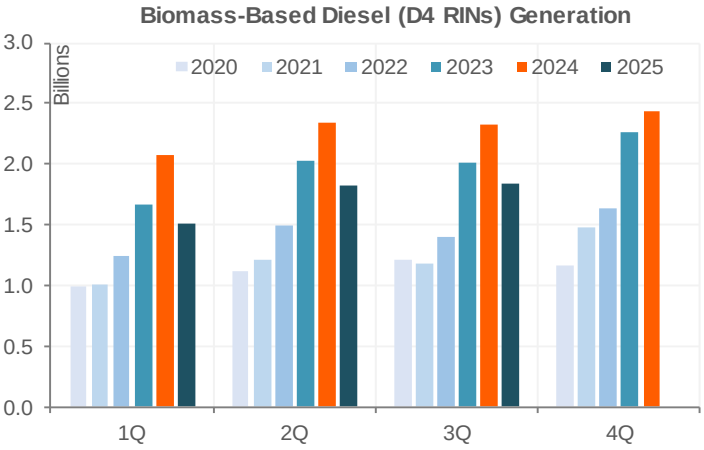


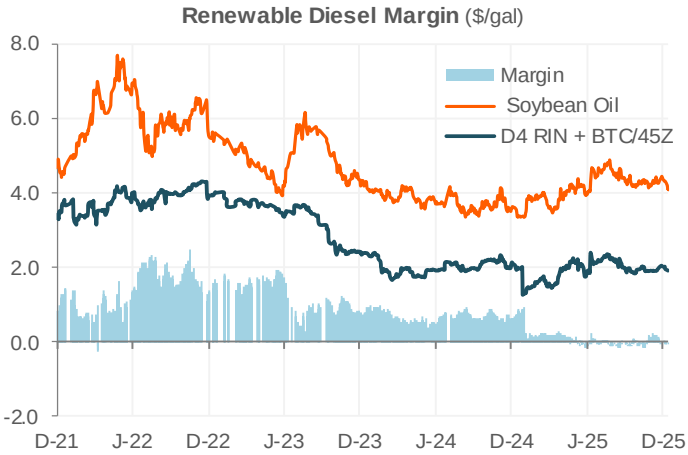
- D6 RIN Generation currently sits at 10.9 billion units YTD, which is 2% below last year's 11.1 billion.
- As it pertains to profitability, a divergence between ethanol and corn prices seems to be taking place and allowing the implied margin to widen.





- D4 RIN Generation continues to struggle to match last year and is currently at 5.2 billion units YTD (vs last year's 6.7 billion, a 30% decline).
- As it pertains to profitability, the renewable diesel margin depends on the price of diesel relative to soybean oil plus the D4 RIN price and gov't subsidies (prev. BTC which awarded $1/gal, now 45Z which is more stringent).





Source: PSC Research, Bloomberg, EIA, EPA

GLOBAL OIL MARKETS – ENTER AUTUMN STILL GOING STRONG

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A Year-End Tug-of-War over Oil Prices

Geo-Pol Supply-Risk versus Softening S/D Fundamentals

Prices fell through near-term support and set a new low for the year Tuesday. Technically, says our PSC wizard, Craig Johnson, WTI would now target $50/b after settling near $55/b yesterday. Brent broke below $60/b and is so far failing to recover and climb over that psychological hurdle today.

We saw this week what happens if geo-political supply risk fades and bearish fundamental factors take over. And for the record, our winter balances are bearish too; it's only if we look at crude oil balances in the ex-China pricing centers through Q1-2026 that we are less bearish/more bullish. And as such bearish signals are flashing everywhere, physical markets continue to soften, refining margins are deflating (some), the US is normalizing and stocks in key places here are rising.

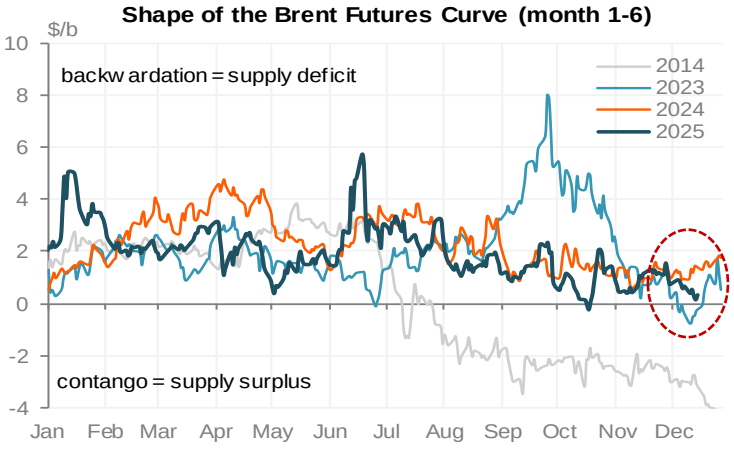
Oil markets still do not trade fundamentals only

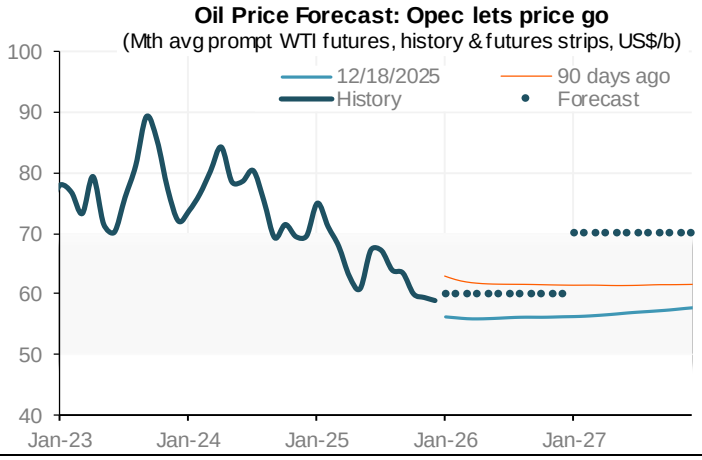
There remains macro risk, geo-pol risk and positioning risk we'd agree that in an era of peace and stability, today's supply surplus would lead prices down, with Q1 trading $50/b WTI, cp.

E.g., today prices bounced as a deal w/ Russia looks more removed – a deal on Ukraine is still not close enough for Putin and Trump to meet. The US is in the maritime blockade game again for the first time since the 1960s. And positioning is way short.

Inside, more on futures structure and Mideast crude export volatility

Click *here* for our most recent *Flowing Oil Data Digest* (Our weekly publication about all things global oil)





We expect to see ***a moderately less-bearish oil price outcome*** through next year – we raise our forecast by $5/b to $60/b WTI through 2026 (+$3/b for Brent)

- **Opec underscores that big supply slugs are coming,** but markets signal that for now at least, **balances remain tight**
- **Demand data have been ok,** even if growth hasn't been lights out anywhere, refining margins are buoyant
- **We still expect *relative* weakness through H1 2026** and probably less depressingly large inv. surpluses
- Two-way risk, before +ves (rates) deliver a growth cycle

$/bl		2019	2020	2021	2022	2023	2024	Q1-'25	Q2-'25	Q3-'25	Q4-'25	2025	Q1-'26	Q2-'26	2026	2027	LT
WTI	**PSC**	**$57**	**$39**	**$68**	**$94**	**$79**	**$76**	**$71**	**$64**	$65	$60	**$65**	$60	$60	**$60**	**$70**	**$70**
	latest/ old fcst									*$60*	$55	$62	$55	$55	$58	$70	$70
	Consensus										$60	$65	$57	$59	$59	$61	$67
	Strip										$60	$65	$56	$56	$56	$57	$59
Brent	**PSC**	**$64**	**$43**	**$71**	**$99**	**$84**	**$80**	**$75**	**$67**	$68	$63	**$68**	$63	$63	**$63**	**$73**	**$73**
	latest/ old fcst									*$63*	$58	$65	$58	$58	$61	$73	$73
	Consensus										$63	$68	$60	$60	$62	$64	$70
	Strip										$67	$67	$61	$59	$60	$60	$63
memo: Brent y/y % chg		-11%	-33%	64%	40%	1%	-5%	-8%	-22%	-13%	-15%	-15%	-16%	-6%	-8%	16%	

Source: PSC Research, Opec, IEA, Bloomberg

OIL MARKET WATCH & LATEST US WEEKLY DATA

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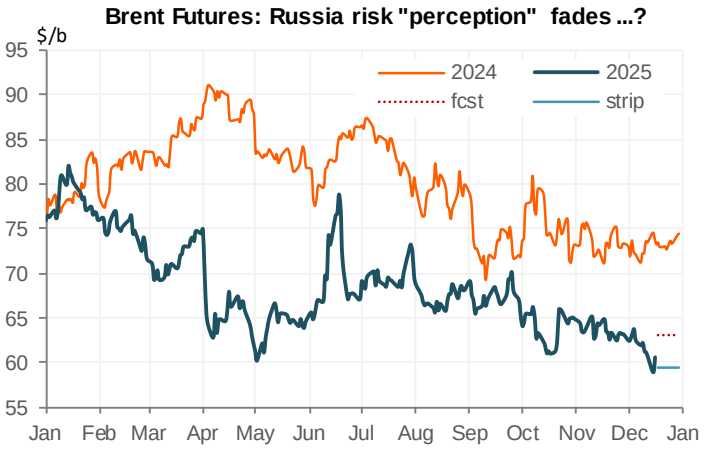
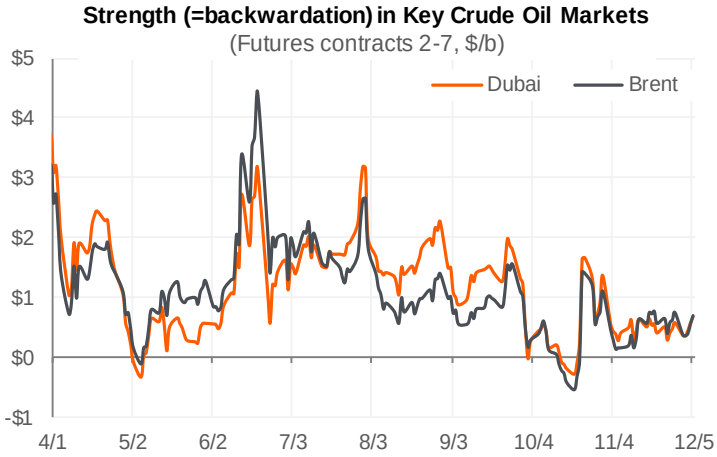
Prices and structure reflated a touch, if only tentatively; refining margins remain buoyant …

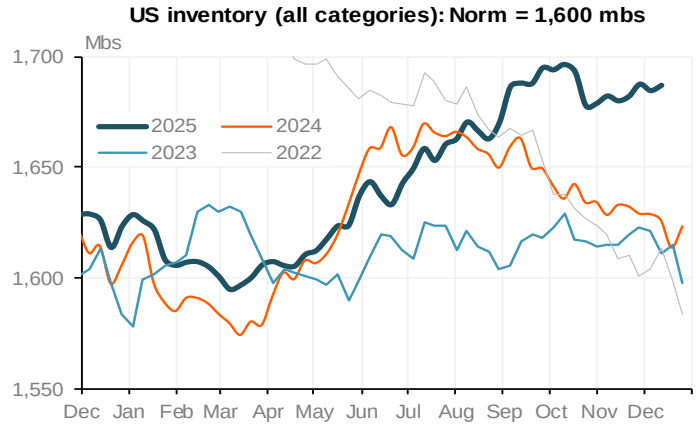


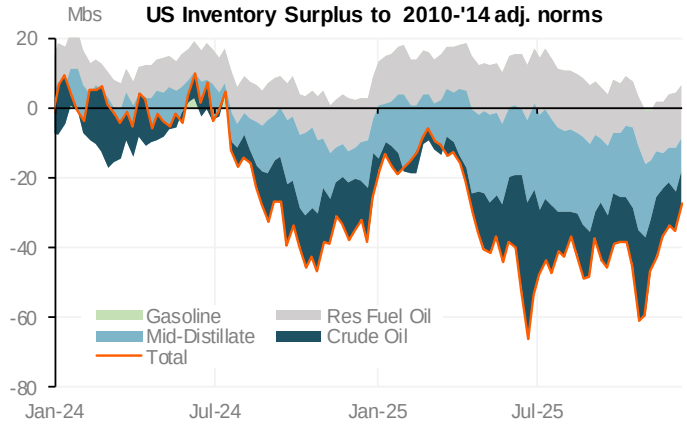


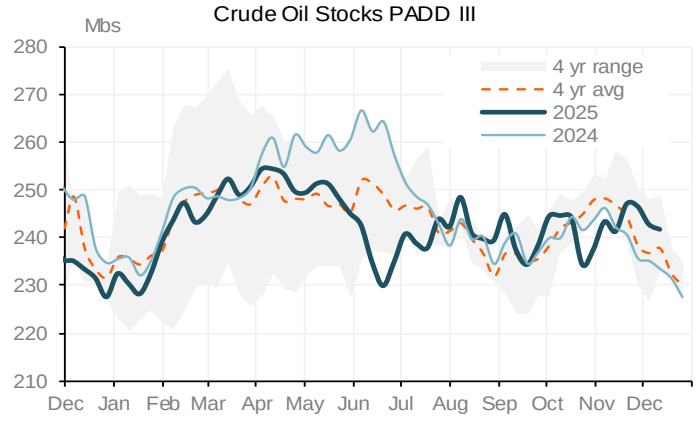
Weekly DOEs: Feel iffy

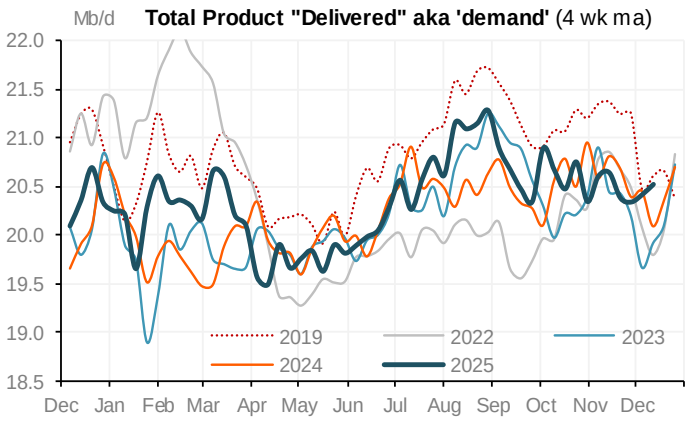
Key data this week: Smaller than normal inventory builds is the one reason we would not label this weekly batch of data outright *negative. Sadly,* we don't like the mini-trends and nuances behind the headline numbers. For instance, crude oil stocks barely drew despite a very large wk/wk increase in exports. Fact is that for a few weeks now, the crude oil trade has been sub-par as if to indicate that the world has plenty enough crude and needs less US exports… Then there's the rapid build up of gasoline inventory – as if US refiners are running too hot. Even middle distillate stocks are building/normalizing. In all, by our measure of inventories, the US inventory deficit of crude oil and key-products has shrunk by some 30-mbs in the last 7-wks. *I.E.: US balances are less tight and inventories went from low to normal*.

Click here for our Weekly US Oil-Data Report









Source: PSC Research, EIA

GLOBAL OIL BALANCES: BACK TO BEING THE MOST BULLISH

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What differentiates our balances and forecasts from those of the EIA, IEA and OPEC

To compare all four balances, we use the line "call on Opec" and more specifically y/y changes in the call-on-Opec. Because the IEA and Opec Secretariat are not allowed to forecast Opec crude oil production their balances do not have the same bottom line of "implied stock changes" for 2025.

We build our own balances, but that's the exception. For many on Wall Street the agency balances are good enough. From the IEA we take only historical demand supply & inventory data for OECD countries. They're the authority on energy statistics for OECD members (aka the DM economies).

Macro uncertainty & Opec abandoning price-management: A new regime is coming

We'll see how things change as the many macro issues & geopolitics & Opec decisions unfold

- We are outliers with massive Opec growth in 2025 with the group preparing to accelerate the pace of its 2.2 mb/d unwind
- Our non-Opec supply growth is the lowest of anyone
- For now, of the four here assembled, the EIA is the most bearish for 2025. The IEA also models -0.9 mb/d growth for their call on Opec while we model -0.5 mb/d
- On the demand side, our modelling of a demand growth recovery in H2 of next year renders us the most positive of the group for 2026 while the IEA are bearish, and the EIA are neutral…

PSC										
Supply						y/y	y/y	y/y	y/y	y/y
Mb/d	**2022**	**2023**	**2024**	**2025**	**2026**	**2022**	**2023**	**2024**	**2025**	**2026**
Global	**100.2**	**102.3**	**103.3**	**105.8**	**108.0**	**3.9**	**2.1**	**1.0**	**2.5**	**2.2**
Non-OPEC *	66.8	68.7	69.5	70.9	71.5	1.8	1.9	0.8	1.4	0.6
US	19.4	20.8	21.6	22.3	22.8	1.1	1.4	0.8	0.6	0.5
Crude oil	12.0	12.9	13.2	13.5	13.5	0.7	0.9	0.3	0.3	0.0
Nopexus	45.0	45.5	45.5	46.3	46.3	0.6	0.5	0.0	0.8	0.0
Opec	33.5	33.6	33.8	34.9	35.9	2.1	0.2	0.2	1.0	1.0
Opec Crude	28.5	28.4	28.3	29.2	30.2	2.1	-0.1	-0.1	0.9	0.9
Opec non-crude oil	5.0	5.3	5.5	5.6	5.7	0.0	0.2	0.3	0.1	0.1
Implied Stk Change	**0.0**	**-0.5**	**-0.5**	**0.9**	**1.6**					
Call on Opec crude oil	28.5	28.9	28.8	28.3	29.1	0.5	0.4	-0.2	-0.5	0.8
Demand										
Global	**100.3**	**102.8**	**103.8**	**104.8**	**106.4**	**2.3**	**2.6**	**1.0**	**1.0**	**1.5**
OECD	45.6	45.6	45.7	45.8	46.5	1.1	0.0	0.1	0.1	0.7
of which US	20.0	20.3	20.5	20.7	20.9	0.4	0.3	0.2	0.2	0.2
Non-OECD	54.6	57.2	58.1	59.0	59.9	1.2	2.6	0.9	1.0	0.8
China	15.7	17.3	16.9	17.3	17.4	-0.3	1.5	-0.3	0.4	0.1
India	5.4	5.7	5.9	6.1	6.3	0.4	0.3	0.3	0.2	0.3
Other EM	33.5	34.3	35.2	35.6	36.1	1.1	0.8	0.9	0.4	0.5

EIA										
Supply						y/y	y/y	y/y	y/y	y/y
Mb/d	**2022**	**2023**	**2024**	**2025**	**2026**	**2022**	**2023**	**2024**	**2025**	**2026**
Global	**100.4**	**102.5**	**103.2**	**106.2**	**107.4**	**4.8**	**2.1**	**0.7**	**3.0**	**1.3**
Non-OPEC *	67.3	69.8	70.4	72.5	73.7	2.0	2.5	0.7	2.1	1.1
US *	20.0	20.3	20.5	20.6	20.6	0.1	0.3	0.2	0.1	0.0
Crude oil	12.0	12.9	13.2	13.6	13.5	0.7	0.9	0.3	0.4	-0.1
Nopexus*	47.3	49.5	50.0	51.9	53.1	1.9	2.2	0.5	2.0	1.2
Opec	33.1	32.7	32.7	33.6	33.8	0.4	-0.4	0.0	0.9	0.1
Opec Crude	27.7	27.2	27.1	27.9	27.9	2.6	-0.6	-0.1	0.8	0.0
Opec non-crude oil	5.4	5.5	5.6	5.7	5.9	-2.2	0.1	0.1	0.1	0.1
Implied Stk Change	**0.8**	**0.7**	**0.4**	**2.2**	**2.3**					
Call on Opec crude oil	26.9	26.5	26.7	25.7	25.6	2.7	-0.4	0.2	-1.0	0.0
Demand										
Global	**99.6**	**101.8**	**102.8**	**103.9**	**105.2**	**2.6**	**2.2**	**1.0**	**1.1**	**1.2**
OECD	45.7	45.7	45.9	45.8	46.0	0.9	0.0	0.1	0.0	0.2
of which US	20.0	20.3	20.5	20.6	20.6	0.1	0.3	0.2	0.1	0.0
Non-OECD	53.9	56.1	56.9	58.1	59.2	1.7	2.2	0.9	1.2	1.1
China	15.1	16.2	16.3	16.6	16.9	0.0	1.1	0.1	0.3	0.3
India	5.2	5.4	5.6	5.6	5.8	0.4	0.2	0.2	0.1	0.2
Other EM	33.6	34.5	35.1	35.9	36.5	1.4	0.9	0.6	0.8	0.6

IEA										
Supply						y/y	y/y	y/y	y/y	y/y
Mb/d	**2022**	**2023**	**2024**	**2025**	**2026**	**2022**	**2023**	**2024**	**2025**	**2026**
Global	**100.0**	**102.3**	**103.1**	**106.0**		**4.3**	**2.3**	**0.9**	**2.9**	
Non-OPEC *	66.9	69.3	70.4	72.0	73.3	1.9	2.4	1.0	1.7	1.2
US	19.2	20.8	21.8	22.5	22.9	12.7	1.6	0.9	0.7	0.3
Crude oil	12.0	12.9	13.2	13.6	13.6	11.9	0.9	0.3	0.3	0.0
Nopexus	47.6	48.5	48.6	49.5	50.4	-10.8	0.8	0.1	0.9	0.9
Opec	34.3	34.1	32.8	34.0		2.4	-0.2	-1.3	1.3	
Opec crude oil	28.8	28.5	27.2	28.4		2.3	-0.3	-1.3	1.2	
Opec non-crude oil	5.5	5.5	5.5	5.7	5.9	0.1	0.1	0.0	0.1	0.2
Implied Stk Change	**-0.3**	**0.1**	**0.0**	**2.1**						
Call on Opec crude oil	27.9	27.3	27.2	26.3	25.6	-2.8	-0.6	-0.1	-0.9	-0.6
Demand										
Global	**100.2**	**102.1**	**103.1**	**103.9**	**104.8**	**-0.8**	**1.9**	**1.0**	**0.8**	**0.9**
OECD	45.7	45.7	45.8	45.8	45.7	1.2	0.0	0.1	0.0	-0.1
of which US	20.2	20.4	20.6	20.7	20.8	0.4	0.2	0.2	0.2	0.0
Non-OECD	54.5	56.4	57.3	58.1	59.0	-1.9	1.9	0.8	0.8	0.9
China	15.2	16.5	16.6	16.7	16.9	-1.3	1.3	0.1	0.1	0.2
India	5.3	5.4	5.7	5.8	5.9	-0.1	0.1	0.2	0.1	0.2
Other EM	34.0	34.5	35.0	35.6	36.2	-0.5	0.5	0.5	0.6	0.6

OPEC										
Supply						y/y	y/y	y/y	y/y	y/y
Mb/d	**2022**	**2023**	**2024**	**2025**	**2026**	**2022**	**2023**	**2024**	**2025**	**2026**
Global	**100.0**	**101.9**	**102.5**			**4.5**	**1.9**	**0.6**		
Non-OPEC *	67.0	69.5	70.4	71.7		1.9	2.5	1.0	1.3	
US	19.4	21.0	21.8	22.1		1.3	1.6	0.8	0.3	
Crude oil	11.9	12.9	13.2	13.3		0.6	1.0	0.3	0.1	
Nopexus	47.6	48.5	48.7	49.7		0.6	0.9	0.2	1.0	
Opec	33.1	32.5	32.1			2.6	-0.6	-0.4		
Opec crude oil	27.7	27.1	26.6			2.5	-0.6	-0.5		
Opec non-crude oil	5.4	5.4	5.5	5.6		0.1	0.0	0.1	0.1	
Implied Stk Change	**0.2**	**-0.4**	**-1.3**							
Call on Opec crude oil	27.5	27.5	27.9	27.8		0.7	0.0	0.4	-0.1	
Demand										
Global	**99.9**	**102.4**	**103.8**	**105.1**		**2.7**	**2.5**	**1.5**	**1.3**	
OECD	45.7	45.7	45.8	46.0		1.3	0.0	0.1	0.1	
of which US	20.2	20.4	20.6	20.8		0.4	0.2	0.2	0.2	
Non-OECD	54.2	56.6	58.0	59.2		1.4	2.5	1.3	1.2	
China	15.0	16.4	16.7	16.9		-0.4	1.3	0.3	0.2	
India	5.1	5.3	5.6	5.7		0.4	0.2	0.2	0.1	
Other EM	34.0	34.9	35.8	36.7		1.5	1.0	0.8	0.9	

Note: *includes processing gains

Source: PSC Research, Opec, IEA, EIA, JODI

INDICATORS: US OIL & GAS ACTIVITY, RIGS AND FRACK-SPREADS

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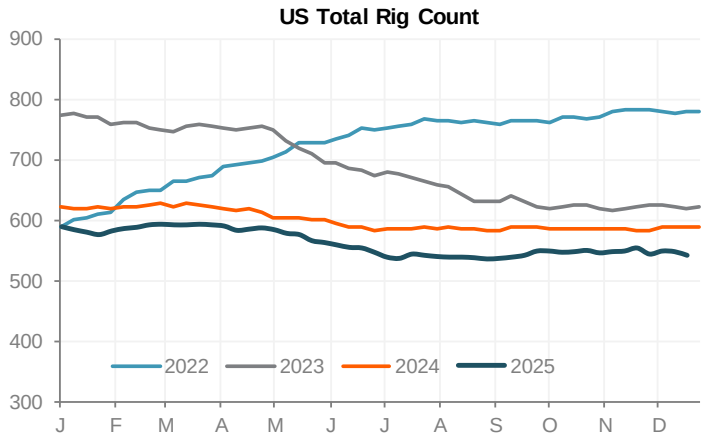
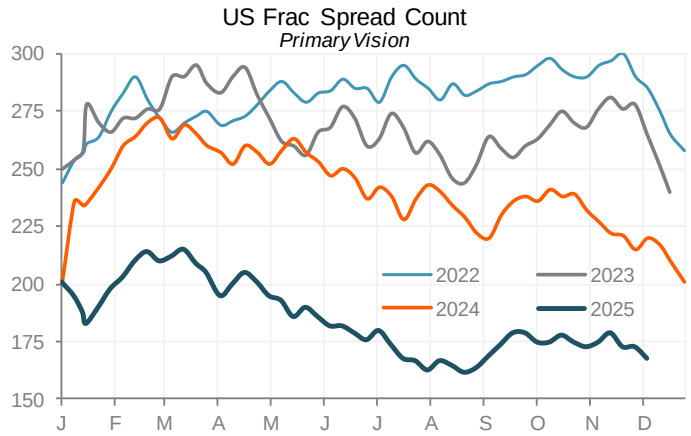
US Rig Count falls to 542

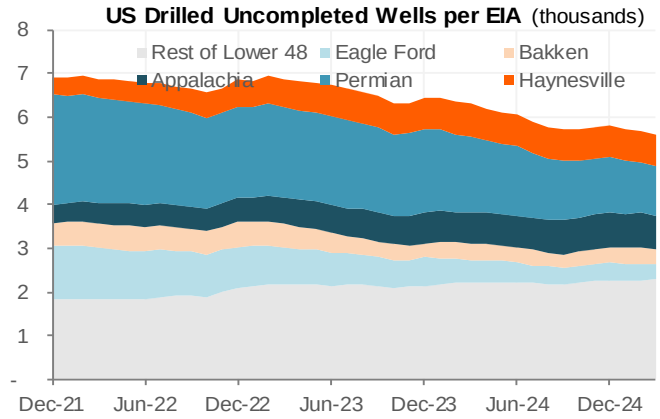
The U.S. Rig Count declined by 6 rigs last week, dropping the count to 542 rigs. Drilling rigs targeting crude oil shed 8 rigs wk/wk to total 406 rigs. The oil rig count is down 77 rigs from a year ago. *This is not bullish…*

Drilling rigs targeting natural gas were unchanged wk/wk with the count remaining at 127 rigs; up 25 rigs y/y. The frac spread remained weak at 160.



North America (Weekly)	Last Update	Last	Prior	Chg
United States	12/19/2025	542	548	(6)
Canada	12/19/2025	185	192	(7)
North American		727	740	(13)
Global (Monthly)				
United States	11/30/2025	549	548	1
Latin America	11/30/2025	134	129	5
Europe	11/30/2025	120	124	(4)
Africa	11/30/2025	101	97	4
Middle East	11/30/2025	513	505	8
Far East	11/30/2025	205	204	1
Total International		1,073	1,059	14
Total World		1,812	1,800	12
Saudi Arabia Rig	11/30/2025	103	103	-
OPEC Countries	3/31/2024	386	383	3
Non-OPEC Countries	3/31/2024	585	575	10
United States (Weekly)				
Total Oil	12/19/2025	406	414	(8)
Total Natural Gas	12/19/2025	127	127	-
Indland	12/19/2025	3	3	-
Offshore	12/19/2025	15	17	(2)
Land	12/19/2025	524	528	(4)
Shale (Weekly)				
Permian	12/19/2025	246	249	(3)
Eagle Ford	12/19/2025	39	39	-
Williston (Bakken)	12/19/2025	29	29	-
Marcellus	12/19/2025	24	24	-
Utica Basin	12/19/2025	15	15	-
Haynesville	12/19/2025	42	45	(3)
Cana Wood	12/19/2025	17	17	-
Granite	12/19/2025	15	14	1
DJ-Niobrara	12/19/2025	7	8	(1)
Total		434	440	(6)





Source: PSC Research, EIA, Primary Vision, Baker Hughes

INDICATORS: AIRLINE TRAFFIC

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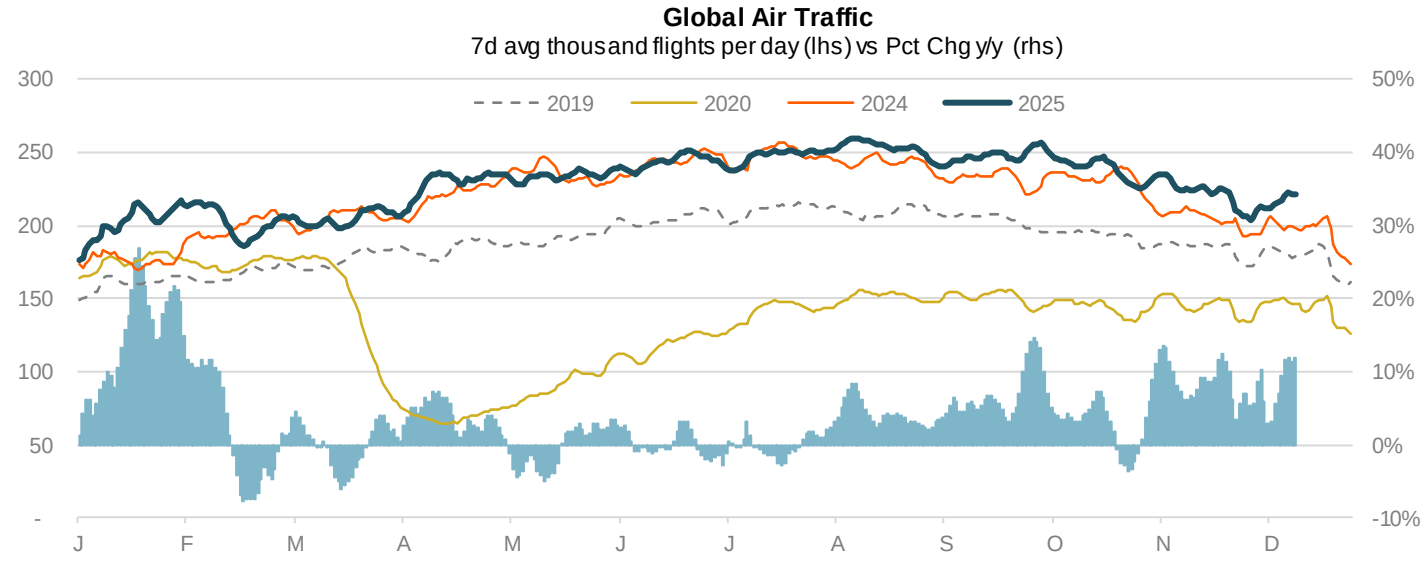
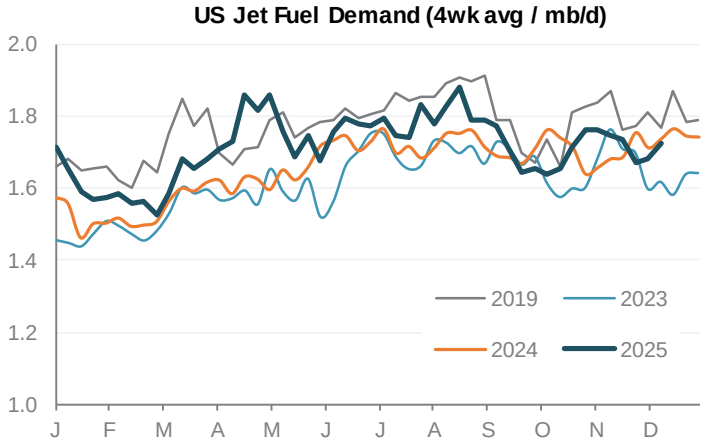
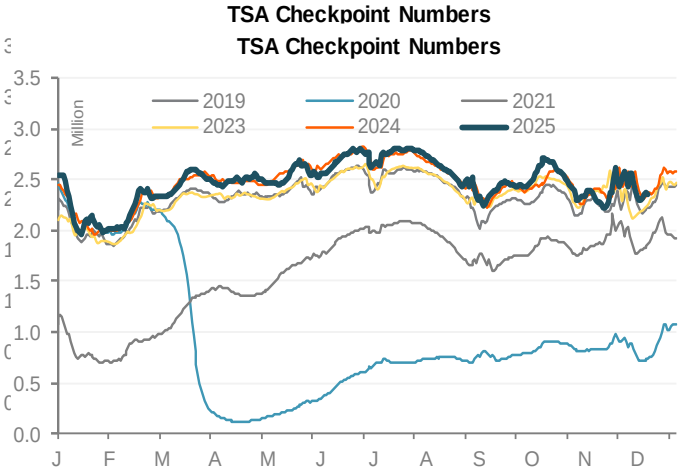
Global Air Traffic: Year-end bounce

- Global air traffic has increased counter-seasonally heading into year-end
- Post-Thanksgiving traffic in the US is tracking at or just below last year's levels
- In China, domestic capacity still accounts for 82% of all seats, with international making up 18%. December domestic capacity has increased by +7% y/y while international capacity rose +8% y/y *per OAG*
- China's international traffic will most likely be negatively impacted by a recent dispute between China and Japan
    - >40% of the scheduled flights from China to Japan have been cut in December
    - This will negatively impact both countries as Japan is the number one international tourist destination for travelers from China
- China's Ministry of Culture and Tourism recently announced plans to boost inbound international travelers by continuing to ease entry into the country and making domestic tourist travel more accessible. The plan calls for an increase in air routes and direct flights (mainly from countries partaking in the Belt and Road initiative)







Source: PSC Research, Bloomberg, FlightTracker24, TSA. OAG, Civil Aviation Administration of China

INDICATORS: RAIL FREIGHT

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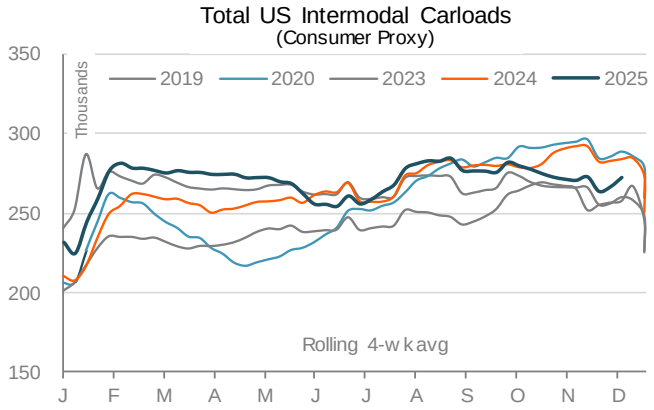
Rail traffic slowing into year-end

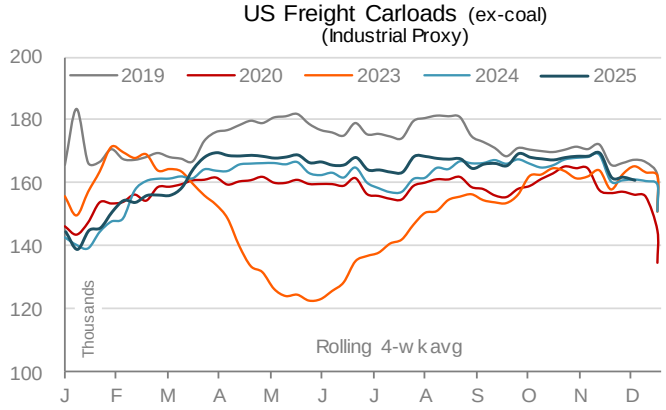
Total rail traffic dropped by -1.6% (y/y, 4wk avg), as intermodal, petroleum and motor vehicle carload weakness weighed on traffic

- Coal carloads picked up (+5.7% y/y, 4wk avg basis) after posting weekly declines since the end of August. This strength looks to continue as natural gas prices remain around $4/mmbtu
- Intermodal volumes continue to weaken which jives with the drop in container imports we've seen the past few months
- Grain, Crushed Stone, Coal and Metallic Ores were the only four carload categories to show an increase vs last year's 4wk average.

Description	2024 4-Wk Avg	11/21/25	11/28/25	12/5/25	12/12/25	W/W Chg	2025 4-Wk Avg	v 2024
Railroad Farm Freight	941	822	583	881	944	7.2%	808	-14.2%
Lumber Freight Carloads	2,615	2,799	2,106	2,591	2,583	-0.3%	2,520	-3.6%
Coke Freight Carloads	2,706	2,642	2,825	2,580	2,655	2.9%	2,676	-1.1%
Pulp & Paper Freight	4,414	3,977	3,837	4,253	4,029	-5.3%	4,024	-8.8%
Metals Freight Carloads	7,201	6,992	7,071	7,401	7,461	0.8%	7,231	0.4%
Intermodal Trailer Freight	10,658	8,756	7,291	10,508	10,265	-2.3%	9,205	-13.6%
Forest Products	8,194	7,759	6,848	8,017	7,553	-5.8%	7,544	-7.9%
Petroleum Freight Carloads	11,110	10,587	9,576	11,591	10,826	-6.6%	10,645	-4.2%
Motor Vehicles Freight	14,684	15,734	11,458	15,115	15,839	4.8%	14,537	-1.0%
Crushed Stone Freight	17,294	20,845	13,919	18,652	17,801	-4.6%	17,804	3.0%
Grain US Freight Carloads	22,412	25,893	21,019	25,098	22,944	-8.6%	23,739	5.9%
Metallic Ores and Metals	18,881	20,737	16,312	19,706	19,269	-2.2%	19,006	0.7%
Chemicals Freight Carloads	32,632	32,699	29,583	32,548	32,013	-1.6%	31,711	-2.8%
Coal Freight Carloads	57,396	62,956	56,972	61,026	61,733	1.2%	60,672	5.7%
US Freight Carloads	218,278	234,592	197,955	228,823	224,620	-1.8%	221,498	1.5%
US Freight Carloads (ex-Coal)	160,883	171,636	140,983	167,797	162,887	-2.9%	160,826	0.0%
Major Category US Freight Carloads**	163,297	178,864	149,263	172,145	169,599	-1.5%	167,468	2.6%
Intermodal Freight	283,572	281,518	233,480	280,176	294,284	5.0%	272,365	-4.0%
Intermodal Containers	272,914	272,762	226,189	269,668	284,019	5.3%	263,160	-3.6%
Total Traffic	501,850	516,110	431,435	508,999	518,904	1.9%	493,862	-1.6%

**Coal, Chemicals, Metallic Ores and Metals, Crushed Stone, Motor Vehicle





Source: PSC Research, Association of American Railroads, FTR

INDICATORS: CONTAINER INDICES & SUEZ CANAL TRANSIT

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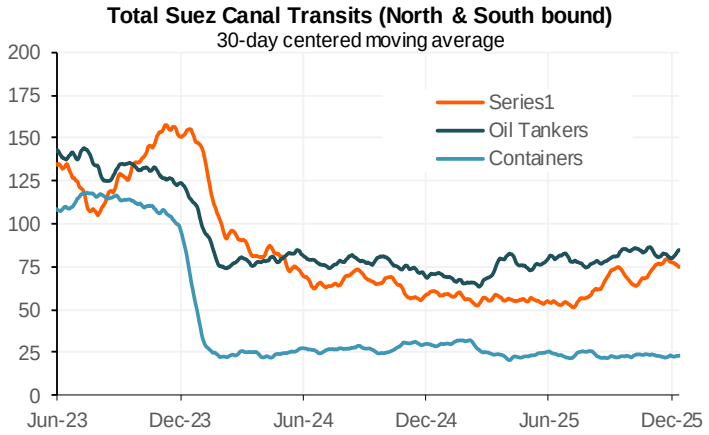
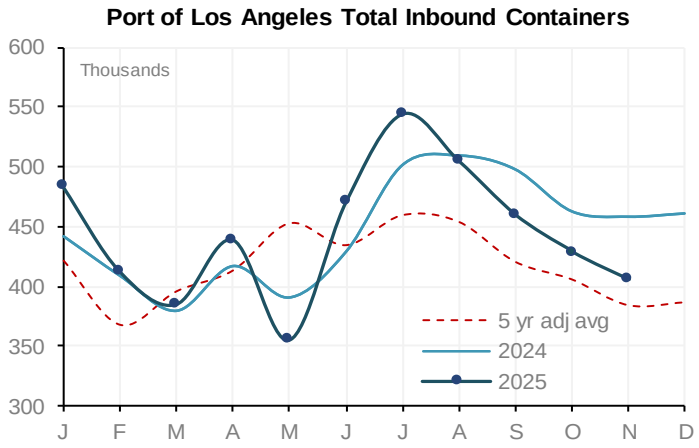
Containers: Return to the Red Sea

- The Israel-Hamas ceasefire has precariously held for almost two-months now so it's no surprise there are rumblings of a return to Red Sea/Suez Canal transits and away from the longer Cape of Good Hope route
- The rebound in Red Sea transits will take time as companies remain hesitant to return to the region
- *Bloomberg* reported that Maersk is 'closely monitoring' developments but currently has no specific timeline as to when they will return to sailing through the Red Sea though the company said conditions in the region appear to be improving

- ZIM's CEO stated that a return is increasingly likely in the near-term while CMA CGM is reportedly gearing up for a full return this month
- A shift back to the shorter Red Sea/Suez Canal route vs the Cape of Good Hope would rapidly increase excess capacity with the potential for significant port congestion mainly at European ports
- Potential port congestion should absorb some of this additional capacity but given the lack of scrapping and the number of newbuilds continuing to hit the water, will likely lead to an over-supplied market and put downward pressure of freight rates
- In November, US Total Container Throughput declined-8% m/m and was down 11.5% y/y. YTD total containers throughput at the port of LA is now just +1% over last year's total through November.
- According to *Descartes*, the Port of LA saw the largest m/m decline in container imports, dropping 9.6%.





Source: PSC Research, Bloomberg, Freightos, WCI, Descartes,

US Coal Markets: Fundamentals & Fuel Parity

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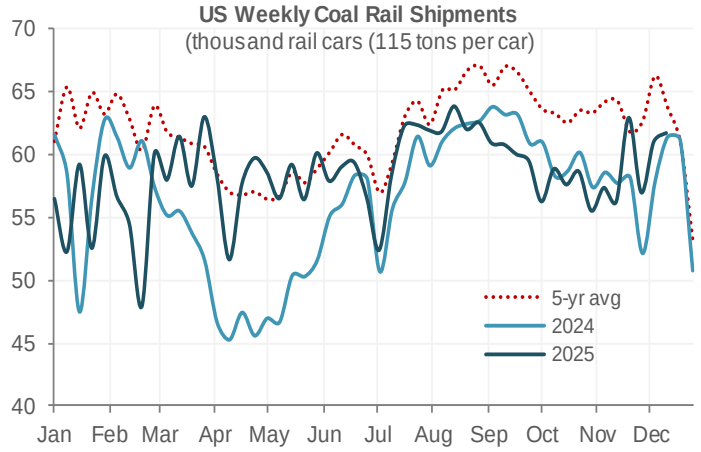
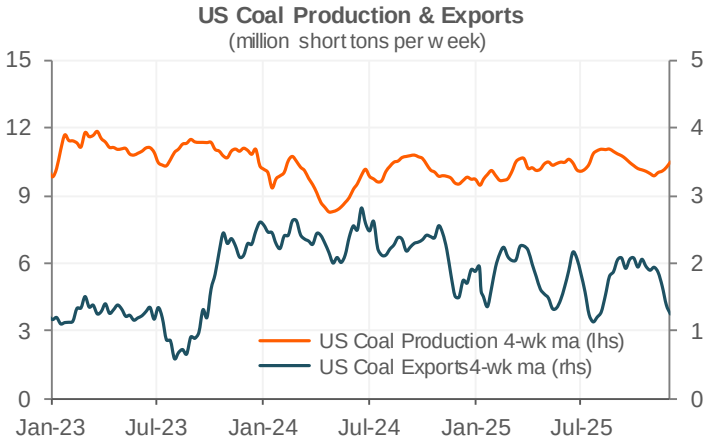
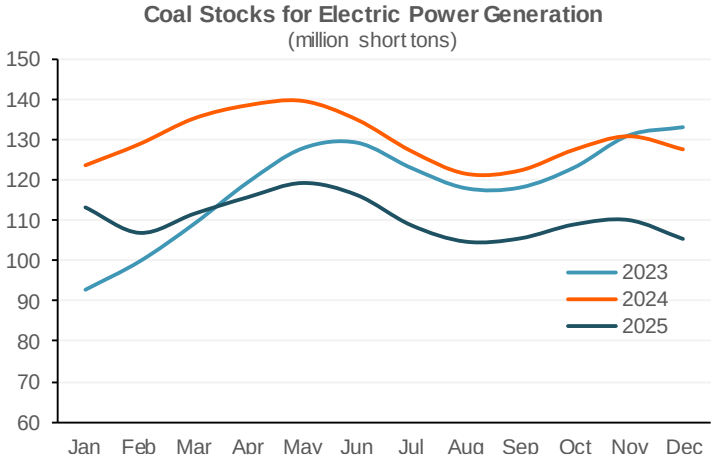
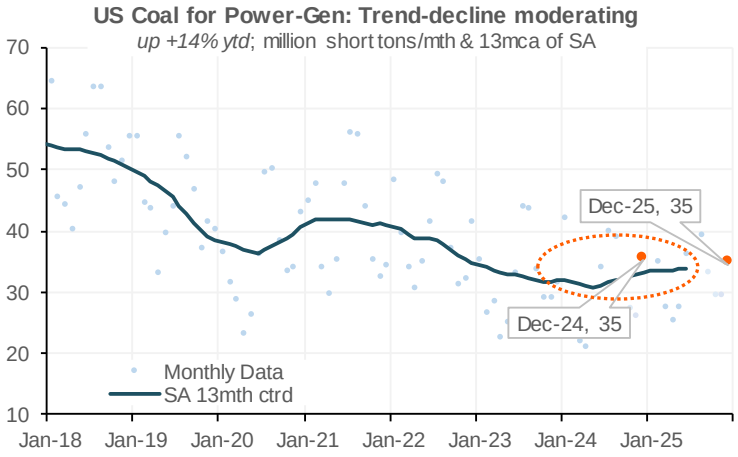
US coal production keeps on trending sideways, whilst demand is slowly grinding higher. To be fair, exports have fallen back toward 1 mst/wk (top right); but that seams seasonal and that relatively low production appears to have torqued inventories down again in December … **constructive**



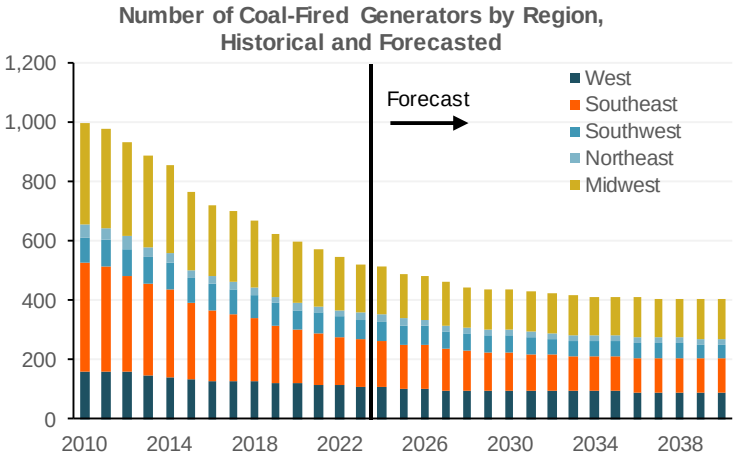


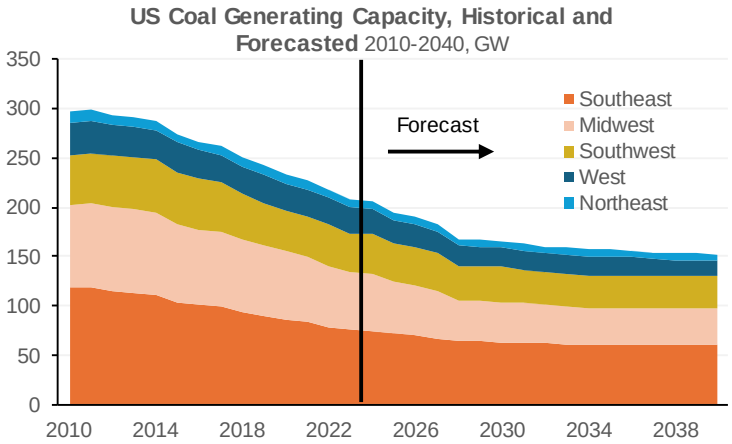
Production trending sideways and exports slipping (top right). Rail traffic is slipping more seasonally (top left) and/or reflects that down-trend in exports





Monthly coal-fired power-gen data, into November, are noisy, but there was a historic, age-old, downtrend, *which since early last year turned up and continues to grind higher …*





Source: PSC Research. Bloomberg Consensus Forecasts, EIA

GLOBAL COAL MARKETS: US COAL IN BROADER CONTEXT

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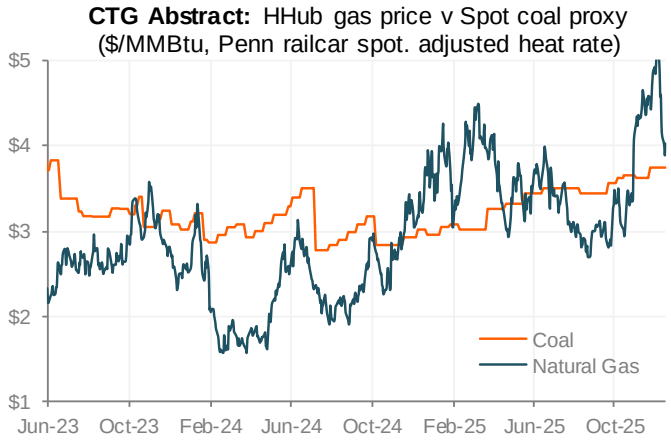
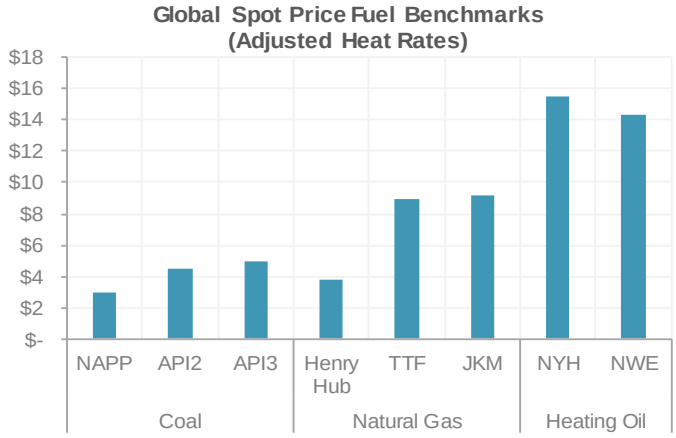
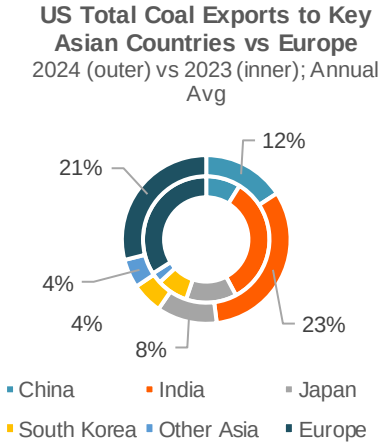
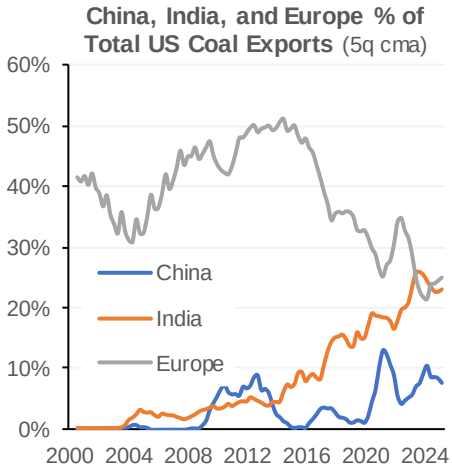
US coal prices have stepped up toward $4/MMBtu gas equivalent, just as front month Hhub gas futures have collapsed back down to those levels for us that story of the interplay between gas and coal carries right on over into next year. In fact, we pegged gas prices below coal for next summer (top left).

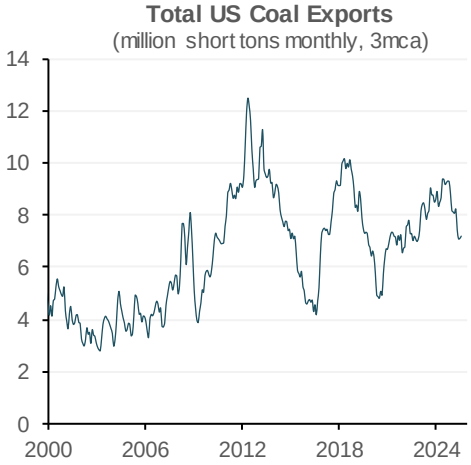




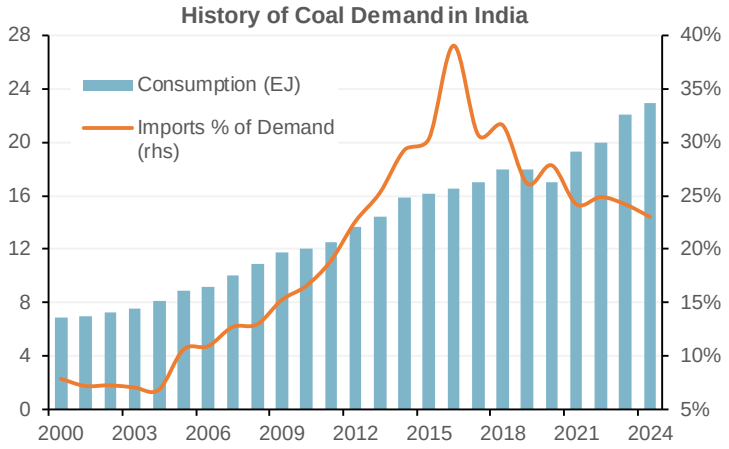
Background / Historic / big picture context: Coal exports hovered around 1/3rd the size of the shrunken domestic market in 2024, they're comprised of metallurgical and steam coal in roughly equal proportions.

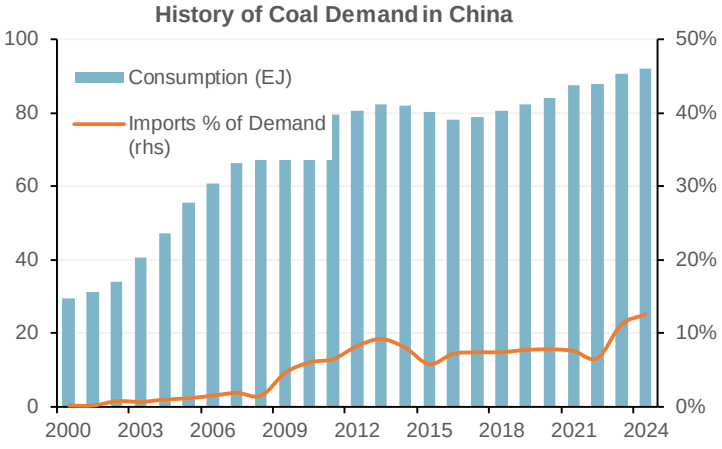






India and Europe were the top two destinations for US coal in 2024 (though China is/was an increasingly hefty customer (~12%). India's coal use is growing fast, unlike Europe's, and it's import dependent too.





Source: PSC Research. EIA, Ei Statistical Review of World Energy

EQUITY STATISTICS: REDEFINING 'ENERGY'

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The global portfolio of energy is changing and so too are investor allocations for investment strategies to meet the needs of the energy transition to a low carbon environment.

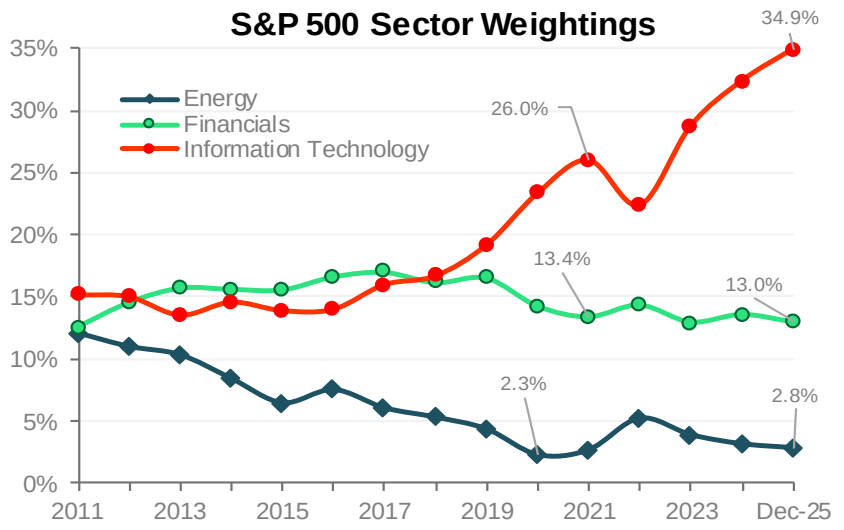
Europe and China thus far have been driven by policy.

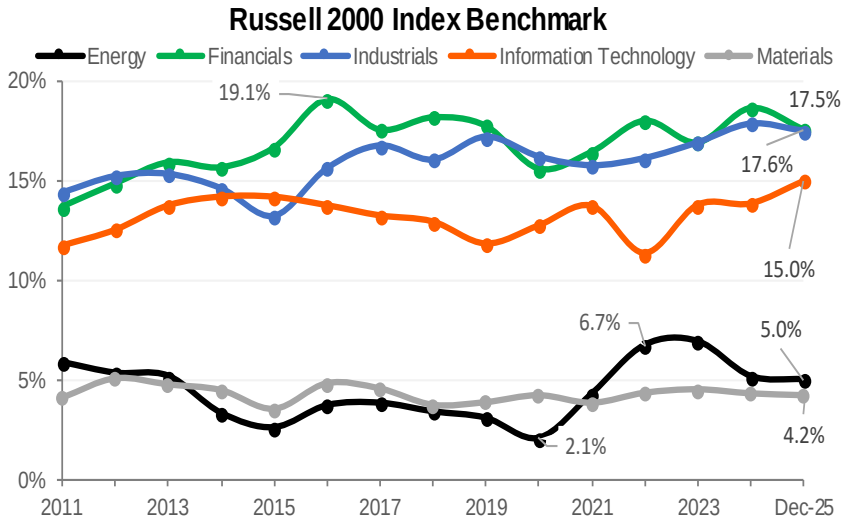
The US remains dominated by business economics.

For everyone, the economics keep on changing under the influence of evolving technology; new policies (at all levels, down from the federal); then there is corporate goal-setting and lastly the unending parade of regulatory changes.

We believe investors should consider a broader scope of more holistically defining Energy.

In fact, we built energy statistics around 14-subsectors. Across them we have over 600 equity tickers, which in aggregate have a gross US $ 13.9 Trillion of market cap.





Energy Equities Statistics: We built tables for 13 sub-sectors

We organized energy equities by sector and functional relationship to the energy transition and we've compiled statistics for 13-subsectors outlined on the next page.

The statistics themselves include: Market Capitalization, Fundamental Earnings Indicators, Technical Indicators, and Performance screens.

- Fundamental Earnings Indicators will be GREEN or RED if the Consensus forecasts have changed +/- 10% over the past 30 days
- Technical Indictors use the 50-day and 200-day moving averages.
    - If current share price is > 50 dma, the 50 dma column will be GREEN.
    - If the current share price is < 50 dma, the 50 dma column will be RED
    - If current share price is between the 200 dma and the 50 dma, the column will be GREEN.
    - If the current share price is < 200 dma, the 200 dma column will be RED

Source: PSC Research, Bloomberg Holdings

EQUITIES OF THE ENERGY SPECTRUM

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– WE KEEP EXTENSIVE STATISTICS FOR >600 TICKERS IN 14 SUB-SECTORS

Sector & Segment	# Tckrs	mkt cap $b
Solar		
Polysilicon	8	$ 28.2
Wafer	2	$ 0.1
Cells	5	$ 1.8
Inverters and Modular-Level Power Elec (MPLE)	5	$ 88.0
Modules	11	$ 55.8
Systems & Services	7	$ 204.0
Other Parts & Services	15	$ 20.5
Total	53	$ 398.3
Wind		
Wind Energy Systems	14	$ 213.7
Wind Equipment Manufacturers	16	$ 110.5
Project Development	4	$ 118.7
Total	34	$ 442.9
Storage		
Fuel Cells	14	$ 55.0
Battery Storage Manufacturers	32	$ 318.0
Total	46	$ 373.0
Transport		
Low Carbon Vehicles	10	$ 22.0
NEV Manufacturers	9	$ 1,754.6
Conventional Car Manufacturers	19	$ 926.6
Low Carbon Equipment	7	$ 125.2
Total	45	$ 2,828.5
Hydrogen		
Supply segment	8	$ 1.2
Diverse Mid-Stream	6	$ 4.5
Grid Oriented Mid-Stream	4	$ 20.0
H2 Consumption & transport	3	$ 2.5
Total	21	$ 28.1
Metals, Rare Earths & Coal		
Cobalt Refiners	3	$ 17.8
Lithium Producers	12	$ 198.6
Materials & Chemical Manufacturers	9	$ 398.2
Rare Earths & Strategic Metals	15	$ 143.5
Uranium	5	$ 53.3
Diversified Minerals & Mining	3	$ 258.7
Metals - Precious	8	$ 283.0
Metals - Aluminum	3	$ 16.9
Metals - Base, Steel & Iron	13	$ 400.3
Coal	7	$ 18.0
Total	78	$ 1,788.3
US Power		
US Independent Power Producers (IPP)	3	$ 92.4
US Regulated Utilities	10	$ 408.7
US Integrated Power	11	$ 471.4
US Renewable Power Generation	15	$ 187.9
Total	39	$ 1,160.5
Green Power ex US		
Power Gen - Diversified Renewables	31	$ 317.8
Power Gen - Solar	10	$ 10.5
Power Gen - Wind	7	$ 8.1
Total	48	$ 336.4

Sector & Segment	# Tckrs	mkt cap $b
Power Infrastructure Services		
Infrastructure Construction/Maintenance	17	$ 233.3
Total	17	$ 233.3
Transports		
North American Airlines	8	$ 121.0
North American Rails	5	$ 400.5
Oil & Product Tankers	14	$ 32.5
LNG & LPG Tankers	4	$ 6.8
Cruise Lines	3	$ 124.1
Shipping	15	$ 156.0
Total	49	$ 840.8
Midstream & MLPs		
Midstream Infrastructure	12	$ 551.9
MLP - Midstream - Liquids	5	$ 69.2
MLP - Midstream - Natural Gas	3	$ 138.0
MLP - Midstream - Processing	1	$ 16.0
MLP - Midstream - Propane	4	$ 4.3
MLP - Midstream - Coal	2	$ 4.3
MLP - Midstream - E&P	1	$ 1.1
MLP - Midstream - Shipping	3	$ 3.1
Other Midstream	4	$ 70.8
Total	35	$ 858.8
Integrated Oils & Refiners		
Global Integrated Oil Companies	18	$ 3,380.7
US Independent Refiners	5	$ 160.3
US Marketing & Retail	6	$ 79.8
US Ethanol	4	$ 3.9
Biofuels	14	$ 217.2
Total	47	$ 3,841.8
E&P and Oil Sands		
International Exploration	16	$ 193.4
Multi Basin Producer	2	$ 80.0
Permian Oil-Leveraged	7	$ 64.4
Bakken Oil-Leveraged	2	$ 7.5
Other Basins	6	$ 11.4
Natural Gas Producers	8	$ 114.3
Canadian Producers	8	$ 54.5
Total	49	$ 525.5
Oil Field Services and Equipment		
Integrated Oil Service Providers	3	$ 125.6
Oil Field Services & Equipment	14	$ 20.4
Proppants	1	$ 0.2
Equipment & Construction	23	$ 79.6
Land Rigs	4	$ 6.6
Offshore Rigs	3	$ 4.4
Offshore Support Vessels	3	$ 10.2
Seismic	3	$ 2.3
Total	54	$ 249.2
Total	**615**	**$ 13,905**

Source: PSC Research. Bloomberg Consensus Forecasts

ENERGY EQUITIES: SCREENS WITH GORY DETAIL

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					Fundamental Indicators			Technical Indicators		Performance		
Name	Ticker	Cur Price	Mkt cap ($B)	Cur Qtr EPS	Cur Yr EPS	Fwd Yr EPS	Growth	50 dma	200d dma	5 Day	QTD	YTD
COMTEC SOLAR SYSTEMS GROUP	712 HK	0.10	$ 0.0					0.13	0.10	-13%	-26%	4%
CSG HOLDING CO LTD - B	200012 CH	0.10	$ 1.5		0.10	0.18	72%	1.81	1.85	-3%	-5%	-22%
DAQO NEW ENERGY CORP-ADR	DQ US	31.97	$ 2.1	(0.26)	(2.73)	(0.32)	88%	30.54	22.06	-7%	14%	64%
GCL TECHNOLOGY HOLDINGS LTD	3800 HK	1.09	$ 4.7		(0.05)	0.04	170%	1.25	1.11	4%	-17%	1%
OCI HOLDINGS CO LTD	010060 KS	106,600	$ 1.4		(5,626)	14,838	364%	105,834	87,287	7%	21%	81%
REC SILICON ASA	RECSI NO	1.05	$ 0.0					1.24	1.72	-7%	-30%	-71%
TBEA CO LTD-A	600089 CH	21.62	$ 15.5		1.30	1.55	19%	21.79	15.07	-3%	21%	73%
WACKER CHEMIE AG	WCH GR	67.65	$ 3.0	1.64	(0.79)	0.54	169%	66.77	66.80	1%	6%	0%
Polysilicon			**$ 28.2**									
DANEN TECHNOLOGY CORP	3686 TT	18.00	$ 0.0					18.82	16.93	-3%	-5%	-10%
SOLARGIGA ENERGY HOLDINGS L	757 HK	0.08	$ 0.0					0.09	0.08	-2%	-23%	-9%
Wafer			**$ 0.1**									
EGING PHOTOVOLTAIC TECHNO-A	600537 CH	3.91	$ 0.7					4.12	3.52	-1%	-2%	27%
MOTECH INDUSTRIES INC	6244 TT	19.45	$ 0.2					16.96	18.43	11%	11%	-5%
TAINERGY TECH CO LTD	4934 TT	13.85	$ 0.1					13.87	12.10	2%	-19%	3%
UNITED RENEWABLE ENERGY CO L	3576 TT	7.56	$ 0.4					6.59	7.36	18%	10%	-25%
WEBSOL ENERGY SYSTEM LTD	WESL IN	82.94	$ 0.4					114.44	123.77	-10%	-33%	-50%
Cells			**$ 1.8**									
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
MAXEON SOLAR TECHNOLOGIE	MAXN	3.24	$ 0.1					3.32	3.46	-8%	-3%	-57%
RISEN ENERGY CO LTD-A	300118 CH	12.80	$ 2.1		(0.73)	0.38	152%	11.12	10.46	20%	22%	7%
S-ENERGY CO LTD	095910 KS	1,116	$ 0.0					1,522	1,601	-3%	-34%	-22%
SHUNFENG INTERNATIONAL CLEAN	1165 HK	0.02	$ 0.0					0.02	0.02		-4%	-8%
ZHEJIANG SUNFLOWER GREAT-A	300111 CH	6.83	$ 1.2					7.80	4.69	0%	-11%	111%
Modules			**$ 55.8**									
ABB LTD-REG	ABBN SW	57.90	$ 134.3	0.69	2.58	2.89	12%	57.80	51.69	-1%	1%	20%
CANADIAN SOLAR INC	CSIQ US	23.33	$ 1.6	(0.56)	(1.50)	0.69	146%	22.01	13.63	-7%	79%	110%
NEXTPOWER INC-CL A	NXT US	85.83	$ 12.7	0.91	4.28	4.60	8%	92.18	65.32	-8%	16%	135%
EMEREN GROUP LTD	SOL US	#N/A Real T	$ 0.1					1.84	1.74	#VALUE!	3%	#VALUE!
SHENZHEN KSTAR SCIENCE AND-A	002518 CH	45.37	$ 3.8	0.32	1.06	1.51	42%	44.61	31.06	-2%	10%	103%
SUNGROW POWER SUPPLY CO LT-	300274 CH	161.31	$ 47.5	2.01	7.14	8.39	18%	174.68	103.44	-7%	0%	124%
SUNRUN INC	RUN US	17.08	$ 4.0	0.04	1.42	0.28	80%	19.16	12.61	-9%	-1%	85%
Sytems & Services			**$ 204.0**									
ARRAY TECHNOLOGIES INC	ARRY	8.38	$ 1.3	(0.01)	0.67	0.84	25%	8.34	7.16	-9%	3%	39%
CHANGZHOU ALMADEN STOCK CO	002623 CH	17.88	$ 0.5		0.18	0.80	344%	19.44	18.31	-1%	-10%	20%
SHOALS TECHNOLOGIES GROUP -A	SHLS US	8.42	$ 1.4	0.14	0.39	0.49	27%	9.04	6.09	-12%	14%	52%
WOLFSPEED INC	WOLF	17.15	$ 0.4	(0.49)	(1.91)	(1.51)	21%			-14%	-40%	
EAST GROUP CO LTD-A	300376 CH	[illegible]	$ 2.0					5.66	4.41	-1%	15%	68%
FTC SOLAR INC	FTCI	9.01	$ 0.1	(0.22)	(2.70)	(0.04)	98%	8.90	5.69	-10%	28%	64%
GIGASOLAR MATERIALS CORP	3691 TT	76.00	$ 0.2					70.77	71.60	1%	-1%	-1%
HANGZHOU FIRST APPLIED MAT-A	603806 CH	13.17	$ 4.9	0.23	0.49	0.68	40%	14.96	14.10	-3%	-16%	-9%
JIANGSU LINYANG ENERGY CO -A	601222 CH	5.52	$ 1.6		0.37	0.43	15%	5.91	5.99	-2%	-10%	-18%
JIANGSU XIUQIANG GLASSWORK-A	300160 CH	5.47	$ 0.6					5.80	5.74	0%	-5%	2%
JOLYWOOD SUZHOU SUNWATT CO	300393 CH	6.52	$ 1.0					6.91	6.10	-5%	1%	4%
SHENZHEN TOPRAYSOLAR CO -A	002218 CH	4.23	$ 0.8					3.95	3.58	0%	17%	7%
SINO-AMERICAN SILICON PRODUC	5483 TT	102.00	$ 2.1	1.48	6.90	10.21	48%	112.07	108.55	-1%	-12%	-20%
SOLARWINDOW TECHNOLOGIES IN	WNDW US	0.43	$ 0.0					0.52	0.43	-4%	-35%	52%
XINYI SOLAR HOLDINGS LTD	968 HK	2.94	$ 3.5		0.12	0.21	73%	3.44	3.08	-1%	-15%	-5%
Other Parts & Services			**$ 20.5**									
			$ 398.3									

Please contact your salesperson for any of these screens

- We keep equity statistics for all of our 13-sub sectors and their respective segments and are happy to share them …

Available Energy Equity Screens

Solar
Battery Storage & Fuel Cells
Transport
US Power Generation
Transports
Mid-stream Infrastructure
Oil Field Services

Wind
Hydrogen
Metals & Rare Earths
Global Power Generation
Integrated Oil, Refining & Marketing
Oil & Gas E&Ps, Oil Sands

Source: PSC Research. Bloomberg Consensus Forecasts

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